On localization phenomena in plasticity models

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Introduction

Part 1

Existence results for a one-dimensional model of a plastic cylindrical bar

CHAPTER 1

Introduction

CHAPTER 2

Small deformation model

In this chapter we develop and analyze a small deformation model of a bar whose sections can only translate and change their size. In particular bending is not allowed. We split our discussion into three parts. The first section is devoted to the development of the model, the second to an existence and uniqueness results. The last section is devoted to the explicit solution of two boundary value problems describing the tensile testing of the bar. In both the cases we show that the small deformation model is not able to predict the necking phenomenon.

1. Formulation

1.1. Motivation from the multidimensional kinematics. In this section we describe the kinematics of the bar from a multidimensional perspective. This will serve as a motivation for the development of the purely one dimensional model discussed in the forthcoming sections.

Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0, L)$ to R. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2'y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p\colon (0, L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2'y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2}\kappa (\operatorname{tr} E_e)^2 + \mu |\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\operatorname{tr} E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa \left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2 \mu |u_2'|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_\Sigma \psi_e dy = \frac{1}{2}A\kappa\,(e_1 + (d-1)e_2)^2 + \frac{d-1}{d}A\mu(e_1 - e_2)^2 + \frac{1}{2}J\mu|u_2'|^2$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

1.2. Kinematics. We now develop a purely one dimensional model of the bar taking inspiration from the multidimensional kinematics.
Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is described by $u_i \colon (0, L) \to \mathbb{R}$ for $i \in \{1, 2\}$. Here u_1 is the longitudinal displacement of the sections in the axis direction and u_2 is the quantity describing the transversal expansion of the sections. The configuration of the microscopic structure of the bar is described by the plastic distortion $p \colon (0, L) \to \mathbb{R}$. We define $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$ the elastic distortions.

1.3. Statics. We denote with $\tilde{v}$ and $\tilde{q}$ the virtual rates of u and p respectively. To describe the coupling between the macroscopic configuration u and the microscopic configuration p we choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_0^L \zeta\tilde{v}_2' dx + \int_0^L \left(\sigma_1(\tilde{v}_1' - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d-1)^{-1}\tilde{q})\right) dx + \int_0^L \tau\tilde{q}dx.$$

The first integral describes a purely macroscopic response.. Here ζ is the transversal stress. The second integral describes the elastic response that couples the macroscopic and the microscopic configurations. Here σ_1 is the longitudinal elastic stress and σ_2 the longitudinal elastic force. The last integral describes the microscopic response and τ is the microscopic force. The external longitudinal and transversal forces distributed along the length of the bar and the external longitudinal and transversal forces acting on the extremities of the bar are included choosing the external power

$$\tilde{\mathcal{P}} = \int_0^L f \cdot \tilde{v}dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Proposition 2.1. The local form of the equilibrium condition is equivalent to

$$\sigma_1' + f_1 = 0 \quad \text{in } (0, L) \tag{2.1a}$$

$$h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\} \tag{2.1b}$$

$$-\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L) \tag{2.1c}$$

$$h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\} \tag{2.1d}$$

$$\sigma_1 - (d-1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L) \tag{2.1e}$$

Proof. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1'\tilde{v}_1 dx + \sigma_1(L)\tilde{v}_1(L) - \sigma_1(0)\tilde{v}_1(0) = \int_0^L f_1\tilde{v}_1 dx + h_1(L)\tilde{v}_1(L) - h_1(0)\tilde{v}_1(0).$$

We get (2.1a) and (2.1b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta'\tilde{v}_2 dx + \zeta(L)\tilde{v}_2(L) - \zeta(0)\tilde{v}_2(0) + \int_0^L \sigma_2\tilde{v}_2 dx = \int_0^L f_2\tilde{v}_2 dx + h_2(L)\tilde{v}_2(L) - h_2(0)\tilde{v}_2(0).$$

We get (2.1c) and (2.1d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d-1)^{-1}\sigma_2 - \tau)\tilde{q}dx = 0.$$

This implies (2.1e). □

1.4. Constitutive relations. We choose the free energy for unit axial length

$$\psi = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 + \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2.$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. We also choose the dissipation potential for unit axial length

$$\phi = R(\dot{p})$$

where $R\colon \mathbb{R} \to [0,\infty]$ satisfy $R(0) = 0$. A possible choice could be $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength. Consistently with these choices we impose the following constitutive relations

$$\begin{aligned}\zeta &= \mu l^2 u_2' \\ \sigma_1 &= \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\ \sigma_2 &= (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\ \tau &\in Bp + \partial R(\dot{p}).\end{aligned}$$

Proposition 2.2. Every process that obeys the constitutive relations satisfy the dissipation inequality.

Proof. Let $\tau = Bp + \xi$ with $\xi \in \partial R(\dot{p})$. The dissipation per unit axial length is given by

$$\zeta\dot{u}_2' + \sigma_1\dot{e}_1 + \sigma_2\dot{e}_2 + \tau\dot{p} - \dot{\psi} = \xi\dot{p}.$$

By definition of subdifferential we have $0 = R(0) \geq R(\dot{p}) - \xi\dot{p}$ so that $\xi\dot{p} \geq R(\dot{p}) \geq 0$. This implies that the dissipation is nonnegative. □

We notice that introducing the constitutive relations into (2.1e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is the elastic force acting on the microscopic configuration and we notice that it depends on the shear modulus only.

2. Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions. For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

2.1. Function spaces and hypotheses. We define the spaces $\boldsymbol{U} = H^1((0,L), \mathbb{R}\times\mathbb{R})$ and $\boldsymbol{P} = L^2((0,L))$. Let also $\boldsymbol{Z} = \boldsymbol{U}\times\boldsymbol{P}$. We will write $z = (u,p) \in \boldsymbol{Z}$ with $u \in \boldsymbol{U}$ and $p \in \boldsymbol{P}$ and $w = (v,q) \in \boldsymbol{Z}$ with $v \in \boldsymbol{U}$ and $q \in \boldsymbol{P}$.

The existence and uniqueness result will be obtained under the following hypotheses.

- (G) $L > 0$ and $d = 2$ or $d = 3$
- (ψ) κ, μ, l and B are positive constants
- (R) $R\colon \mathbb{R} \to \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$
- (f) $f \in H^1((0,T), \boldsymbol{U}^*)$ and $f(0) = 0$

2.2. Formulation of the problem. We introduce the linear operator $\mathcal{A}\colon \boldsymbol{Z} \to \boldsymbol{Z}^*$ defined by

$$\begin{aligned}\langle \mathcal{A}z, w\rangle = &\int_0^L l^2\mu u_2' v_2' dx + \int_0^L \left[\kappa(u_1' + (d-1)u_2) + \mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)\right](v_1' - q)dx + \\ &+ \int_0^L \left[(d-1)\kappa(u_1' + (d-1)u_2) - \mu\left(u_1' - u_2 - \frac{d}{d-1}q\right)\right]\left(v_2 + (d-1)^{-1}p\right)dx + \\ &+ \int_0^L Bpq\,dx\end{aligned}$$

and the functional $\mathcal{R}\colon \boldsymbol{Z} \to \mathbb{R} \cup \{\infty\}$ defined by

$$\mathcal{R}(w) = \int_0^L R(q)dx.$$

We define also $\mathcal{F}\colon [0,T] \to \boldsymbol{Z}^*$ by

$$\langle \mathcal{F}(t), w\rangle = \int_0^T f \cdot v dx.$$

Now the boundary value problem associated to the model introduced in Section 1 can be formulated as follows.

Problem 2.3. Find $z = (u, p) \in \mathrm{AC}([0,T], \boldsymbol{Z})$ such that $z(0) = 0$ and for almost every $t \in (0,T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t)\rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geq \langle \mathscr{F}(t), w - \dot{z}(t)\rangle \quad \text{for all } w \in \boldsymbol{Z}.$$

2.3. Existence and uniqueness theorem. To prove the existence and uniqueness result we make use of the following general theorem.

Theorem 2.4. Let $\boldsymbol{Z}$ be a reflexive and separable Banach space. Let $\mathcal{A}\colon \boldsymbol{Z} \to \boldsymbol{Z}^*$ be a linear and continuous functional which is also coercive, meaning that

$$\frac{\langle \mathcal{A}z, z\rangle}{\|z\|_Z} \to \infty \quad \text{as } \|z\|_Z \to \infty.$$

Let $\mathcal{R}\colon \boldsymbol{Z} \to \mathbb{R} \cup \{\infty\}$ be convex, proper, l.s.c., 1-homogeneous and Lipchitz continuous on its domain. Let $\mathcal{F} \in H^1((0,T), \boldsymbol{Z}^*)$ such that $\mathcal{F}(0) = 0$. Then there exists a unique $z \in \mathrm{AC}([0,T], \boldsymbol{Z})$ such that for almost every $t \in (0,T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t)\rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geq \langle \mathscr{F}(t), w - \dot{z}(t)\rangle \quad \text{for all } w \in \boldsymbol{Z}.$$

Proof. See [22]. □

The main result of this section is the following theorem.

Theorem 2.5. Assume the hypotheses of Section 2.1. Then there exists a unique solution to Problem 2.3.

Proof. We verify the hypothesis of Theorem 2.4.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z\rangle \geq \alpha\|z\|_Z^2$. If we call $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned}
\langle \mathcal{A}z, z\rangle = 2\int_0^L \psi dx &= \kappa\|e_1 + (d-1)e_2\|_2^2 + \mu\|e_1 - e_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 = \\
&= \kappa\|u_1' + (d-1)u_2\|_2^2 + \mu\left\|u_1' - u_2 - \frac{d}{d-1}p\right\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 \geq \\
&\geq (1-\theta)\,\kappa\|u_1'\|_2^2 + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa\|u_2\|_2^2 + (1-\tau)\mu\|u_1' - u_2\|_2^2 + \\
&\qquad l^2\mu\|u_2'\|_2^2 + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2 \\
&\geq \left[(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu\right]\|u_1'\|_2^2 + \\
&\quad + \left[\left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa + \left(1 - \frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + \\
&\qquad + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2
\end{aligned}$$

for all θ, τ and η in $(0,\infty)$. We restrict to θ, τ and η in $(0,1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned}\langle \mathcal{A}z, z\rangle \geq C_1 \left[(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu\right] \|u_1\|^2_{1,2}+\\ + \left[C_2 l^2\mu + \left(1-\frac{1}{\theta}\right)(d-1)^2\kappa + \left(1-\frac{1}{\eta}\right)(1-\tau)\mu\right] \|u_2\|^2_{1,2}+\\ + \left[\left(1-\frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2 \mu + B\right]\|p\|^2_2\end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|^2_{1,2}$, $\|u_2\|^2_{1,2}$ and $\|p\|^2_2$ are positive and we get the desired coercivity.

Step 2. Since $\mathcal{R}(w) = Y\|q\|_1$ we see immediately that $\mathcal{R}$ is convex, proper, continuous and 1-homogeneous.

Step 3. Since $f \in H^1((0,T), \boldsymbol{U}^*)$ and $f(0) = 0$ then $\mathcal{F} \in H^1((0,T), \boldsymbol{Z}^*)$ and $\mathcal{F}(0) = 0$. □

3. Some analytical solutions

3.1. Traction with free end sections. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $h_2(L,t) = h_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (2.1a) and (2.1b) we have that $\sigma_1 = F$ on all $(0,L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned}\sigma_2 &= (d-1)\left(F - \mu(e_1 - e_2)\right) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) =\\ &= (d-1)F - d\frac{\mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 =\\ &= \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.\end{aligned}$$

Now from equation (2.1c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L,t) = h_2(0,t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = F.$$

When $F \leq Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

3.2. Traction with clamped end sections.

Setup of the problem. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $u_2(L,t) = u_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 3.1 we find the equation

$$l^2 \mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa + \mu} F + \frac{d^2}{d-1} \frac{\kappa\mu}{\kappa + \mu} p + d^2 \frac{\kappa\mu}{\kappa + \mu} u_2. \tag{2.2}$$

Solution in the elastic range. In the elastic range $p = 0$ everywhere in $(0,L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa - \mu}{d^2 \kappa\mu} F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}} \frac{x - \frac{L}{2}}{l} \right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa - \mu}{d^2 \kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l} \right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l} \right)} - 1 \right) F.$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress is the elastic range is given by

$$m = \frac{d}{d-1} \frac{\mu}{\kappa + \mu} (F - d\kappa e_2) = \frac{d}{d-1} \frac{\mu}{\kappa + \mu} \left(1 - \frac{(d-1)\kappa - \mu}{d\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l} \right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l} \right)} - 1 \right) \right) F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first.

Solution in the plastic range. From the above considerations we are let to seek for a solution where $p > 0$ in the interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \in [0, L/2]$ is a function of time.

Remark 2.6. Since the solution is symmetric with respect to $x = L/2$ from now on we focus on the $(0, L/2)$ half of the bar only.

On $(0, L/2 - \epsilon)$ the solution is given by $p = 0$ and u_2 of the form

$$u_2 = \frac{(d-1)\kappa - \mu}{d^2 \kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l} \right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l} \right)} - 1 \right) F + C_1 \sinh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}} x \right) \tag{2.3}$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$

$$\frac{d}{d-1} \frac{\mu}{\kappa + \mu} (F - d\kappa u_2 - \frac{d\kappa}{d-1} p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1} \frac{\mu}{\kappa+\mu} F - \frac{d^2}{d-1} \frac{\kappa\mu}{\kappa+\mu} u_2 - Y}{\frac{d^2}{(d-1)^2} \frac{\mu\kappa}{\kappa+\mu} + B}.$$

Substituting that identity into equation (2.2) we have obtain a differential equation in y of the form

$$y'' = ay + b$$

where a and b depends on time only. A solution which is symmetric with respect to $x = L/2$ is of the form

$$y = -\frac{b}{a} + C_2 \cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right). \tag{2.4}$$

Remark 2.7. Since the computations became to long to be done by hand we perform them with an automatic algebraic manipulator (the MATLAB Symbolic Math Toolbox). Since most of the resultant expressions are too long we do not report them.

We now look for a solution $u_2 \in C^2([0, L])$. When $\epsilon = L/2$ the solution is simply given by

$$u_2 = \frac{b}{a}\left(\frac{\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right)}{\cosh\left(\sqrt{a}\frac{L}{2}\right)} - 1\right)$$

When $\epsilon \in (0, L/2)$ we consider (2.3) on $(0, L/2 - \epsilon)$ and (2.4) on $(L/2 - \epsilon, L/2)$ and we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$. Solving the resultant linear system we find the constants C_1 and C_2.

Determination of ϵ. We are left with the determination of $\epsilon \in [0, L/2]$ as a function of time. To this aim we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the value of F for which that Mandel stress reach Y. We find

$$F_Y = \frac{d-1}{d}\frac{k+\mu}{\mu}\left(-\frac{(d-1)\kappa - \mu}{d\mu}\left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right) + 1\right)^{-1} Y.$$

For $F \leq F_Y$ we have $\epsilon = 0$. To determine ϵ when $F > F_Y$ we consider the equation $m(L/2 - \epsilon) = Y$. That equation cannot be solved analytically in ϵ. Instead we can find explicitly F as a function $f =: [0, \epsilon_0) \to \mathbb{R}$ of ϵ. Here $[0, \epsilon_0)$ is the maximal interval where f is well defined. After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load F reach the value

$$f(L/2) = \frac{d-1}{d}\frac{\kappa+\mu}{\mu}Y,$$

the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. We conclude that

$$\epsilon = \begin{cases} 0 & \text{if } F \leq F_Y \\ f^{-1}(F) & \text{if } F_Y < F \leq f(L/2) \\ L/2 & \text{if } F > f(L/2). \end{cases}$$

Absence of necking. From the analytical expression one can see that for $F > f(L/2)$ the solution u_2 depends linearly on F. In particular as $F \to \infty$ we have that $u_2 \to -\infty$ uniformly on all compact sets contained in $(0, L)$. We conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section seems to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

The limit of thin bar. To get further insight on the solution we look at the limit $l \to 0^+$.

Proposition 2.8. In the limit $l \to 0^+$ the solution u_2 that we have found converges pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 3.1. This convergence is also uniform on compact subsets of $(0, L) \times [0, T]$.

Proof. It is a direct computation. □

This results confirm the absence of necking in the small deformation model. Indeed it says that a thin bar deforms almost uniformly.

Some plots. For completeness we report some plots illustrating the solution that we have found. We fix $d = 3$, $L = 1$, $l = 0.1$, $\kappa = 100$, $\mu = 100$, $B = 0.1$, $Y = 1$. In Figure 1 on the left we port the graph of f. We see that ϵ_0, which correspond to the abscissa of the vertical asymptote, is slightly greater then $L/2$. In Figure 2 we report the plot of u_2 and p when $\epsilon = 0.3$. In Figure 3 we report a picture of the bar when $\epsilon = 0.493$. On the top there is the result obtained with a purely elastic bar while on the bottom the result predicted including the plastic behavior. The displacements of cross sections is magnified with a factor 7. The color represents the variable p.

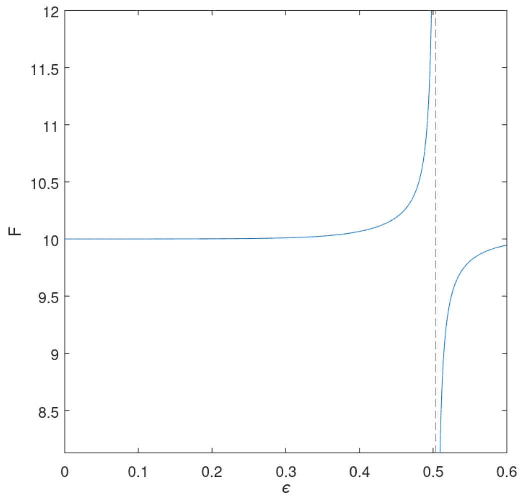

Figure 1. Graph of the function f.

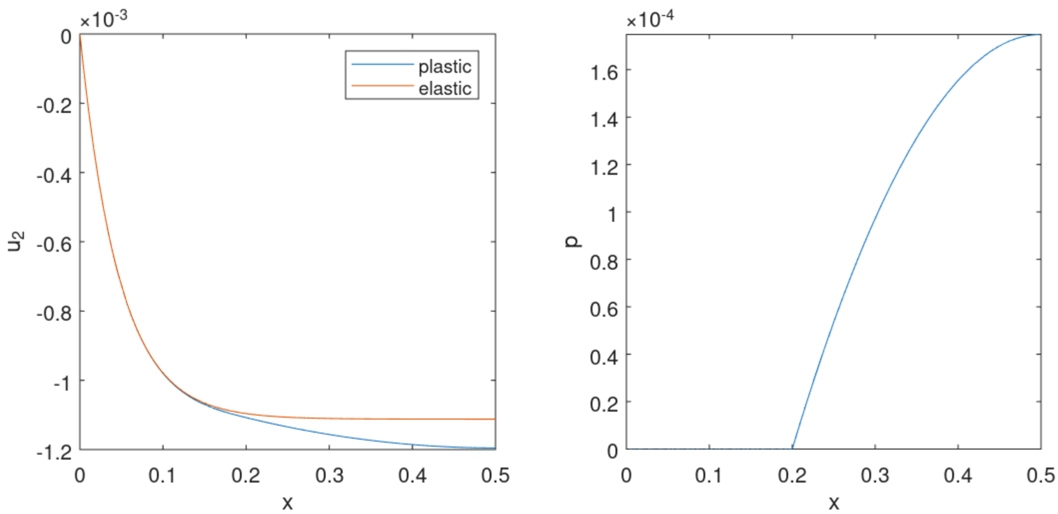

Figure 2. Plot of u_2 (left) and p (right) for $\epsilon = 0.3$.

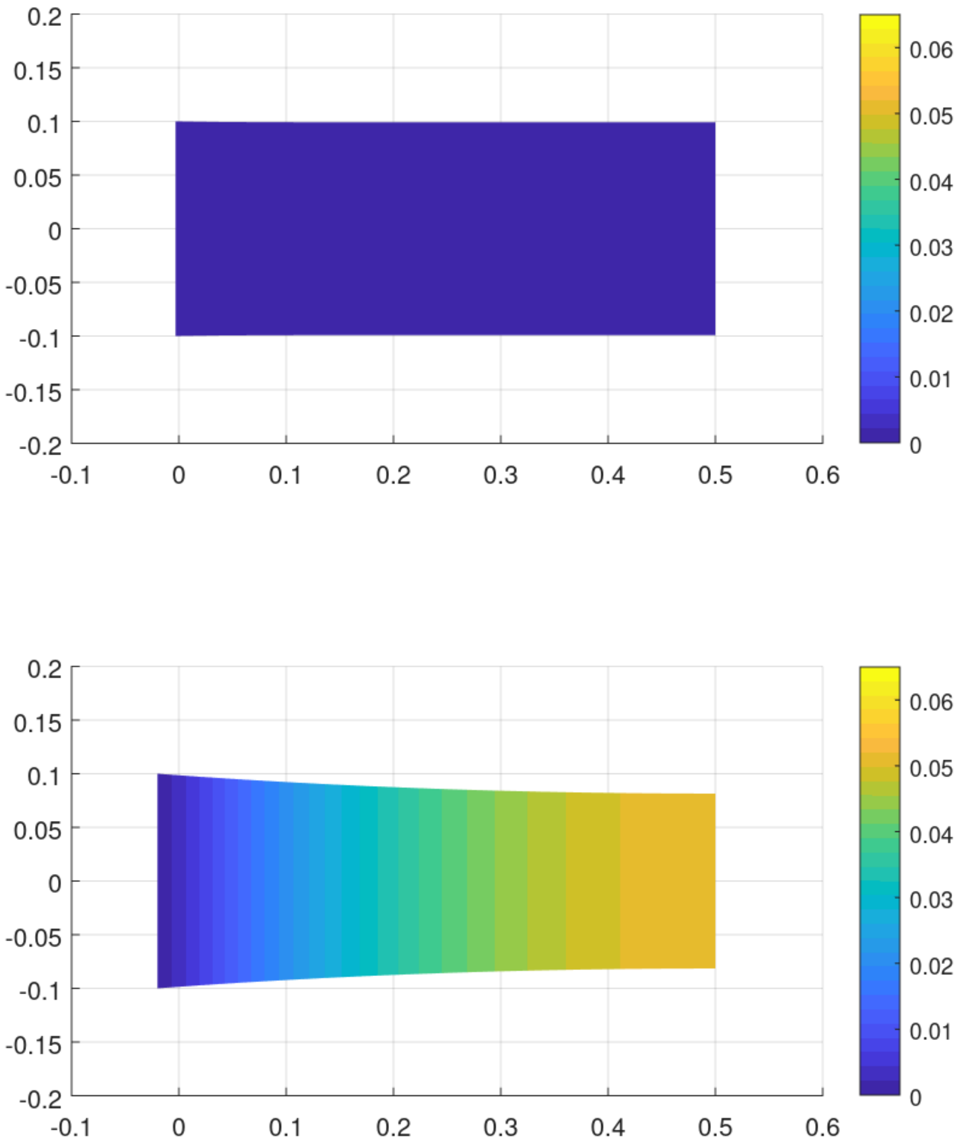

FIGURE 3. Picture of the half $(0, L/2)$ of the bar when $\epsilon = 0.493$. The bar on the bottom is plastic, the one on the top is purely elastic and subject to the same load. The transversal displacement is magnified with a factor 7. The color represents p.

CHAPTER 3

Large deformation model

As we showed in the previous chapter, a small strain model of a bar is not able to predict necking. This leads us to consider a geometrically exact model that allows large deformations.

1. Formulation

1.1. Motivation form the multidimensional kinematics. Let $\Omega = (0, L) \times \Sigma \subset \mathbb{R}^d$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ the reference configuration of the cylindrical bar. The generic material point will be denoted as $(X, Y) \in \Omega$ with $X \in (0, L)$ and $Y \in \Sigma$. We consider deformations of the form

$$(X, Y) \mapsto \begin{pmatrix} \chi_1(X) \\ \chi_2(X)Y \end{pmatrix}$$

where $\chi_1 \colon (0, L) \to \mathbb{R}$ satisfy $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. The corresponding deformation gradient is

$$F = \begin{pmatrix} \chi_1' & 0 \\ \chi_2' Y & \chi_2 I \end{pmatrix}.$$

We define $\epsilon_1 = \log \chi_1'$ and $\epsilon_2 = \log \chi_2$. The plastic distortion is assumed to be of the form

$$F_p = \begin{pmatrix} P & 0 \\ 0 & P^{-\frac{1}{d-1}} I \end{pmatrix}$$

where $P \colon (0, L) \to (0, \infty)$. This choice implies plastic incompressibility. We define $p = \log P$. The elastic distortion is then given by

$$F_e = F F_p^{-1} = \begin{pmatrix} \chi_1' P^{-1} & 0 \\ \chi_2' P^{-1} Y & \chi_2 P^{\frac{1}{d-1}} I \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ GY & E_2 I \end{pmatrix}.$$

We define

$$e_1 = \log E_1 = \epsilon_1 - p \quad \text{and} \quad e_2 = \log E_2 = \epsilon_2 + (d-1)^{-1} p.$$

Now the spatial velocity gradient is given by

$$L = \dot{F} F^{-1} = \begin{pmatrix} \dot{\epsilon}_1 & 0 \\ \left(\frac{\dot{\chi}_2'}{\chi_1'} - \frac{\chi_2'}{\chi_1'} \dot{\epsilon}_2 \right) & \dot{\epsilon}_2 I \end{pmatrix}$$

and the plastic rate is

$$L_p = \dot{F}_p F_p^{-1} = \begin{pmatrix} \dot{p} & 0 \\ 0 & -\frac{\dot{p}}{d-1} I \end{pmatrix}.$$

Then the elastic rate is

$$L_e = \dot{F}_e F_e^{-1} = L - F_e L_p F_e^{-1} = \begin{pmatrix} \dot{e}_1 & 0 \\ \left(\frac{\dot{G}}{E_1} - \frac{G}{E_1} \dot{e}_2 \right) Y & \dot{e}_2 I \end{pmatrix}.$$

Consider the case $d = 2$. In this case the first principal invariant of $C_e = F_e^T F_e$ is

$$I_e = \operatorname{tr}(C_e) = E_1^2 + G^2 |Y|^2 + E_2^2$$

and the determinant of F_e is

$$J_e = \det F_e = E_1 E_2.$$

$$\psi_e = C \left(\frac{I_e}{J_e} - 2 \right) + w(J) = C \left(\frac{E_1^2 + G^2 |Y|^2 + E_2^2}{E_1 E_2} - 2 \right) + w(E_1 E_2)$$

1.2. Kinematics. Let $L > 0$ the length of the cylindrical bar. The macroscopic configuration of the bar is described by $\chi_1\colon (0,L) \to \mathbb{R}$ with $\chi_1' > 0$ and $\chi_2\colon (0,L) \to (0,\infty)$. Here χ_1 describes the positions of the sections and χ_2 is the transversal deformation factor of the sections. The configuration of the microscopic structure of the bar, that is the plastic distortion, is described by $P\colon (0,L) \to (0,\infty)$. We define the elastic distortions as $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$.

1.3. Statics. We assume that the internal power is given by

$$\mathcal{W} = \int_0^L \left(\zeta \dot{G} + \sigma_1 \dot{E}_1 + \sigma_2 \dot{E}_2\right) dX + \int_0^L \left(\tau \dot{P} + \kappa \dot{P}'\right) dX.$$

Here ζ is the transversal elastic stress, σ_1 is the longitudinal elastic stress and σ_2 is the transversal elastic force while τ is the microscopic force and κ the microscopic stress.

Remark 3.1. In the small strain model of Chapter 2 we did not included the microscopic stress κ. However, we are now including it because we expect that in the large-strain regime, it is necessary to obtain an existence result in Sobolev spaces. This is suggested by the existence theorem for three-dimensional elastoplasticity at large strains presented in [23].

We notice that

$$\begin{aligned} G^{-1}\dot{G} &= (\chi_2')^{-1}\dot{\chi}_2' - P^{-1}\dot{P} \\ E_1^{-1}\dot{E}_1 &= (\chi_1')^{-1}\dot{\chi}_1' - P^{-1}\dot{P} \\ E_2^{-1}\dot{E}_2 &= \chi_2^{-1}\dot{\chi}_2 + (d-1)^{-1}P^{-1}\dot{P}. \end{aligned}$$

Then if we call $\tilde{v}_i$ the virtual rate of χ_i and $\tilde{Q}$ the virtual rate of P we are let to consider the internal virtual power

$$\begin{aligned} \tilde{\mathcal{W}} = \int_0^L & G\zeta\left((\chi_2')^{-1}\tilde{v}_2' - P^{-1}\tilde{Q}\right) dX + \\ &+ \int_0^L \left[E_1\sigma_1\left((\chi_1')^{-1}\tilde{v}_1' - P^{-1}\tilde{Q}\right) + E_2\sigma_2\left(\chi_2^{-1}\tilde{v}_2 + (d-1)^{-1}P^{-1}\tilde{Q}\right)\right] dX + \\ &+ \int_0^L \left[\tau\tilde{Q} + \kappa\tilde{Q}'\right] dX. \end{aligned}$$

The external longitudinal and transversal forces distributed along the length of the bar and the external longitudinal and transversal forces acting on the extremities of the bar are included choosing the external power

$$\tilde{\mathcal{P}} = \int_0^L f\cdot\tilde{v}\,dx + h(L)\cdot\tilde{v}(L) - h(0)\cdot\tilde{v}(0).$$

Proposition 3.2. The local form of the equilibrium condition is

$$\begin{gathered} (P^{-1}\sigma_1)' + f_1 = 0 \quad \text{in } (0,L) \\ h_1(X) = P(X)^{-1}\sigma_1(X) \quad \text{for } X \in \{0,L\} \\ -P^{\frac{1}{d-1}}\sigma_2 + (P^{-1}\zeta)' + f_2 = 0 \quad \text{in } (0,L) \\ h_2(X) = P(X)^{-1}\zeta(X) \quad \text{for } X \in \{0,L\} \\ \chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\zeta - \tau + \kappa' = 0 \quad \text{for } X \in \{0,L\} \\ \kappa(X) = 0 \quad \text{for } X \in \{0,L\} \end{gathered}$$

Proof. Rearranging the terms we have

$$\begin{aligned} \tilde{\mathcal{W}} = \int_0^L & \left[P^{-1}\sigma_1\tilde{v}_1' + P^{\frac{1}{d-1}}\sigma_2\tilde{v}_2 + P^{-1}\zeta\tilde{v}_2'\right] dX + \\ &+ \int_0^L \left[\left(\tau - \chi_1' P^{-2}\sigma_1 + (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 - \chi_2' P^{-2}\zeta\right)\tilde{Q} + \kappa\tilde{Q}'\right] dX. \end{aligned}$$

By integration by parts and application of the fundamental lemma of the calculus of variation we get the thesis. □

1.4. Constitutive relations. We choose a free energy for unit axial length of the form

$$\psi = W(E_1, E_2, G) + H(P, P')$$

where $W\colon (0,\infty)\times(0,\infty)\times\mathbb{R}\to\mathbb{R}$ is elastic free energy density and $H\colon (0,\infty)\times\mathbb{R}\to\mathbb{R}$ is the hardening free energy density. The plastic response is obtained by introducing a dissipation potential per unit axial length of the form

$$\phi = R(P^{-1}\dot{P})$$

where $R\colon \mathbb{R}\to[0,\infty]$ satisfies $R(0)=0$. Consistently with that choice of ψ and ϕ we impose the following constitutive relations

$$\sigma_i = \frac{\partial W}{\partial E_i} \quad \text{and} \quad \zeta = \frac{\partial W}{\partial G}$$

$$\tau \in \frac{\partial H}{\partial P} + P^{-1}\partial R(P^{-1}\dot{P}) \quad \text{and} \quad \kappa = \frac{\partial H}{\partial P'}.$$

Proposition 3.3. The constitutive relations (3.2) satisfy the dissipation inequality.

Proof. Let $\tau = \frac{\partial H}{\partial P} + P^{-1}\xi$ with $\xi \in \partial R(\dot{P}P^{-1})$. Then dissipation per unit axial length is given by

$$\zeta\dot{G} + \sigma_2\dot{E}_2 + \sigma_2\dot{E}_2 + \tau\dot{P} + \kappa\dot{P}' - \dot{\psi} = \xi P^{-1}\dot{P}.$$

By definition of subdifferential we have $0 = R(0) \geq R(P^{-1}\dot{P}) - \xi P^{-1}\dot{P}$ so that $\xi P^{-1}\dot{P} \geq R(P^{-1}\dot{P}) \geq 0$. This implies that the dissipation is nonnegative. □

Particular constitutive relations. Particular constitutive relations are the following:

$$W(E_1,E_2,G) = \begin{cases} \alpha_1|E_1|^{q_W} + \alpha_2|E_2|^{q_W} + \alpha_3 l^2|G|^{q_W} + \alpha_4\frac{1}{|E_1E_2|^{s_W}} & \text{of } E_1,E_2>0 \\ \infty & \text{otherwise} \end{cases}$$

$$H(P,P') = \begin{cases} \alpha_5|P|^{q_H} + \alpha_6\frac{1}{|P-p|^{s_H}} + \alpha_7|P'|^{r_H} & \text{if } P>p \\ \infty & \text{otherwise} \end{cases}$$

$$R(P,\dot{P}) = \begin{cases} \alpha_8\frac{|\dot{P}|}{P} & \text{if } P\geq p \\ \infty & \text{otherwise,} \end{cases}$$

where $\alpha_1,\dots,\alpha_8 > 0$, $s_W, s_H \in (0,\infty)$, $q_W, q_H, r_H \in (1,\infty)$ and $0 < p < 1$. This constitutive relations are motivated by the existence theorem that we prove in Section 2.

Remark 3.4. Let the dissipation distance density $D\colon \mathbb{R}\to[0,\infty]$ be defined by

$$D(P_0,P_1) = \inf\left\{\int_0^1 R(P(t),\dot{P}(t))dt \;\middle|\; P\in C^1([0,1]) \text{ and } P(i)=P_i \text{ for } i\in\{0,1\}\right\}.$$

Then

$$D(P_0,P_1) = \begin{cases} \alpha_8|\log P_1 - \log P_0| & \text{if } P_0,P_1\geq p \\ \infty & \text{otherwise.} \end{cases}$$

Indeed notice that for each $P\in C^1([0,1],(0,\infty))$ connecting P_0 with P_0, it holds

$$\int_0^1 R(P(t),\dot{P}(t))dt \geq \alpha_8\int_0^1\left|\frac{\dot{P}(t)}{P(t)}\right|dt \geq \alpha_8\left|\int_0^1\frac{\dot{P}(t)}{P(t)}dt\right| = \alpha_8|\log P_1 - \log P_0|.$$

This implies $D(P_0,P_1)\geq \alpha_8|\log P_1 - \log P_0|$. If $P_i < p$ for some $i\in\{0,1\}$, then every C^1 curve P connecting P_0 with P_1, has $\int_0^1 R(P(t),\dot{P}(t))dt = \infty$. If instead $P_0,P_1\geq p$ then we can take $P(t) = P_0^tP_1^{1-t}\geq p$ and in this case

$$\int_0^1 R(P(t),\dot{P}(t))dt = \alpha_8\int_0^1\left|\frac{\dot{P}(t)}{P(t)}\right|dt = \alpha_8|\log P_1 - \log P_1|.$$

This implies $D(P_0,P_1) = \alpha_8|\log P_1 - \log P_0|$

2. Existence result

In this section we prove the existence of energetic solutions. A self-contained introduction to this solution concept, including an explanation of the notation employed in this section, can be found in Section 1 of Appendix D. The aim of this section is to verify the hypotheses required to apply Theorem D.4.

2.1. Formulation of the initial boundary value problem. We now describe how the model formulated in the previous section, can be treated within the context of energetic solutions.

Dirichlet conditions. Suppose that at time $t \in [0,T]$ the following Dirichlet conditions are imposed

$$\chi_i(X) = g_i(X,t) \quad \text{for } X \in \Gamma_D^i \text{ and } i \in \{1,2\}, \tag{3.3}$$

where $\Gamma_D^i \subset \{0,L\}$ and $g_i\colon (0,L)\times[0,T] \to \mathbb{R}$ satisfy $g_1' > 0$ and $g_2 > 0$. To avoid free translations of the bar, it is required that $\Gamma_D^1 \neq \emptyset$, but Γ_D^2 can be empty instead. Conditions (3.3) can be converted into time-independent conditions by expressing χ_i as

$$\chi_1(X) = g_1(y_1(X),t) \quad \text{and} \quad \chi_2(X) = g_2(y_1(X),t)y_2(X),$$

with $y_i\colon (0,L) \to \mathbb{R}$, $y_1' > 0$ and $y_2 > 0$, and requiring that

$$y_1(X) = X \quad \text{for } X \in \Gamma_D^1 \quad \text{and} \quad y_2(X) = 1 \quad \text{for } X \in \Gamma_D^2.$$

Having made that change of variables, the unknown of the problem are y and P. We notice that the elastic distortions E_1, E_2 and G are given by

$$E_1 = g_1'(y_1,t)y_1'P^{-1}, \quad E_2 = g_2(y_1,t)y_2P \quad \text{and} \quad G = g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}.$$

Function spaces. Let $Y = W^{1,q_Y}((0,L),\mathbb{R}^2)$ with $q_Y \in (1,\infty)$ be the space of y and $Z = L^{q_H}((0,L)) \cap W^{1,r_H}((0,L))$ with $q_H \in (1,\infty)$, $r_H \in (1,q_H]$ be the space of P. Let $Q = Y \times Z$. The Dirichlet conditions are imposed requiring that y, P belong respectively to the admissible spaces

$$\mathscr{Y} = \left\{ y \in Y \,\middle|\, y_1(X) = X \text{ for } X \in \Gamma_D^1 \text{ and } y_2(X) = 1 \text{ for } X \in \Gamma_D^2 \right\} \quad \text{and} \quad \mathscr{Z} = Z.$$

Let $\mathcal{Q} = \mathscr{Y} \times \mathscr{Z}$. It is evident that $\mathcal{Q}$ is a weakly closed subset of Q.

Energy functional and dissipation distance. To complete the formulation of the initial boundary value problem, we need to specify the energy functional $\mathcal{E}\colon \mathcal{Q}\times[0,T] \to (-\infty,\infty]$ and the dissipation distance $\mathcal{D}\colon \mathscr{Z}\times\mathscr{Z} \to [0,\infty]$. **Mettere qualcosa, magari nella parte generale, per spiegare perché il modello descritto precedentemente, se espresso mediante le energetic solution, conduce a queste particolari $\mathcal{E}$ e $\mathcal{D}$.** The energy functional is defined by

$$\mathcal{E}(y,P,t) = \int_0^L W\left(g_1'(y_1,t)y_1'P^{-1}, g_2(y_1,t)y_2P, g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}\right)dX + \\ + \int_0^L H(P,P')dX - \int_0^L \left(f_1(t)g_1(y_1,t) + f_2(t)g_2(y_1,t)y_2\right)dX.$$

The dissipation distance can be expressed in term of the dissipation distance density $D\colon \mathbb{R}^2 \to [0,\infty]$, discussed in Remark 3.4, as

$$\mathcal{D}(P_0,P_1) = \int_0^L D(P_0,P_1)dX.$$

2.2. Assumptions and preliminary results. We now discuss the assumptions under which we will prove the validity of hypotheses (E1)-(E4), (D1)-(D2) and (C1)-(C2) of Section 1 in Appendix D. Subsequently, these assumptions will be implicitly supposed to be valid and will not be recalled.

Remark 3.5 (on notation). Sometimes the notation $G = E_3$ and $E = (E_1,E_2,E_3)$ will be used. Given $E,N \in \mathbb{R}^3$, then $NE \in \mathbb{R}^3$ is the component-wise multiplication between E and N, i.e., $(NE)_i = N_iE_i$ for $i \in \{1,2,3\}$.

The assumption on the Dirichlet data is

(G) $g_1 \in C^2(\mathbb{R}\times[0,T])$, $g_2 \in C^2(\mathbb{R}\times[0,T],(0,\infty))$ and there exists $0 < \gamma < \Gamma < \infty$ for $i \in \{1,2\}$ such that $\gamma < g_1' < \Gamma$, $\gamma < g_2 < \Gamma$ and $|g_2'| \le \Gamma$.

Only to prove Proposition 3.12 we will assume also the following additional requirement

($\hat{\mathrm{G}}$) $g_2' = 0$.

We do not know whether the proof works without ($\hat{\mathrm{G}}$). Anyway assuming ($\hat{\mathrm{G}}$), it is not restrictive for our purposes.

The elastic free energy density is supposed to satisfy the following assumptions

(W1) there exists $\mathbb{W}\colon \mathbb{R}^4 \to (-\infty,\infty]$ continuous such that

$$W(E_1,E_2,G) = \mathbb{W}(E_1,E_2,G,E_1E_2)$$

for all $E_1,E_2,G \in \mathbb{R}$ and $\mathbb{W}$ is convex in (E_1,G,E_1E_2),

(W2) there exist $C_W > 0$ and $q_W \in (1,\infty)$ such that $W(E_1,E_2,G) \ge C_W(-1+|E_1|^{q_W}+|E_2|^{q_W}+|G|^{q_W})$ for all $E_1,E_2,G \in \mathbb{R}$,

(W3) $W(E_1,E_2,G) = \infty$ if and only if $E_1 \le 0$ or $E_2 \le 0$,

(W4) W is differentiable in $\operatorname{dom} W$ and there exist $C_0^W \in \mathbb{R}$, $C_1^W > 0$ and a modulus of continuity ω^W such that
 (i) for all $E \in \operatorname{dom} W$ and $i \in \{1,2,3\}$ it holds $\left|E_i \frac{\partial W}{\partial E_i}(E)\right| \le C_1^W(C_0^W + W(E))$,
 (ii) let $K(E) = E\cdot\frac{\partial W}{\partial E}(E)$, then $|K(NE)-K(E)| \le \omega^W(|N-1|)(W(E)+C_0^W)$ for all $N \in (0,\infty)^3$.

Assumption (W1) is a sort policonvexity requirement necessary to ensure lower semicontinuity, (W2) is a growth condition, (W3) is an incompenetrability condition for the elastic distortion and (W4) is a further regularity assumption which is needed to get suitable a priori estimates. Later on we will use the following consequence of (W4).

Lemma 3.6. Fix $0 < \delta < 1$ small enough. Then there exists $C > 0$ such that

$$W(NE) + \max_{i\in\{1,2,3\}} \left|E_i \frac{\partial W}{\partial E_i}(NE)\right| \le C(W(E) + C_0^W)$$

for all $E \in (0,\infty)\times(0,\infty)\times\mathbb{R}$ and $N \in (1-\delta,1+\delta)^3$. Here $NE \in \mathbb{R}^3$ denotes the component-wise multiplication, i.e., $(NE)_i = N_iE_i$ for all $i \in \{1,2,3\}$.

Proof. *Step 1.* For all $i \in \{1,2,3\}$, by (W4, i)

$$\left|E_i\frac{\partial W}{\partial E_i}(NE)\right| = |N_i|^{-1}\left|N_iE_i\frac{\partial W}{\partial E_i}(NE)\right| \le |N_i|^{-1}C_1^W(C_0^W + W(NE)) \le \le \frac{C_1^W}{1-\delta}(C_0^W + W(NE)).$$

Step 2. We have that

$$|W(NE) - W(E)| = \left|\int_0^1 \frac{d}{d\lambda}W(\tilde{N}(\lambda)E)d\lambda\right|,$$

where $\tilde{N}(\lambda) = (1-\lambda)+\lambda N$, i.e., $\tilde{N}_i(\lambda) = (1-\lambda)+\lambda N_i \in (1-\delta,1+\delta)$ for all $i \in \{1,2,3\}$. Then

$$|W(NE)-W(E)| = \left|\int_0^1 \frac{\partial W}{\partial E}(\tilde{N}(\lambda)E)\cdot(N-1)d\lambda\right| \le \sum_{i=1}^3\int_0^1 \left|\tilde{N}_i(\lambda)E_i\frac{\partial W}{\partial E_i}(\tilde{N}(\lambda)E)\right|\frac{|N_i-1|}{\tilde{N}_i(\lambda)}d\lambda \le \le \frac{\delta}{1-\delta}C_1^W\int_0^1 (W(\tilde{N}(\lambda)E) + C_0^W)d\lambda.$$

Step 3. Let $\theta(E) = \sup_{N\in(1-\delta,1+\delta)^3} W(NE)$. From Step 2

$$|W(NE)-W(E)| \le \theta(E) - W(E) \le \frac{\delta}{1-\delta}C_1^W(\theta(E)+C_0^W).$$

This implies

$$\left(1-\frac{\delta}{1-\delta}C_1^W\right)\theta(E)\leq W(E)+\frac{\delta}{1-\delta}C_1^W C_0^W.$$

If δ is small enough then

$$\theta(E)\leq\frac{W(E)+\frac{\delta}{1-\delta}C_1^W C_0^W}{1-\frac{\delta}{1-\delta}C_1^W}.$$

Step 4. From Step 3

$$|W(NE)-W(E)|\leq\frac{\delta}{1-\delta}C_1^W\left(\frac{W(E)+\frac{\delta}{1-\delta}C_1^W C_0^W}{1-\frac{\delta}{1-\delta}C_1^W}+C_0^W\right)\leq\left(\frac{1}{\frac{\delta}{1-\delta}C_1^W}-1\right)^{-1}(W(E)+C_0^W).$$

Step 5. From Step 1

$$\max_{i\in\{1,2,3\}}\left|E_i\frac{\partial W}{\partial E_i}(NE)\right|\leq\frac{C_1^W}{1-\delta}(C_0^W+\theta(E))\leq\frac{C_1^W}{1-\delta}\left(\frac{W(E)+\frac{\delta}{1-\delta}C_1^W C_0^W}{1-\frac{\delta}{1-\delta}C_1^W}+C_0^W\right)=$$

$$=\frac{C_1^W}{1-\delta(1+C_1^W)}(W(E)+C_0^W).$$

Step 6. Summing together the results of Step 4 and Step 5 we get the thesis. □

The plastic free energy density satisfies

(H1) $H\colon\mathbb{R}^2\to(-\infty,\infty]$ is continuous and $H(P,\cdot)$ is convex for all $P\in\mathbb{R}$,

(H2) there exist $C_H>0$ and $q_H\in(1,\infty)$, $r_H\in(1,q_H]$ such that $H(P,P')\geq C_H(-1+|P|^{q_H}+|P'|^{r_H})$ for all $P,P'\in\mathbb{R}$,

(H3) there exists $p>0$ such that $H(P,P')=\infty$ if and only if $P\leq p$.

The convexity assumption (H1) is required for lower semicontinuity, (H2) is a growth condition ensuring that the hardening and the plastic gradient regularization is sufficiently strong. Assumption (H3) is a strong incompenetrability condition for the plastic distortion.

To handle correctly the nonlinear coupling between elastic and plastic distortions, we require the following compatibility condition on the Hölder exponents

(Ö) $\frac{1}{q_Y}=\frac{1}{q_W}+\frac{1}{q_H}$.

The requirements on the dissipation potential density are

(d1) $D\colon\mathbb{R}^2\to[0,\infty]$ is lower semicontinuous,

(d2) D is a quasidistance, in the sense that $D(P_0,P_1)=0$ if and only if $P_0=P_1$ and $D(P_0,P_2)\leq D(P_0,P_1)+D(P_1,P_2)$ for all $P_0,P_1,P_2\in\mathbb{R}$,

(d3) D is finite and continuous on $[p,\infty)\times[p,\infty)$, where p is the constant appearing in (H3).

2.3. Verification the properties of the energy functional. We start with the following lemma which is necessary to handle the nonlinear coupling between elastic and plastic distortion.

Lemma 3.7. Let $a,b>0$. Then for all $\delta>0$ and $\rho>1$

$$\frac{a}{b}\geq\rho\delta^{\frac{\rho-1}{\rho}}a^{\frac{1}{\rho}}-(\rho-1)\delta b^{\frac{1}{\rho-1}}$$

Proof. By Young inequality, if $A,B>0$ then

$$AB\leq\frac{A^\rho}{\rho}+\frac{B^{\rho^*}}{\rho^*}$$

and

$$A^{\frac{1}{\rho}}B^{\frac{1}{\rho^*}}\leq\frac{A}{\rho}+\frac{B}{\rho^*}.$$

It follows that

$$A \geq \rho A^{\frac{1}{\rho}} B^{\frac{1}{\rho^*}} - \frac{\rho}{\rho^*} B.$$

If we put $A = a/b$ and B such that

$$B^{\frac{1}{\rho^*}} = \delta^{\frac{1}{\rho^*}} b^{\frac{1}{\rho}}$$

the thesis follows. □

We now prove that (E2) is satisfied. We start with proving the boundedness of sublevels.

Proposition 3.8 (Boundedness of energy sublevels). The sublevels of $\mathcal{E}(\cdot, t)$ are bounded in Q for all $t \in [0, T]$.

Proof. From (W1) and (H1) it follows

$$\mathcal{E}(y, P, t) \geq C\Big(-1 + \|g_1'(y_1, t) y_1' P^{-1}\|_{q_W}^{q_W} + \|g_2(y_1, t) y_2 P\|_{q_W}^{q_W} + \\ + \|g_2'(y_1, t) y_1' y_2 P^{-1} + g_2(y_1, t) y_2' P^{-1}\|_{q_W}^{q_W} + \|P\|_{q_H}^{q_H} + \|P'\|_{r_H}^{r_H}\Big). \tag{3.4}$$

By Hölder inequality

$$\|y_1'\|_{q_Y} = \|y_1' P^{-1} P\|_{q_Y} \leq \|y_1' P^{-1}\|_{q_W} \|P\|_{q_H}$$

so that using Lemma 3.7 with $\rho = q_W/q_Y$ one gets

$$\|y_1' P^{-1}\|_{q_W}^{q_W} \geq \frac{\|y_1'\|_q^{q_W}}{\|P\|_{q_H}^{q_W}} \geq \rho \delta^{\frac{\rho-1}{\rho}} \|y_1'\|_{q_Y}^{q_Y} - (\rho - 1)\delta \|P\|_{q_H}^{q_H}$$

for all $\delta > 0$. We have used that $q_W/(\rho - 1) = q_H$. Moreover

$$\|y_1' P^{-1}\|_{q_W} = \|g_1'(y_1, t)^{-1} g_1'(y_1, t) y_1' P^{-1}\|_{q_W} \leq \gamma^{-1} \|g_1'(y_1, t) y_1' P^{-1}\|_{q_W}.$$

This implies that

$$\|g_1'(y_1, t) y_1' P^{-1}\|_{q_W}^{q_W} \geq \gamma^{q_W} \rho \delta^{\frac{\rho-1}{\rho}} \|y_1'\|_{q_Y}^{q_Y} - \gamma^{q_W} (\rho - 1)\delta \|P\|_{q_H}^{q_H}. \tag{3.5}$$

Similarly by Hölder and Young inequalities, for all $\delta > 0$

$$\|y_2\|_{q_Y}^{q_Y} = \|y_2 P P^{-1}\|_{q_Y}^{q_Y} \leq \|y_2 P\|_{q_W}^{q_Y} \|P^{-1}\|_{q_H}^{q_Y} \leq \frac{1}{\rho \delta^\rho} \|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} \|P^{-1}\|_{q_H}^{q_H} \leq \frac{1}{\rho \delta^\rho} \|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} L p^{-q_H}$$

where now $\rho = 1 + q_W/q_H$. Last inequality holds since $P > p$ a.e. on the sublevels of $\mathcal{E}$. Then

$$\|g_2(y_1, t) y_2 P\|_{q_W}^{q_W} \geq \gamma^{q_W} \rho \delta^\rho \|y_2\|_q^q - \gamma^{q_W} \frac{\rho}{\rho^*} \delta^{\rho + \rho^*} p^{-q_H}. \tag{3.6}$$

Substituting (3.5) and (3.6) into (3.4) with δ sufficiently small, one gets that on sublevels of $\mathcal{E}$ the norms $\|y_1'\|_{q_Y}$, $\|y\|_{q_Y}$, $\|P\|_{q_H}$ and $\|P'\|_{r_H}$ are bounded. By Poincaré inequality it follows that also $\|y_1\|_{1,q_Y}$ is bounded. It remains to prove that $\|y_2'\|_{q_Y}$ is bounded. Let

$$G = g_2'(y_1, t) y_1' y_2 P^{-1} + g_2(y_1, t) y_2' P^{-1}.$$

Then

$$y_2' = \frac{GP - g_2'(y_1, t) y_1' y_2}{g_2(y_1, t)}.$$

Il follows that

$$\|y_2'\|_{q_Y} \leq \frac{1}{\gamma} \|GP\|_{q_Y} + \frac{1}{\gamma} \|g_2'(y_1, t) y_1' y_2\|_{q_Y} \leq \frac{1}{\gamma} \|G\|_{q_W} \|P\|_{q_H} + \frac{\Gamma}{\gamma} \|y_1' y_2\|_{q_Y} \leq \\ \leq \frac{1}{\gamma} \|G\|_{q_W} \|P\|_{q_H} + \frac{\Gamma}{\gamma} \|y_1'\|_{q_Y} \|y_2\|_\infty.$$

Since $L^\infty((0, T))$ is continuously embedded in $W^{1,q_Y}((0, L))$, it follows that for all $\epsilon > 0$ there exists $C > 0$ such that

$$\|y_2\|_\infty \leq C(\|y_2\|_{q_Y} + \epsilon \|y_2'\|_{q_Y}) \quad \text{for all } y_2 \in W^{1,q_Y}((0, L)).$$

On a sublevel of $\mathcal{E}$ the quantities $\|G\|_{q_W}$, $\|P\|_{q_H}$, $\|y_1'\|_{q_Y}$ and $\|y_2\|_{q_Y}$ are bounded, so that there exists $M > 0$ such that

$$\|y_2'\|_{q_Y} \leq M(1 + \epsilon \|y_2'\|_{q_Y}).$$

Choosing ϵ small enough, it follows that $\|y_2'\|_{q_Y}$ is bounded on the sublevels of $\mathcal{E}$. This concludes the proof. □

We now move to the proof of the weak lower semicontinuity of $\mathcal{E}$. We first need the following technical result.

Proposition 3.9 (convergence of elastic distortion). Let $y^k, y \in W^{1,q_Y}((0,L), \mathbb{R}\times\mathbb{R})$ and $P_k, P \in C^0([0,L])$ such that

$$\begin{aligned} y^k &\rightharpoonup y \quad \text{in } W^{1,q_Y}((0,L)) \\ P_k &\to P \quad \text{in } C^0([0,L]). \end{aligned}$$

Assume also that $P_k, P \geq p > 0$ a.e. in $(0,L)$. Let also $\gamma_1^k, \gamma_1, \gamma_2^k, \gamma_2, \rho_2^k, \rho_2 \in C^0([0,L])$ such that

$$\begin{aligned} \gamma_1^k &\to \gamma_1 \quad \text{in } C^0([0,L]) \\ \gamma_2^k &\to \gamma_2 \quad \text{in } C^0([0,L]) \\ \rho_2^k &\to \rho_2 \quad \text{in } C^0([0,L]) \end{aligned}$$

Then, up to extracting a subsequence,

$$\begin{aligned} \gamma_1^k (y_1^k)' P_k^{-1} &\rightharpoonup \gamma_1 y_1' P^{-1} && \text{in } L^1((0,L)) \\ \gamma_2^k y_2^k P_k &\to \gamma_2 y_2 P && \text{in } C^0([0,L]) \\ \rho_2^k (y_1^k)' y_2^k P_k^{-1} + \gamma_2^k (y_2^k)' P_k^{-1} &\rightharpoonup \rho_2 y_1' y_2 P^{-1} + \gamma_2 y_2' P^{-1} && \text{in } L^1((0,L)) \\ \gamma_1^k (y_1^k)' \gamma_2^k y_2^k &\rightharpoonup \gamma_1 y_1' \gamma_2 y_2 && \text{in } L^1((0,L)) \end{aligned}$$

Proof. First notice that $P_k^{-1} \to P^{-1}$ in $L^\infty((0,L))$. Indeed

$$|P_k^{-1} - P^{-1}| = \frac{|P_k - P|}{|P_k P|} \leq \frac{|P_k - P|}{p^2}.$$

Notice also that, since $W^{1,q}((0,L))$ is compactly embedded in $C^0([0,L])$, up to extracting a subsequence $y_2^k \to y_2$ in $C^0([0,L])$. Now the results follows from Lemma 3.10 below. □

Lemma 3.10. Let $f_k, f \in L^q((0,L))$ with $q \in (1,\infty)$, $g_k, g \in L^\infty((0,L))$ and $h_k, h \in C^0([0,L])$ such that

$$\begin{aligned} f_k &\rightharpoonup f \quad \text{in } L^q((0,L)) \\ g_k &\to g \quad \text{in } L^\infty((0,L)) \\ h_k &\to h \quad \text{in } C^0([0,L]). \end{aligned}$$

Then $f_k g_k h_k \rightharpoonup fgh$ in $L^1((0,L))$.

Proof. Let $\phi \in L^\infty((0,L))$. Then

$$\begin{aligned} &\left| \int_0^L (f_k g_k h_k - fgh)\phi dX \right| \leq \\ &\qquad \leq \left| \int_0^L (g_k - g) f_k h_k \phi dX \right| + \left| \int_0^L g f_k (h_k - h)\phi dX \right| + \left| \int_0^L g(f_k - f) h\phi dX \right|. \end{aligned}$$

Since f_k are bounded in $L^q((0,L))$, h_k are bounded in $C^0([0,L])$, then $f_k h_k \phi$ and $g f_k \phi$ are bounded in $L^1((0,L))$. Moreover $gh\phi \in L^{q^*}((0,L))$. Then the thesis follows. □

We are now ready to get the weakly lower semicontinuity of $\mathcal{E}$.

Proposition 3.11 (lower semicontinuity of the energy). For all $t \in [0,T]$ the map $\mathcal{E}(\cdot, t)$ is lower semicontinuous.

Proof. Let $(y^k, P_k) \rightharpoonup (y, P)$ in $Y \times Z$. We have to prove that

$$\mathcal{E}(y, P, t) \leq \liminf_{k\to\infty} \mathcal{E}(y_k, P_k, t).$$

We take a (not relabeled) subsequence such that $\mathcal{E}(y_k, P_k, t)$ has a limit. Then we need to show that

$$\mathcal{E}(y, P, t) \leq \lim_{k\to\infty} \mathcal{E}(y_k, P_k, t). \tag{3.7}$$

If $\mathcal{E}(y_k, P_k, t) \to \infty$ we have nothing to prove. Then we assume that instead it converges to a finite limit. In this case we can assume that $\mathcal{E}(y_k, P_k, t) < \infty$ for all k. This implies that $P_k > p$ a.e. in $(0, L)$, so that also $P \geq p$ a.e. in $(0, L)$.

By Rellich–Kondrachov theorem, up to extracting a subsequence, $y_i^k \to y_i$ for $i \in \{1, 2\}$ and $P_k \to P$ in $C^0([0, L])$. Moreover $g_1'(y_1^k, t) \to g_1'(y_1, t)$, $g_2(y_1^k, t) \to g_2(y_1, t)$ and $g_2'(y_1^k, t) \to g_2'(y_1, t)$ in $C^0([0, L])$. Indeed if $\gamma \in C^1(\mathbb{R})$ then

$$\begin{aligned} |\gamma(y_1^k) - \gamma(y_1)| &= \left|\int_0^1 \frac{d}{d\lambda}\gamma((1-\lambda)y_1^k + \lambda y_1)d\lambda\right| = \left|\int_0^1 \gamma'((1-\lambda)y_1^k + \lambda y_1)(y_1 - y_1^k)d\lambda\right| \leq \\ &\leq \|y_1 - y_1^k\|_\infty \int_0^1 |\gamma'((1-\lambda)y_1^k + \lambda y_1)|d\lambda. \end{aligned} \tag{3.8}$$

Since y_1^k converges to y_1 uniformly, then $y_1^k(t)$ takes values in a bounded set as $k \in \mathbb{N}$ and $t \in [0, T]$. Then the continuity of γ' implies that the last integral in (3.8). We have proved that $\gamma(y_1^k) \to \gamma(y_1)$ in $C^0([0, T])$, and this provides the desired results.

Applying Proposition 3.9, it follows that, up to extracting a subsequence,

$$\begin{aligned} g_1'(y_1^k, t)(y_1^k)'P_k^{-1} &\rightharpoonup g_1(y_1, t)y_1'P^{-1} && \text{in } L^1((0, L)) \\ g_2(y_1^k, t)y_2^k P_k &\to g_2(y_1, t)y_2 P && \text{in } C^0([0, L]) \\ g_2'(y_1^k, t)(y_1^k)'y_2^k P_k^{-1} + g_2(y_1^k, t)(y_2^k)'P_k^{-1} &\rightharpoonup g_2'(y_1, t)y_1'y_2P^{-1} + g_2(y_1, t)y_2'P^{-1} && \text{in } L^1((0, L)) \\ g_1'(y_1^k, t)(y_1^k)'g_2(y_1^k, t)y_2^k &\rightharpoonup g_1'(y_1, t)y_1'g_2(y_1, t)y_2 && \text{in } L^1((0, L)). \end{aligned}$$

By these convergence results, together with $P_k \to P$ in $C^0([0, L])$ and $P_k' \rightharpoonup P'$ in $L^{r_H}((0, T))$, and by the convexity assumptions (W0) and (H0), we can apply Theorem C.25 which leads to (3.7). This concludes the proof. □

Until now we have established (E1) and (E2). We now prove (E3) and (E4).

Proposition 3.12. Then there exists $\mathcal{D} \subset \mathcal{Q}$ such that $\operatorname{dom} \mathcal{E}(\cdot, t) = \mathcal{D}$ for all $t \in [0, T]$. Moreover

(i) for all $q \in \mathcal{D}$ the function $\mathcal{E}(q, \cdot)$ is $C^1([0, T])$,
(ii) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E(C_0^E + \mathcal{E}(q, t))$ for all $q \in \mathcal{Q}$ and $t \in [0, T]$,
(iii) there exists a modulus of continuity ω^E such that for all $q \in \mathcal{D}$ and $t, s \in [0, T]$,

$$|\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| \leq \omega^E(|t - s|) \min\{\mathcal{E}(q, t) + C_0^W, \mathcal{E}(q, s) + C_0^W\}.$$

Proof. Since $g_1' > 0$ and $g_2 > 0$, it is clear that

$$\mathcal{D} = \left\{(y, P) \in Y \times Z \;\middle|\; y_1', y_2, P > 0 \text{ a.e. on } (0, L)\right\}.$$

Proof of (i) Let $E_1 = g_1'(y_1, t)y_1'P^{-1}$, $E_2 = g_2(y_1, t)y_2P$ and $G = g_2(y_1, t)y_2'P^{-1}$ as functions of $(y, y', P, t) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \times t \in [0, T]$. Let $q = (y, P) \in \mathcal{D}$. Then $(E_1, E_2, G) \in \operatorname{dom} W$ a.e. on $(0, L)$ and everywhere on $[0, T]$. Then $t \mapsto W(E_1, E_2, G)$ is differentiable in $[0, T]$ and a.e. in $(0, L)$ and

$$\frac{d}{dt}W(E_1, E_2, G) = E_1\frac{\partial W}{\partial E_1}(E_1, E_2, G)\frac{\dot{g}_1'(y_1, t)}{g_1'(y_1, t)} + E_2\frac{\partial W}{\partial E_2}(E_1, E_2, G)\frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)} + G\frac{\partial W}{\partial G}(E_1, E_2, G)\frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}$$

there. We have to prove that for all $t \in [0, T]$, there exists

$$\partial_t \mathcal{E}(q, t) = \lim_{\epsilon\to 0} \frac{\mathcal{E}(q, t + \epsilon) - \mathcal{E}(q, t)}{\epsilon} = \int_0^L \frac{d}{dt}W(E_1, E_2, G)dX. \tag{3.9}$$

By Lagrange theorem there exists $\eta \in (0,1)$ depending on X, t and ϵ, such that

$$\frac{\mathcal{E}(q,t+\epsilon)-\mathcal{E}(q,t)}{\epsilon} = \int_0^L W(g_1'(y_1,t+\eta\epsilon)y_1'P^{-1}, g_2(y_1,t+\eta\epsilon)y_2P, g_2(y_1,t+\eta\epsilon)y_2'P^{-1})dX.$$

For ϵ small, $g_1'(y_1,t+\eta\epsilon)/g_1'(y_1,t)$ and $g_2(y_1,t+\eta\epsilon)/g_2(y_1,t)$ are close to 1 uniformly on $(0,L)\times[0,T]$. Thus, for ϵ small enough, we can apply Lemma 3.6 and we get

$$\begin{aligned} -C_0^W &\leq W(g_1'(y_1,t+\eta\epsilon)y_1'P^{-1}, g_2(y_1,t+\eta\epsilon)y_2P, g_2(y_1,t+\eta\epsilon)y_2'P^{-1}) \leq \\ &\leq C(W(g_1'(y_1,t)y_1'P^{-1}, g_2(y_1,t)y_2P, g_2(y_1,t)y_2'P^{-1}) + C_0^W) \in L^1((0,L)). \end{aligned}$$

Then, by dominated convergence theorem, we can pass to the limit $\epsilon \to 0$ and we get (3.9).

Proof of (ii). Taking into account the fact that $\frac{\dot{g}_1'}{g_1'}$ and $\frac{\dot{g}_2}{g_2}$ are bounded in $\mathbb{R}\times[0,T]$, assumption (W4, i) yields

$$|\partial_t\mathcal{E}(y,P,t)| \leq 3C_1^W \max\left\{\left\|\frac{\dot{g}_1'}{g_1'}\right\|_\infty, \left\|\frac{\dot{g}_2}{g_2}\right\|_\infty\right\}\int_\Omega (C_0^W + W(E_1,E_2,G))dX.$$

This implies the existence of $C_0^E \in \mathbb{R}$ and $C_1^E > 0$ such that $|\partial_t\mathcal{E}(q,t)| \leq C_1^E(C_0^E + \mathcal{E}(q,t))$ for all $(q,t) \in \mathcal{D}\times[0,T]$.

Proof of (iii). Let

$$K_i = E_i\frac{\partial W}{\partial E_i} \quad \text{for } i \in \{1,2,3\} \quad \text{and} \quad V = \left(\frac{\dot{g}_1'(y_1,t)}{g_1'(y_1,t)}, \frac{\dot{g}_2(y_1,t)}{g_2(y_1,t)}, \frac{\dot{g}_2(y_1,t)}{g_2(y_1,t)}\right)$$

as functions of $(X,t) \in (0,L)\times[0,T]$. Then

$$\frac{d}{dt}W(E) = K\cdot V.$$

Let $E \in \mathbb{R}$ and $q \in \mathcal{D}$ such that $\sup_{t\in[0,T]}\mathcal{E}(q,t)$ and let $t,s \in [0,T]$. By the regularity properties (G) of g, there exists modulus of continuity ω^V and ω^G independent of q such that

$$|V(t)-V(s)| \leq \omega^V(|t-s|) \quad \text{and} \quad \max\left\{\frac{g_1'(y_1,s)}{g_1'(y_1,t)}, \frac{g_2(y_1,s)}{g_2(y_1,t)}\right\} \leq \omega^V(|t-s|)$$

on $(0,L)$. Moreover $|K| \leq 3C_1^W(W(E)+C_0^W)$ and there exists $M > 0$ independent of q, such that $\|V\|_\infty \leq M$. Then by (W2, i) and (W2, ii),

$$\begin{aligned} |\partial_t\mathcal{E}(q,t) - \partial_t\mathcal{E}(q,s)| &\leq \int_0^L |K(t)\cdot V(t) - K(s)\cdot V(s)|dX \leq \\ &\leq \int_0^L |K(t)-K(s)||V(s)|dX + \int_0^L |K(t)||V(t)-V(s)|dX \leq \\ &\leq \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W\omega^V(|t-s|)\right)\int_0^L (W(E(t)) + C_0^W)dX \\ &\leq \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W\omega^V(|t-s|)\right)(\mathcal{E}(q,t)+C_0^W). \end{aligned}$$

This concludes the proof. □

2.4. Verification the properties of the dissipation distance. Notice that (d1) implies that $\mathcal{D}$ is well defined. Property (D1) follows immediately form (d2). We now prove (D2).

Proposition 3.13. Let $P_k \rightharpoonup P$ and $\hat{P}_k \rightharpoonup \hat{P}$ in Z. Then $\mathcal{D}(P,\hat{P}) \leq \liminf_{k\to\infty}\mathcal{D}(P_k,\hat{P}_k)$.

Proof. Let $P_k, P, \hat{P}_k, \hat{P} \in Z$ such that

$$P_k \rightharpoonup P \quad \text{and} \quad \hat{P}_k \rightharpoonup \hat{P} \quad \text{in } Z.$$

By Proposition C.2 it follows that

$$P_k \to P \quad \text{and} \quad \hat{P}_k \to \hat{P} \quad \text{in } C^0([0,L]).$$

Now

$$\liminf_{k\to\infty} \mathcal{D}(P_k,\hat{P}_k) = \liminf_{k\to\infty} \int_0^L D(P_k,\hat{P}_k)dX = \lim_{k\to\infty}\inf_{j\geq k}\int_0^L D(P_j,\hat{P}_j)dX \geq \lim_{k\to\infty}\int_0^L \inf_{j\geq k} D(P_j,\hat{P}_j)dX.$$

By monotone convergence the last limit can be took inside the integral yielding

$$\liminf_{k\to\infty} \mathcal{D}(P_k,\hat{P}_k) \geq \int_0^L \liminf_{k\to\infty} D(P_k,\hat{P}_k)dX \geq \int_0^L D(P,\hat{P})dX.$$

The last inequality holds since D is lower semicontinuous. This concludes the proof. □

2.5. Verification of the compatibility conditions. We now verify (C1) and (C2). In view of Proposition D.5 we need to verify (PC1) and (PC2). Notice that (PC2) has been established in Proposition 3.12 (iii). Thus we need to verify (PC1) only. A stronger version of (PC1) is the following proposition.

Proposition 3.14. Let $(q_k,t_k),(q,t)\in Q\times[0,T]$ such that $(q_k,t_k)\rightharpoonup(q,t)$ in $Q\times[0,T]$ and $q_k\in\mathscr{D}$. Then for all $\hat{q}\in Q$,

$$\limsup_{k\to\infty}\left(\mathcal{E}(\hat{q},t_k)+\mathcal{D}(q_k,\hat{q})\right)\leq\mathcal{E}(\hat{q},t)+\mathcal{D}(q,\hat{q}).$$

Proof. Let $(q_k,t_k)\rightharpoonup(q,t)$ in $Q\times[0,T]$. If $\hat{q}\notin\mathscr{D}$ there is nothing to prove since $\mathcal{E}(\hat{q},t)=\infty$. Then assume that $\hat{q}\in\mathscr{D}$. By Proposition'(3.12) the map $\mathcal{E}(\hat{q},\cdot)$ is continuous. Then $\mathcal{E}(\hat{q},t_k)\to\mathcal{E}(\hat{q},t)$ and it remains to prove only that

$$\limsup_{k\to\infty}\mathcal{D}(q_k,\hat{q})\leq\mathcal{D}(q,\hat{q}).$$

Recall that this is the same as

$$\limsup_{k\to\infty}\mathcal{D}(P_k,\hat{P})\leq\mathcal{D}(P,\hat{P}).$$

Since $\hat{q}\in\mathscr{D}$, then (H3) implies that $\hat{P}>p$ a.e. in $(0,L)$, so that $\hat{P}\geq p$ in $[0,L]$. Since also $q_k\in\mathscr{D}$ then, for the same reason, also $P_k\geq p$ in $[0,L]$. Notice also that by C.2, $P_k\to P$ in $C^0([0,T])$. Then actually

$$\lim_{k\to\infty}\mathcal{D}(P_k,\hat{P})=\lim_{k\to\infty}\int_0^L D(P_p,\hat{P})dX=\mathcal{D}(P,\hat{P}),$$

where last inequality holds because D is finite and continuous on $[p\ infty)\times[p,\infty)$ by (d3). This concludes the proof. □

2.6. Conclusion. The application of Theorem D.4 leads to the desired existence result.

Theorem 3.15. Let $q_0=(y_0,P_0)\in\mathcal{Q}$ such that $\mathcal{E}(q_0,0)<\infty$. Then there exists an energetic solution $q=(y,P)\colon[0,T]\to\mathcal{Q}$ such that $q(0)=q_0$.

3. Tensile test

We consider the following boundary value problem: the bar in not subject to any distributed load, that is $f=0$. Moreover the end sections are left free to slide in the transversal direction, this means that $h_2(X)=0$ for $X\in\{0,L\}$ and $\Gamma_D^2=\emptyset$. The longitudinal position of the end sections is assigned, this means that $\Gamma_D^1=\{0,L\}$. In particular we consider the boundary conditions $\chi_1(0)=0$ and $\chi_1(L)=a(t)L$, where $a\colon[0,T]\to[1,\infty)$ is the imposed elongation.

3.1. Purely elastic model. In the purely elastic model, equations (3.1) reduces to

$$\begin{aligned}\sigma_1'&=0 \quad \text{in } (0,L)\\ -\sigma_2+\zeta'&=0 \quad \text{in } (0,L)\\ \zeta(X)&=0 \quad \text{for } X\in\{0,L\}.\end{aligned}$$

From the first equation it follows that $\sigma_1=F$, where F is a constant, representing the axial reaction force needed to impose the boundary conditions on χ_1. The constitutive relations yields $\sigma_1=W_1(E_1,E_2,G)$, $\sigma_2=W_2(E_1,E_2,G)$ and $\zeta=W_G(E_1,E_2,G)$, where $E_1=\chi_1'$, $E_2=\chi_2$ and $G=\chi_2'$.

We have denoted the partial derivatives of W with respect to its arguments E_1, E_2 and G with the subscripts $1, 2$ and G respectively. Inserting these constitutive relations into (3.10), we get

$$\begin{gathered}\frac{d}{dX}W_1(\chi_1', \chi_2, \chi_2') = 0 \quad \text{in } (0, L)\\ -W_2(\chi_1', \chi_2, \chi_2') + \frac{d}{dX}W_G(\chi_1', \chi_2, \chi_2') = 0 \quad \text{in } (0, L)\\ W_G(\chi_1'(X), \chi_2(X), \chi_2'(X)) = 0 \quad \text{for } X \in \{0, L\}.\end{gathered}$$

We first look for homogeneous solutions of the form $\chi_1^*(X) = aX$, $\chi_2^*(X) = E_2^*$ where $a = a(t)$ is the assigned elongation and $E_2^* \in (0, \infty)$ is unknown. Notice that $E_1^* = a$ and $G^* = (\chi_2^*)' = 0$. Then E_2^* must satisfy the nonlinear equation

$$W_2(E_1^*, E_2^*, 0) = 0.$$

To study the stability of the homogeneous solution, we look for a perturbed solution $\chi_1 = \chi_1^* + u_1$ and $\chi_2 = \chi_2^* + u_2$, we linearize around the homogeneous solution, and we seek for nontrivial perturbations u. The linearized equations are

$$\begin{gathered}W_{11}^* u_1'' + W_{12}^* u_2' + W_{1G}^* u_2'' = 0 \quad \text{in } (0, L)\\ \left.\begin{aligned}- W_{12}^* u_1' - W_{22}^* u_2 - W_{2G}^* u_2' +\\ + W_{1G}^* u_1'' + W_{2G}^* u_2' + W_{GG}^* u_2'' = 0\end{aligned}\right. \quad \text{in } (0, L)\\ W_{1G}^* u_1'(X) + W_{2G}^* u_2(X) + W_{GG}^* u_2'(X) = 0 \quad \text{for } X \in \{0, L\}.\end{gathered}$$

We have denoted with a "*" superscript, the evaluation at $(E_1^*, E_2^*, 0)$. From mechanical ground, it is reasonable to assume that $\sigma_i = W_i$ for $i \in \{1, 2\}$ is even in G and $\zeta = W_G$ is odd. Then $W_{iG}^* = 0$ for $i \in \{1, 2\}$. Moreover we assume that $W_{GG}^* > 0$, $W_{11}^* > 0$, $W_{22}^* > 0$. Then

$$\begin{gathered}W_{11}^* u_1'' + W_{12}^* u_2' = 0 \quad \text{in } (0, L)\\ -W_{12}^* u_1' - W_{22}^* u_2 + W_{GG}^* u_2'' = 0 \quad \text{in } (0, L)\\ u_2'(X) = 0 \quad \text{for } X \in \{0, L\}.\end{gathered}$$

Differentiating the second equation and eliminating u_1'' from the first equation, we get

$$\begin{gathered}W_{GG}^* u_2''' + \left(\frac{(W_{12}^*)^2}{W_{11}^*} - W_{22}^*\right) u_2' = 0 \quad \text{in } (0, L)\\ u_2'(X) = 0 \quad \text{for } X \in \{0, L\}.\end{gathered}$$

Let $\xi = u_2'$ and

$$\lambda = \frac{\frac{(W_{12}^*)^2}{W_{11}^*} - W_{22}^*}{W_{GG}^*}.$$

Then $\xi'' + \lambda\xi = 0$ with $\xi(X) = 0$ for $X \in \{0, L\}$, admits a nontrivial solution $\xi(X) = \sin(\sqrt{\lambda}X)$, if and only if $\sqrt{\lambda}L = k\pi$ for some $k \in \mathbb{N} = \{1, 2, \dots\}$. This means that the homogeneous solution is stable until E_1^* is such that $\lambda(E_1^*) = \pi^2/L^2$.

Remark 3.16. Let $F^* = W_1^*$ be the axial force obtained in the homogeneous solution. Then

$$\frac{dF^*}{da} = W_{11}^* + W_{12}^* \frac{dE_2^*}{da}.$$

On the other end we know that $W_2^* = 0$ so that, by differentiating with respect to a,

$$W_{12}^* + W_{22}^* \frac{dE_2^*}{da} = 0 \quad \text{that implies} \quad \frac{dE_2^*}{da} = -\frac{W_{12}^*}{W_{22}^*}.$$

Then

$$\frac{dF^*}{da} = W_{11}^* - \frac{(W_{12}^*)^2}{W_{22}^*} = -W_{GG}^* \frac{W_{11}^*}{W_{22}^*}\lambda.$$

Since at the first stages of deformation, $\frac{dF^*}{dE_1^*} > 0$, then $\lambda < 0$. If $\lambda = \pi^2/L^2$ instead $\frac{dF^*}{dE_1^*} < 0$. This means that instability can occur only after that a local maximum value of F^* is attained.

An example. Now we show that there exists a particular constitutive model that is likely to have unstable homogeneous solutions. Consider

$$W(E_1, E_2, G) = \begin{cases} \mathbb{W}(E_1, E_2, G, E_1E_2) & \text{if } E_1 > 0 \text{ and } E_2 > 0 \\ \infty & \text{otherwise,} \end{cases}$$

with

$$\mathbb{W}(E_1, E_2, G, J) = A\frac{|E_1 - 1|^p + |E_2 - 1|^p + |lG|^p}{J^\alpha} + B|J-1|^r$$

where $A, B > 0$, $r > 1$, $0 < \alpha \leq p - 1$ and $p > 1 + \frac{\alpha}{r}$. This particular W satisfies assumptions (W1), (W2), (W3) and (W4) of previous section. In particular it satisfies the convexity condition (W1) but it is not convex in (E_1, E_2, G), this is crucial to get instability. Indeed if W was convex, then the energy would have a unique minimizer and the homogeneous solution would be always stable.

We start with the check of the assumptions (W1) and (W2). Hypotheses (W1) is a consequence of the following lemma.

Lemma 3.17. Let $w\colon (0,\infty) \to \mathbb{R}$ strictly convex and let $g\colon (0,\infty)\times(0,\infty) \to \mathbb{R}$ given by $g(r,y) = \frac{r^p}{y^\alpha} + w(y)$ for some $p > 1$ and $0 < \alpha \leq p-1$. Let also $n \in \mathbb{N}$ and $f\colon \mathbb{R}^n \times (0,\infty) \to \mathbb{R}$ given by $f(x,y) = g(|x|, y)$. Then f is strictly convex in $\mathbb{R}^n \times (0,\infty)$.

Proof. We first prove that g is strictly convex. The hessian matrix of g is given by

$$\nabla^2 g(r,y) = \begin{pmatrix} p(p-1)\frac{r^{p-2}}{y^\alpha} & -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} \\ -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} & \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) \end{pmatrix}.$$

Then

$$\begin{aligned}\det \nabla^2 g(r,p) &= p(p-1)\frac{r^{p-2}}{y^\alpha}\left(\alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y)\right) - p^2\alpha^2\frac{r^{2p-2}}{y^{\alpha+2}} = \\ &= p\alpha(p-\alpha-1)\frac{r^{2p-2}}{y^{\alpha+2}} + p(p-1)\frac{r^{p-2}}{y^\alpha}w''(y) > 0.\end{aligned}$$

and

$$\operatorname{tr}\nabla^2 g(r,y) = p(p-1)\frac{r^{p-2}}{y^\alpha} + \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) > 0.$$

This proves that $\nabla^2 g$ is everywhere positive-definite. This implies that g is strictly convex. Then for any $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^n \times (0,\infty)$ and $0 < \lambda < 1$ it holds

$$\begin{aligned} f((1-\lambda)(x_0,y_0) + \lambda(x_1,y_1)) &= \frac{|(1-\lambda)x_0 + \lambda x_1|^p}{y^\alpha} + w(y) \leq \\ &\leq \frac{[(1-\lambda)|x_0| + \lambda|x_1|]^p}{\big[(1-\lambda)y_0 + \lambda y_1)\big]^\alpha} + w((1-\lambda)y_0 + \lambda y_1)) = \\ &= g((1-\lambda)(|x_0|, y_0) + \lambda(|x_1|, \lambda_1)) < (1-\lambda)g(|x_0|, y_0) + \lambda g(|x_1|, y_1) = \\ &= (1-\lambda)f(x_0,y_0) + \lambda f(x_1,y_1). \end{aligned}$$

This is exactly the thesis. □

We now prove the growth condition (W2) for $q_W = p/\big(1 + \frac{\alpha}{r}\big) > 1$. Lemma 3.7 yields for all $\delta > 0$ and $\rho > 1$ that

$$\begin{aligned}\mathbb{W}(E_1, E_2, G, J) \geq A\rho\delta^{\frac{\rho-1}{\rho}}|E_1 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|E_2 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|lG|^{\frac{p}{\rho}} - \\ - (2A+1)(\rho-1)\delta J^{\frac{\alpha}{\rho-1}} + B|J-1|^r.\end{aligned}$$

Choosing ρ such that $\frac{\alpha}{\rho-1} = r$ we get $\frac{p}{\rho} = q_H$. Moreover choosing $\delta > 0$ sufficiency small we have

$$\inf_{J>0}\left[-(2A+1)(\rho-1)\delta J^r + B|J-1|^r\right] > -\infty.$$

This is implied immediately from the following lemma.

Lemma 3.18. Let $r \geq 1$. Then there exists $C > 0$ such that $|x-1|^r \geq C|x|^r - 1$ for all $x \in \mathbb{R}$.

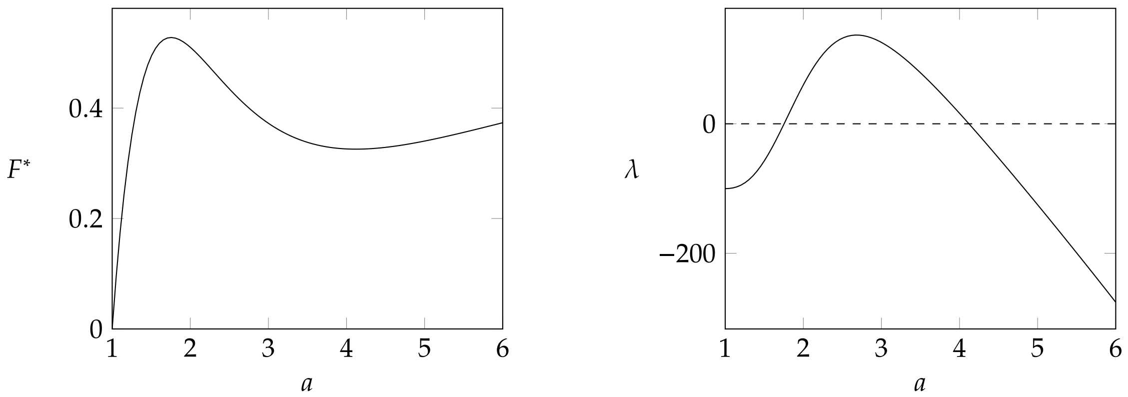

FIGURE 1. Graphs of F^* and λ as functions of the imposed elongation a for the homogeneous deformation in the purely elastic model when $A = 1$, $B = 10^{-4}$, $p = 2$, $\alpha = 1$ and $r = 2$. Notice that λ became positive when a is sufficiently large, indicating that the uniform solution is likely to became unstable if L is large enough.

PROOF. By the equivalence of the norms there exists $K > 0$ such that

$$(|x-1|^r + 1)^{1/r} \geq K(|x-1| + 1) \geq K|x|.$$

Now the thesis follows immediately. □

Thus we have that W satisfies the required assumptions. Now we show that W provides homogeneous solutions that are likely to become unstable. Indeed choosing the following particular values of the parameter appearing in the definition of W

$$A = 1,\, B = 10^{-4},\, p = 2,\, \alpha = 1 \text{ and } r = 2, \tag{3.15}$$

we get the graphs of F^* and λ as functions of a in Figure 1. In particular notice that λ became positive as E_1^* is increased. This implies that, if L is sufficiently large, then there exists a such that $\lambda = \pi^2/L^2$.

REMARK 3.19. We notice that the particular choice of the parameters in (3.15) is not mechanically reasonable for metals, and was motivated only by the need to show that the mechanical model that we have developed in this chapter is able to capture instability phenomena. The inaccuracy of (3.15) stems to the fact that, since B is small compared to A, the nearly-compressibility behavior of metals is not correctly reproduced. Moreover (3.15) implies that cross-sections growth in traction, this is clearly wrong for metals.

3.2. Plastic model. I still have to complete this part. Up to now I have develop the constitutive model bellow. It satisfies the assumptions of the existence theory developed in previous section. Since in the purely elastic model the lost of stability of the uniform solution is accompanied by the non-monotonicity of the tensile force F^* as a function of a, we expect that this is also true in the plastic model. Thus I tried to develop a model that exhibits a non-monotone relationship between F^* and a. It turned out that a simple quadratic hardening free energy density H does not satisfy this requirement. For that reason I looked for other possibilities and at the end the Armstrong–Frederick appeared as a reasonable choice.

An example. Consider the elastic free energy density proposed previously for the purely elastic model. Consider also the following hardening free energy densities,

(i) a simple quadratic energy density, with an additional penalization as P approaches to $0 < p < 1$,

$$H(P, P') = \begin{cases} C(P-1)^2 + \frac{D}{P-p} + E|lP'|^q & \text{if } P > p \\ \infty & \text{otherwise,} \end{cases}$$

(ii) and a more refined model

$$H(P,P') = \begin{cases} C\left(P^q + \frac{q}{\delta}e^{-\delta(P-1)}\right) + \frac{D}{P-p} + E|lP'|^q & \text{if } P > p \\ \infty & \text{otherwise,} \end{cases}$$

where $C, D, E, \delta > 0$ and $q > 1$. These constitutive relations both satisfies the assumptions (H1), (H2) and (H3).

Remark 3.20. The expression of H in (ii) is obtained as a modification of the Armstrong–Frederick kinematical hardening model. According to the Armstrong–Frederick model, the backstress $\beta = \frac{\partial H}{\partial P}$ satisfies

$$\dot{\beta} = \delta(C\dot{P} - |\dot{P}|\beta) \tag{3.16}$$

for some $\delta, C > 0$, see [35]. In tension $\dot{P} \geq 0$ and (3.16) becomes

$$\delta\dot{P} = \frac{\dot{\beta}}{C-\beta} = -\frac{d}{dt}\log(C-\beta).$$

This suggest the relation

$$\log\frac{C-\beta}{C} = -\delta(P-1) \quad \text{so that} \quad \beta = C(1-e^{-\delta(P-1)}). \tag{3.17}$$

The main feature of this relation is that β is bounded from above by C and $P \to C^-$ as $P \to \infty$. Now (3.17) corresponds to the energy density

$$H = C\left(P + \frac{1}{\delta}e^{-\delta(P-1)}\right)$$

and formula in (ii) is a modification of this as the linear growth in P has been substituted with a superlinear growth in order to satisfy (H2). The superlinear growth makes β unbounded as $\beta \to \infty$, but the main features of the original Armstrong–Frederick model are preserved. **Ma poi per garantire che i sottolivelli siano limitati, non andebbe a limite anche bene che H abbia crescita solo lineare in P? Tanto poi il termine col gradiente compensa questa cosa.** In particular H in (ii) is non-convex in P.

Let also

$$R(P,\dot{P}) = \begin{cases} Y\frac{|\dot{P}|}{P} & \text{if } P \geq p \\ \infty & \text{otherwise,} \end{cases}$$

with $Y > 0$. In Remark 3.4 we already showed that the corresponding dissipation distance density $D\colon \mathbb{R} \to [0,\infty]$,

$$D(P_0,P_1) = \begin{cases} \alpha_8|\log P_1 - \log P_0| & \text{if } P_0, P_1 \geq p \\ \infty & \text{otherwise,} \end{cases}$$

satisfies the assumptions (d1), (d2) and (d3).

In Figure 2 is reported the graph of the tensile force F^* along the homogeneous solution as a function of a. The solid line correspond to H in (ii) while the dashed line to H in (i).

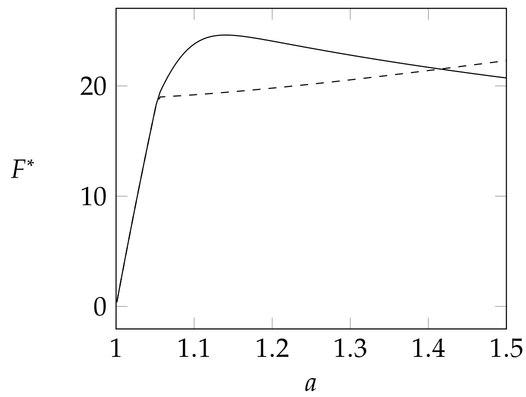

Figure 2. Graphs of F^* and λ as functions of the elongation a along the homogeneous solution in the plastic model. The solid line is referred to the model H in (ii) and the dashed line to H in (i).

Part 2

Existence results for a model describing pressure sensitive plastic materials

CHAPTER 4

Introduction

CHAPTER 5

Small deformation model

1. Formulation

In this section we formulate the mechanical model. The description is made very quickly and we will expand it to give a better justification. In this section the emphasis is on the mechanical modeling, the correct functional setting will be investigated in the subsequent sections.

1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u\colon \Omega \to \mathbb{R}^d$. We set $E = \operatorname{sym}\operatorname{grad} u$ and $M = \operatorname{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $u = \overline{u}$. The configuration of the microscopic structure is described by the plastic distortion $H_p\colon \Omega \to \mathrm{Lin}$. The skew symmetric part $W_p = \operatorname{skew} H_p$ describes the rotation of the microscopic structure and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \operatorname{sym} H_p\colon \Omega \to \mathrm{Sym}$ which describes the deformation of microscopic structure. We can regard E_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$. The volumetric deformation of the microscopic structure is described by $\lambda_p = \operatorname{tr} E_p$ while the deviatoric deformation by $U_p = \operatorname{dev} E_p$. We call $M_p = \operatorname{grad} U_p$ and we set $\mathbb{U}_p = (U_p, l\,M_p) \in \mathrm{DevPla}$ where $l > 0$ is an internal length scale. Here $\mathrm{DevPla} = \mathrm{Sym} \otimes (\mathrm{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic strains and their gradients. In what follows we will denote with $\circ$ the natural scalar product on DevPla. We assume that the material is not dilatant (this is a simplification and should be generalized to describe the dilatant behavior observed in most materials exhibiting a pressure dependent yield strength) so we add the constrain $\lambda_p = 0$. To impose this constrain we add the Lagrange multiplier $p\colon \Omega \to \mathbb{R}$ with the meaning of a plastic pressure. The constrain can be written in weak form as

$$\int_\Omega \lambda_p \tilde{q}\, dx = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

1.2. Statics. We denote with $\tilde{v}$, $\tilde{L}_p$ $\tilde{V}_p$ and $\tilde{\mu}_p$ the virtual rates of u, E_p, U_p and λ_p respectively. The virtual rate of E_e is $\tilde{L}_e = \operatorname{grad}\tilde{v} - \tilde{L}_p$ where $\tilde{L} = \operatorname{grad}\tilde{v}$. Due to the presence of the Dirichlet condition we require that $\tilde{v}$ vanish on Γ_D. Instead we do not require that $\tilde{\mu}_p = 0$ because the constrain $\lambda_p = 0$ is imposed by mean of the Lagrange multiplier p. To describe the coupling between the macroscopic deformation u and the microscopic configuration E_p we choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_\Omega \left(T : \tilde{L} + K \,\vdots\, \operatorname{grad}^2 \tilde{v} \right) dx + \int_\Omega T_e : \tilde{L}_e dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad} \tilde{V}_p \right) dx.$$

The first integral is related to the macroscopic response. Here $T \in \mathrm{Sym}$ is the macrostress and $K \in \mathrm{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order of $\tilde{v}$ is due to the need of a sufficient regularization in view of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e \in \mathrm{Sym}$ is the elastic stress. The third integral is associated with the microscopic response. Here $T_p \in \mathrm{DevSym}$ is the microforce and $K_p \in \mathrm{DevSym} \otimes \mathbb{R}$ is the microstress. In Figure 1 we report a picture that represents schematically the meaning of the internal virtual power that we have chosen. We define $\tilde{\mathbb{V}}_p = (\tilde{V}_p, l \operatorname{grad} \tilde{V}_p)$ and $\mathbb{T}_p = (T_p, l^{-1} K_p)$ so that

$$T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad} \tilde{V}_p = \mathbb{T}_p \circ \tilde{\mathbb{V}}_p.$$

The action of the plastic pressure is obtained by considering the reaction virtual power

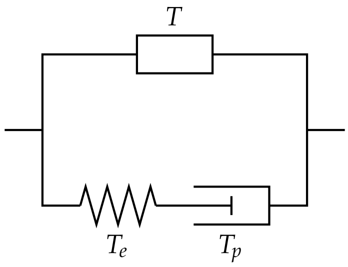

FIGURE 1. Rheological model describing the choice of the internal virtual power $\tilde{\mathcal{W}}$. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the microscopic response. We notice that the microscopic response besides the usual dissipative plastic microforce force can include conservative microforces to model hardening. A more detailed discussion on constitutive relations is in Section 1.3.

$$\tilde{\mathcal{R}} = -\int_\Omega p \tilde{\mu}_p dx.$$

We notice that this power vanish on virtual rates that satisfy the constrain that is $\tilde{\mu}_p = 0$. The external actions of the volume force f on Ω and the traction h on Γ_N are included choosing the external virtual power

$$\tilde{\mathcal{P}} = \int_\Omega f \cdot \tilde{v} dx + \int_{\Gamma_N} h \cdot \tilde{v} dx.$$

For simplicity we assume that no external torque is applied to the body and that there is no external action on the microscopic configuration. These external actions could be added to describe more accurately the boundary conditions imposed on the body.

LEMMA 5.1. The local form of the equilibrium condition is

$$\begin{aligned}
&\operatorname{div}(T + T_e - \operatorname{div} K) + f = 0 \quad \text{in } \Omega \\
&(T + T_e - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) = h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega \\
&\{\!\{(Kn)\tau\}\!\} = 0 \quad \text{in } \partial^{d-2}\Omega \\
&-\operatorname{dev} T_e + T_p - \operatorname{div} K_p = 0 \quad \text{in } \Omega \\
&p = -\frac{1}{d} \operatorname{tr} T_e \quad \text{in } \Omega \\
&K_p n = 0 \quad \text{in } \partial^{d-1}\Omega.
\end{aligned}$$

PROOF. By integration by parts we can write

$$\begin{aligned}
&\int_\Omega \left(T : \operatorname{grad} \tilde{v} + K \,\vdots\, \operatorname{grad}^2 \tilde{v} \right) dx = \\
&\qquad = -\int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} \left[(T - \operatorname{div} K)n \cdot \tilde{v} + Kn : \operatorname{grad} \tilde{v} \right] dx.
\end{aligned}$$

Now denoting with $P = I - n \otimes n$ the projector on $\partial^{d-1}\Omega$ we have

$$Kn : \operatorname{grad} \tilde{v} = Kn : \left(\frac{\partial \tilde{v}}{\partial n} \otimes n + \operatorname{grad}_P \tilde{v} \right) = (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} + Kn : \operatorname{grad}_P \tilde{v}$$

and

$$Kn : \operatorname{grad}_P \tilde{v} = (Kn)P : \operatorname{grad}_P \tilde{v} = \operatorname{div}_P(P(Kn)^T \tilde{v}) - \operatorname{div}_P((Kn)P) \cdot \tilde{v}.$$

But by integration by parts

$$\int_{\partial^{d-1}\Omega} \operatorname{div}_P(P(Kn)^T \tilde{v}) dx = \int_{\partial^{d-2}\Omega} \{\!\{P(Kn)^T \tilde{v} \cdot \tau\}\!\} dx = \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.$$

Here $\{\!\{\cdot\}\!\}$ denotes sum of its argument evaluated on both sides of $\partial^{d-2}\Omega$ and τ denotes the unit normal vector to $\partial^{d-2}\Omega$ and outer tangent to $\partial^{d-1}\Omega$. We have obtained

$$\int_\Omega \left(T : \operatorname{grad} \tilde{v} + K \vdots \operatorname{grad}^2 \tilde{v} \right) dx =$$

$$= - \int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n - \operatorname{div}_P((Kn)P)] \cdot \tilde{v} dx +$$

$$+ \int_{\partial^{d-1}\Omega} (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} dx + \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.$$

Moreover the remaining terms of the virtual power can be written as

$$- \int_\Omega \operatorname{div} T_e : \tilde{v} dx + \int_{\partial^{d-1}\Omega} T_e n \cdot \tilde{v} dx + \int_\Omega (T_p - \operatorname{dev} T_e - \operatorname{div} K_p) : \tilde{V}_p dx +$$

$$+ \int_{\partial^{d-1}\Omega} K_p n : \tilde{V}_p dx - \int_\Omega \frac{1}{d} \operatorname{tr} T_e \tilde{\mu} dx.$$

Now the results follows from the fundamental lemma of the calculus of variations. □

1.3. Constitutive relations. We assume a constitutive relation for the free energy density of the form

$$\psi = \frac{1}{2} \mathfrak{C} E_e : E_e + \frac{1}{2} \mathfrak{A} M \vdots M + \frac{1}{2} \mathfrak{B} \mathbb{U}_p \circ \mathbb{U}_p.$$

with $\mathfrak{C}$, $\mathfrak{A}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym, $\mathrm{Lin} \otimes \mathbb{R}$ and DevPla respectively. Moreover we choose a dissipation potential density of the form

$$\phi = \frac{1}{2} \mathfrak{D} \dot{E} : \dot{E} + \frac{1}{2} \mathfrak{E} \dot{M} \vdots \dot{M} + \frac{1}{2} \mathfrak{F} \dot{\mathbb{U}}_p \circ \dot{\mathbb{U}}_p + Y(p) G(\dot{U}_p).$$

Here $\mathfrak{D}$, $\mathfrak{E}$, $\mathfrak{F}$ are symmetric and positive definite linear operators on Sym, $\mathrm{Lin} \otimes \mathbb{R}$ and DevPla respectively. Moreover $G\colon \mathrm{DevSym} \to [0, \infty]$ is the convex and subdifferentiable plastic dissipation potential satisfying $G(0) = 0$ and $Y\colon \mathbb{R} \to [0, \infty)$ is the pressure dependent yield strength. Consistently with these choices of the free energy density and of the dissipation potential density we impose the following constitutive relations. The macroscopic response is described by

$$T = \mathfrak{D} \dot{E} \quad \text{and} \quad K = \mathfrak{A} M + \mathfrak{E} \dot{M}.$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite linear operators on Sym and $\mathrm{Lin} \otimes \mathbb{R}$ respectively. The elastic response is given by

$$T_e = \mathfrak{C} E_e$$

and the viscoplastic response by

$$\mathbb{T}_p \in \mathfrak{B} \mathbb{U}_p + \mathfrak{F} \dot{\mathbb{U}}_p + Y(p) \partial_{\dot{U}_p} G(\dot{U}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla, $G\colon \mathrm{DevSym} \to \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p \geq 0$ and $Y\colon \mathbb{R} \to [0, \infty)$ is the pressure dependent yield strength.

Remark 5.2. In the above expression we are viewing $Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) \in \mathrm{DevSym}$ as an element of DevPla by considering DevSym as a subspace of DevPla.

1.4. Refinements of the model. To include the dilatancy in the model we could do the following. We put the unilateral constrain $\lambda_p \geq F(U_p)$, meaning that when the microscopic structure suffer shear distortions it must suffer also an expansion. This constrain is still imposed through the Lagrange multiplier p representing the microscopic pressure. To make the expansion irreversible (once the microscopic structure expanded it can not be compacted again) we add a microscopic force ξ conjugate to $\tilde{\mu}_p$ in the internal virtual power and in the dissipation potential density we add $\iota_{[0,\infty)}(\dot{\xi})$. In this way ξ opposes to the compaction of the microscopic structure.

2. Mathematical analysis

In what follows we write simply U and λ in place of U_p and λ_p respectively. Moreover for simplicity we assume $\Gamma_D = \partial\Omega$ and $\overline{u} = 0$ since the extension to the general case in trivial.

2.1. Hypotheses. To get the global existence result we assume the following hypotheses.

(S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2,3\}$ is open, bounded and Lipschitz
(b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$

(Y) (a) $Y \in C^{0,1}(\mathbb{R}, [0,\infty))$

(G) (a) $G \in C^{1,1}(\mathrm{DevSym}, \mathbb{R})$ is convex
(b) $\partial G \in L^\infty(\mathrm{DevSym}, \mathrm{DevPla})$
(c) $\partial G(\dot U) : \dot U \geq 0$ for all $\dot U \in \mathrm{DevSym}$

(L) (a) $f \in C^1([0,T], L^2(\Omega)^d)$
(b) $h \in C^1([0,T], L^2(\Gamma_N)^d)$

(E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in $\mathrm{Sym} \otimes \mathrm{Sym}$, $\mathfrak{B}$ and $\mathfrak{F}$ are in $\mathrm{DevPla} \otimes \mathrm{DevPla}$, $\mathfrak{A}$ and $\mathfrak{E}$ are in $(\mathrm{Lin} \otimes \mathbb{R}) \otimes (\mathrm{Lin} \otimes \mathbb{R})$
(b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geq \gamma|E|^2$ and $\mathfrak{D}E : E \geq \delta|E|^2$ for all $E \in \mathrm{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}\mathbb{U} \circ \mathbb{U} \geq \beta|\mathbb{U}|^2$ and $\mathfrak{F}\mathbb{U} \circ \mathbb{U} \geq \phi|\mathbb{U}|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M \,\vdots\, M \geq \alpha|\operatorname{sym} M|^2$ and $\mathfrak{E}M \,\vdots\, M \geq \epsilon|\operatorname{sym} M|^2$ for all $M \in \mathrm{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

2.2. Function spaces. We now introduce the function spaces that we use to formulate the problem. Let

$$\mathcal{X} = \left\{ y = (u, U, \lambda) \,\middle|\, \begin{matrix} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \mathrm{DevSym}) \\ \lambda \in L^2(\Omega) \end{matrix} \right\} \quad \text{and} \quad \mathcal{Q} = \left\{ p \in L^2(\Omega) \right\}.$$

The rates of $y = (u, U, \lambda) \in \mathcal{X}$ will be denoted by $\dot y = (\dot u, \dot U, \dot\lambda) \in \mathcal{X}$. The variations of $\dot y = (\dot u, \dot\xi, \dot\lambda) \in \mathcal{X}$ and $p \in \mathcal{Q}$ will be denoted as $z = (v, V, \mu) \in \mathcal{X}$ and $q \in \mathcal{Q}$. The test functions will be denoted as $\tilde z = (\tilde v, \tilde V, \tilde\mu)$.

2.3. Formulation of the problem. We define $\mathcal{F}\colon \mathcal{X} \times \mathcal{Q} \times \mathcal{X} \times [0,T] \to \mathcal{X}' \times \mathcal{Q}'$ by

$$\begin{aligned}\langle \mathcal{F}(\dot y, p, y, t), (\tilde z, \tilde q)\rangle = &\int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde v - \tilde V - \frac{\tilde\mu}{d}I\right) dx + \\ &+ \int_\Omega (\mathfrak{A} \operatorname{grad}^2 u + \mathfrak{E} \operatorname{grad}^2 \dot u) \,\vdots\, \operatorname{grad}^2 \tilde v dx + \int_\Omega \mathfrak{D}E\dot u : E\tilde v dx + \\ &+ \int_\Omega \left[(\mathfrak{B}\mathbb{U} + \mathfrak{F}\dot{\mathbb{U}}) \circ \tilde{\mathbb{V}} + Y(p)\partial G(\dot U) : \tilde V\right] dx - \\ &- \int_\Omega p\tilde\mu dx - \int_\Omega \lambda\tilde q dx - \\ &- \int_\Omega f(t) \cdot \tilde v dx - \int_{\Gamma_N} h(t) \cdot \tilde v dx.\end{aligned}$$

Lemma 5.3. The map $\mathcal{F}$ is well defined.

Proof. The linear terms are trivially well defined so we need to check only

$$\int_\Omega Y(p)\partial G(\dot U) : \tilde V dx.$$

But

$$\int_\Omega |Y(p)||\partial G(\dot{\mathbb{U}})||\tilde{\mathbb{V}}| dx \leq \|\partial G\|_\infty \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\tilde{\mathbb{V}}| dx \lesssim (1 + \|p\|_2)\|\tilde V\|_{1,2}.$$

This concludes the proof. □

Now we can formulate the evolution problem as follows.

Problem 5.4. Find $y \in H^1((0,T),\mathcal{Z})$ and $p \in L^2([0,T],\mathcal{Q})$ such that $y(0)=0$ and

$$\mathcal{F}(\dot{y}(t),p(t),y(t),t)=0$$

for almost every $t \in (0,T)$.

2.4. Global existence. This section is devoted to prove the following existence of a global in time solution to Problem 5.4.

Theorem 5.5. Assume the hypotheses of Section 2.1. Then there exists $y \in H^1((0,T),\mathcal{Z})$ and $p \in C^0([0,T],\mathcal{Q})$ solution to Problem 5.4.

The proof is based on the Rothe method so we first prove the existence to a discrete in time version of the problem and then we pass to the vanishing time step limit.

Existence of the discrete in time solution. Let $N \in \mathbb{N}$. We define $\tau_N = N^{-1}T$ and $t_n^N = n\tau_N$ for $n \in \{0,\dots,N\}$. We consider the discrete in time version of Problem 5.4.

Problem 5.6 (Discrete problem). Let $y_0^N = 0 \in \mathcal{Z}$. Find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ for $n \in \{1,\dots,N\}$ such that

$$\mathcal{F}\left(\frac{y_n^N - y_{n-1}^N}{\tau_N}, p_n^N, y_n^N, t_n^N\right) = 0 \tag{5.1}$$

for all $n \in \{1,\dots,N\}$.

We first prove the existence of a solution to Problem 5.6. Reasoning by induction we reduce to find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ that satisfy (5.1) given $y_{n-1}^N \in \mathcal{Z}$. We immediately observe that in view of the constrain $\lambda_n^N = 0$ for all n. To simplify the notation we call $y = y_{n-1}^N$, $\dot{y} = \tau^{-1}(y_n^N - y_{n-1}^N)$ and $t = t_n^N$. Then $y_n^N = y + \tau\dot{y}$. It is enough to prove that there exists $\dot{y} \in \mathcal{Z}$ and $p \in Q$ such that

$$\mathcal{F}(\dot{y},p,y+h\dot{y},t)=0. \tag{5.2}$$

We define the operator $\mathcal{A}\colon \mathcal{Q}\times\mathcal{Z} \to \mathcal{Z}'$ by

$$\langle \mathcal{A}(p,\dot{y}),\tilde{z}\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx +$$
$$+\int_\Omega (\mathfrak{E}+\tau\mathfrak{A})\operatorname{grad}^2\dot{u} \,\vdots\, \operatorname{grad}^2\tilde{v}dx + \int_\Omega \left[(\mathfrak{F}+\tau\mathfrak{B})\dot{\mathbb{U}}\circ\tilde{\mathbb{V}} + Y(p)\partial G(\dot{U}) : \tilde{V}\right]dx$$

the linear operator $\mathcal{B}\colon \mathcal{Q} \to \mathcal{Z}'$ by

$$\langle \mathcal{B}p,\tilde{z}\rangle = -\int_\Omega p\tilde{\mu}dx$$

and the operator $\mathcal{C}\colon \mathcal{Z} \to \mathcal{Q}'$ by

$$\langle \mathcal{C}(\dot{y}),\tilde{q}\rangle = \int_\Omega (\lambda+\tau\dot{\lambda})\tilde{q}dx = 0.$$

Moreover we introduce the linear form $l \in \mathcal{Z}'$

$$\langle l,\tilde{z}\rangle = \int_\Omega f(t)\cdot\tilde{v}dx + \int_{\Gamma_N} h(t)\cdot\tilde{v}dx -$$
$$-\int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx -$$
$$-\int_\Omega \mathfrak{A}\operatorname{grad}^2 u \,\vdots\, \operatorname{grad}^2\tilde{v}dx - \int_\Omega \mathfrak{B}\mathbb{U}\circ\tilde{\mathbb{V}}dx.$$

Now equation (5.2) can be written as

$$\begin{cases} \mathscr{A}(p,\dot{y}) + Bp = l(t) \\ \mathscr{C}(\dot{y}) = 0. \end{cases}$$

We have already noticed that $\lambda = 0$ and $\dot{\lambda} = 0$. We call

$$\mathscr{N} = \left\{ \dot{y} = (\dot{u},\dot{U},0) \;\middle|\; \begin{matrix} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega,\mathrm{DevSym}) \end{matrix} \right\}.$$

Now we study the first equation in (5.2). We first study the solvability of that equation in $\dot{y} \in \mathcal{N}$ given $p \in \mathcal{Q}$. If we test the first equation with $\tilde{z} \in \mathcal{N}$ we get

$$\langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle = \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \mathcal{N}.$$

Now we define the space

$$\mathcal{X} = \left\{ \xi = (\dot{u}, \dot{U}) \;\middle|\; \substack{\dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \mathrm{DevSym})} \right\}$$

and we can consider the natural bijection $\mathcal{J}$ from $\mathcal{X}$ to $\mathcal{N}$. In particular if $\xi = (\dot{u}, \dot{U})$ we define

$$\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{u}, \dot{U})$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V})$ in $\mathcal{X}$ we set

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle \mathcal{A}(p, \mathcal{J}(\xi)), \mathcal{J}(\tilde{\zeta}) \rangle = \int_\Omega \tau \mathfrak{C} \left(E\dot{u} - \dot{U} \right) : \left(E\tilde{v} - \tilde{V} \right) dx + \int_\Omega \mathfrak{D} E\dot{u} : E\tilde{v} dx +$$
$$+ \int_\Omega (\mathfrak{E} + \tau \mathfrak{A}) \operatorname{grad}^2 \dot{u} \,\vdots\, \operatorname{grad}^2 \tilde{v} dx + \int_\Omega \left[(\mathfrak{F} + \tau \mathfrak{B}) \dot{\mathbb{U}} \circ \tilde{\mathbb{V}} + Y(p) \partial G(\dot{U}) : \tilde{V} \right] dx.$$

Now for each $p \in \mathcal{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle l, \mathcal{J}(\tilde{\zeta}) \rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

We want to apply the Browder–Minty theorem (Theorem C.20). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

Lemma 5.7. The operator $\mathcal{T}_p \colon \mathcal{X} \to \mathcal{X}'$ is bounded and continuous.

Proof. We have

$$|\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle| \lesssim \left(\|E\dot{u}\|_2 + \|\dot{U}\|_2 \right) \left(\|E\tilde{v}\|_2 + \|\tilde{V}\|_2 \right) +$$
$$+ \|\operatorname{grad}^2 \dot{u}\|_2 \|\operatorname{grad}^2 \tilde{v}\|_2 + \|\dot{\mathbb{U}}\|_2 \|\tilde{\mathbb{V}}\|_2 + (1 + \|p\|_2) \|\tilde{\mathbb{V}}\|_2.$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \to \xi$ in $\mathcal{X}$. Then $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2. We have

$$\left| \int_\Omega Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}})) \circ \tilde{\mathbb{V}} dx \right| \le \|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2 \|\tilde{\mathbb{V}}\|_2$$

and

$$\|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2^2 = \int_\Omega Y(p)^2 (\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))^2 dx \to 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2 and we can apply the dominated convergence theorem to pass to the limit. □

Lemma 5.8. The operator $\mathcal{T}_p \colon \mathcal{X} \to \mathcal{X}'$ is coercive uniformly in p.

Proof. For any $\xi \in \mathcal{X}$ we have

$$\langle \mathcal{T}_p(\xi), \xi \rangle = \int_\Omega \tau \mathfrak{C} \left(E\dot{u} - \dot{U} \right) : \left(E\dot{u} - \dot{U} \right) dx + \int_\Omega \mathfrak{D} E\dot{u} : E\dot{u} dx +$$
$$+ \int_\Omega (\mathfrak{E} + \tau \mathfrak{A}) \operatorname{grad}^2 \dot{u} \,\vdots\, \operatorname{grad}^2 \dot{u} dx + \int_\Omega \left[(\mathfrak{F} + \tau \mathfrak{B}) \dot{\mathbb{U}} \circ \dot{\mathbb{U}} + Y(p) \partial G(\dot{U}) \circ \dot{U} \right] dx.$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{U} : I = 0$ since $\dot{U} \in \mathrm{Dev}$ we have

$$\langle \mathcal{T}_p(\xi), \xi \rangle \ge \tau \gamma \left\| E\dot{u} - \dot{U} \right\|_2^2 + \tau \alpha \|\operatorname{grad} E\dot{u}\|_2^2 + (\phi + \tau \beta) \|\dot{\mathbb{U}}\|_2^2.$$

By the Young inequality

$$\left\| E\dot{u} - \dot{U} \right\|_2^2 \ge (1 - \theta) \|E\dot{u}\|_2^2 + \left(1 - \frac{1}{\theta} \right) \|\dot{U}\|_2^2$$

for any $\theta > 0$. Then

$$\langle \mathcal{T}_p(\xi), \xi\rangle \geq (1-\theta)\tau\gamma\|E\dot{u}\|_2^2 + \tau\alpha\|\operatorname{grad} E\dot{u}\|_2^2 + \\ + \left(1-\frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi+\tau\beta)\|\dot{\mathbb{U}}\|_2^2.$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1-\frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi+\tau\beta)\|\dot{\mathbb{U}}\|_2^2 \geq C_1\|\dot{\mathbb{U}}\|_2^2 \geq C_2\|\dot{U}\|_{1,2}^2.$$

The thesis follows by application of Korn inequality. □

Lemma 5.9. The operator $\mathcal{T}_p\colon \mathcal{X} \to \mathcal{X}'$ is strongly monotone uniformly in p, i.e., there exists $M > 0$ such that

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle \geq M\|\xi_2-\xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$.

Proof. For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle = \\ = \int_\Omega \tau\mathfrak{C}\left(E(\dot{u}_2-\dot{u}_1) - (\dot{U}_2-\dot{U}_1)\right) : (E(\dot{u}_2-\dot{u}_1) - (U_2-U_1))\,dx + \\ + \int_\Omega \mathfrak{D}(E(\dot{u}_2-\dot{u}_1)) : (E(\dot{u}_2-\dot{u}_1))dx + \\ + \int_\Omega (\mathfrak{E}+\tau\mathfrak{A})(\operatorname{grad}^2\dot{u}_2 - \operatorname{grad}^2\dot{u}_1) \,\vdots\, (\operatorname{grad}^2\dot{u}_2 - \operatorname{grad}^2\dot{u}_1)dx + \\ + \int_\Omega \left[(\mathfrak{F}+\tau\mathfrak{B})(\dot{\mathbb{U}}_2-\dot{\mathbb{U}}_1)\circ(\dot{\mathbb{U}}_2-\dot{\mathbb{U}}_1) + Y(p)(\partial G(\dot{U}_2) - \partial G(\dot{U}_1)) : (\dot{U}_2-\dot{U}_1)\right] dx \geq \\ \geq (\delta+\tau\gamma)\|E(\dot{u}_2-\dot{u}_1) - (\dot{U}_2-\dot{U}_1)\|_2^2 + (\phi+\tau\beta)\|\dot{\mathbb{U}}_2-\dot{\mathbb{U}}_1\|_2^2.$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 5.8 we have

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle \geq (1-\theta)(\delta+\tau\gamma)\|E(\dot{u}_2-\dot{u}_1)\|_2^2 + (\alpha+\tau\epsilon)\|\operatorname{grad} E(\dot{u}_2-\dot{u}_1)\|_2^2 + \\ + \left(1-\frac{1}{\theta}\right)(\delta+\tau\gamma)\|\dot{U}_2-\dot{U}_1\|_2^2 + \\ + (\phi+\tau\beta)\|\dot{\mathbb{U}}_2-\dot{\mathbb{U}}_1\|_2^2.$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using second order Korn inequality we conclude. □

Now applying Theorem C.20 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle l, \mathcal{J}(\tilde{\zeta})\rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique. Indeed by strong monotonicity if ξ_1 and ξ_2 are two solutions we have

$$M\|\xi_2-\xi_1\|_{\mathcal{X}}^2 \leq \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2-\xi_1\rangle = 0.$$

We call $\Xi\colon \mathcal{Q} \to \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ. Now to conclude the existence of a solution to Problem 5.6 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0,0,\tilde{\mu})\rangle + \langle Bp, (0,0,\tilde{\mu})\rangle = \langle l, (0,0,\tilde{\mu})\rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_\Omega \operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right]\frac{\tilde{\mu}}{d}dx - \int_\Omega p\tilde{\mu}dx = \int_\Omega \operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right]\frac{\tilde{\mu}}{d}dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$p = \frac{1}{d}\operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right] - \frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] = \mathcal{S}(p) \tag{5.3}$$

where we have introduced the operator $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem C.15). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ. We first need the following lemma.

Lemma 5.10. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

Proof. By (Y, a) and (G, c) we have

$$|\langle \mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y}), \tilde{z}\rangle| \leq \int_\Omega |Y(p) - Y(q)||\partial G(\dot{U})||\tilde{V}|dx \leq$$
$$\leq \mathrm{Lip}(Y)\|\partial G\|_\infty \int_\Omega |p-q||\tilde{V}|dx \leq L\|p-q\|_2\|\tilde{V}\|_2$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y})\|_{\mathcal{X}'} \leq L\|p-q\|_{\mathcal{Q}}$$

This concludes the proof. □

Now we can prove the continuity of Ξ.

Lemma 5.11. The operator Ξ is continuous and has bounded range.

Proof. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geq n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \leq \|\langle l, \mathcal{J}(\cdot)\rangle\|_{\mathcal{X}'}.$$

But by Lemma 5.8

$$\frac{\langle \mathcal{T}_p(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \to \infty$$

and this is a contradiction. Now we prove that Ξ is Lipschitz continuous. Indeed let $p_i \in \mathcal{Q}$ and $\xi_i = \Xi(p_i)$ with i equals to 1 or 2. Since $\mathcal{T}_{p_1}(\xi_1) = \mathcal{T}_{p_2}(\xi_2)$, $\mathcal{T}_p$ is strictly monotone uniformly in p and $\mathcal{A}$ is uniformly Lipschitz in p we have

$$M\|\xi_2 - \xi_1\|^2 \leq \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_1}(\xi_1), \xi_2 - \xi_1\rangle = \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_2}(\xi_2), \xi_2 - \xi_1\rangle \leq$$
$$\leq L\|p_2 - p_1\|_{\mathcal{Q}}\|\xi_2 - \xi_1\|_{\mathcal{X}}.$$

This concludes the proof. □

The following lemma establish the regularity of $\mathfrak{J}$.

Lemma 5.12. The map $\mathcal{J}\colon \mathcal{X} \to \mathcal{Z}$ is continuous and bounded.

Proof. It is trivial since $\mathfrak{J}(\xi) = (\dot{u}, \dot{U}, 0)$. □

Now we are ready to prove the desired properties of $\mathcal{S}$.

Lemma 5.13. The map $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$ is continuous and has a bounded and compact range.

Proof. We notice that since Ξ is continuous and has compact range then also $\mathcal{S}$ is continuous and has compact range. We notice that by Rellich–Kondarchov theorem then $\dot{u} \mapsto E\dot{u}$ from H^2 to L^2 and $\dot{U} \mapsto \dot{U}$ from H^1 to L^2 are compact. Then $\mathcal{S}$ is compact. Since $\mathcal{S}$ has bounded range and is compact then it has compact range. □

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

Proposition 5.14. Problem 5.6 admits at least a solution.

Now we redefine the operator $\mathcal{A}\colon \mathcal{Z} \to \mathcal{Z}'$ by

$$\langle Ay, \tilde{z}\rangle = \int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right) dx + \int_\Omega \mathfrak{A}\,\mathrm{grad}\, u \,\vdots\, \mathrm{grad}\,\tilde{v} dx + \int_\Omega \mathfrak{B}\mathbb{U} \circ \tilde{\mathbb{V}} dx$$

and we define operator $\mathcal{E}\colon \mathcal{Z} \to \mathcal{Z}'$ by

$$\langle \mathcal{E}\dot{y}, \tilde{z}\rangle = \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v} dx + \int_\Omega \mathfrak{E}\,\mathrm{grad}\,\dot{u} \,\vdots\, \mathrm{grad}\,\tilde{v} dx + \int_\Omega \mathfrak{F}\dot{\mathbb{U}} \circ \tilde{\mathbb{V}} dx.$$

We define also $\mathcal{D}\colon \mathcal{Q} \times \mathcal{Z} \to \tilde{\mathcal{Z}}'$ by

$$\langle \mathcal{D}(p, \dot{y}), \tilde{z}\rangle = \int_\Omega Y(p)\partial G(\dot{U}) : \tilde{V} dx.$$

The operator $\mathcal{B}$ is defined exactly as before and and $\mathcal{C}\colon \mathcal{Z} \to \mathcal{Q}'$ is redefined by

$$\langle \mathcal{C}y, \tilde{q}\rangle = \int_\Omega \lambda \tilde{q} dx = 0.$$

Let also $l\colon [0,T] \to \mathcal{Z}'$ redefined by

$$\langle l(t), \tilde{z}\rangle = \int_\Omega f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx.$$

Proposition 5.14 ensures that that for each N there exists $\left\{(\dot{y}_n^N, p_n^N, y_n^N)\right\}_{n=0}^N$ such that $(\dot{y}_0^N, p_0^N, y_0^N) = 0$, $y_n^N = y_{n-1}^N + \tau \dot{y}_n^N$ and

$$\begin{cases} \mathcal{A}y_n^N + \mathcal{E}\dot{y}_n^N + \mathcal{D}(p_n^N, \dot{y}_n^N) + \mathcal{B}p_n^N = l_n^N \\ \mathcal{C}y_n^N = 0 \end{cases} \tag{5.4}$$

for all $n \in \{1, \dots, N\}$. Here $l_n^N = l(t_n^N)$.

Construction of the continuous in time solution. Now we want to construct the continuous in time solution passing to the limit $N \to \infty$. We define

$$\begin{aligned} l^N &\in L^2((0,T), \mathcal{Z}') && \text{piecewise constant interpolation of } \{l_n^N\}_{n=1}^N \\ \overline{y}^N &\in L^2((0,T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{y_n^N\}_{n=1}^N \\ y^N &\in H^1((0,T), \mathcal{Z}) && \text{piecewise linear interpolation of } \{y_n^N\}_{n=1}^N \\ \dot{y}^N &\in L^2((0,T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{\dot{y}_n^N\}_{n=1}^N \\ p^N &\in L^2((0,T), \mathcal{Q}) && \text{piecewise constant interpolation of } \{p_n^N\}_{n=1}^N. \end{aligned}$$

We will need the following lemma.

Lemma 5.15. We have that $l^N \to l$ in $L^2((0,T), \mathcal{Z}')$ and in particular

$$\|l^N\|^2_{L^2((0,T),\mathcal{Z}')} = \sum_{n=1}^N \tau_N \|l_n^N\|^2_{\mathcal{Z}'}$$

is bounded.

Proof. For all $t \in [t_{n-1}^N, t_n^N]$ we have

$$l^N(t) = l_n^N = l(t_n^N) = l(t) + \int_t^{t_n^N} \dot{l}(s) ds.$$

By Hölder inequality

$$\|l^N(t) - l(t)\|_{\mathcal{Z}'} \leq \int_t^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'} ds \leq \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'} ds \leq \sqrt{\tau_N} \left[\int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|^2_{\mathcal{Z}'} ds\right]^{1/2}.$$

Taking the square and integrating for $t \in [t_{n-1}^N, t_n^N]$ we have

$$\int_{t_{n-1}^N}^{t_n^N} \|l^N(t) - l(t)\|^2_{\mathcal{Z}'} dt \leq \tau_N^2 \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|^2_{\mathcal{Z}'} ds.$$

Adding this inequalities for $n \in \{1, \dots, N\}$ we have

$$\|l^N - l\|^2_{L^2((0,T),\mathcal{Z}')} \leq \tau_N^2 \|\dot{l}\|^2_{L^2((0,T),\mathcal{Z}')} \to 0.$$

This concludes the proof. □

We now obtain a priori estimates on $\overline{y}^N$, y^N and $\dot{y}^N$.

LEMMA 5.16. The functions $\overline{y}^N$ and y^N are bounded in $L^\infty((0,T),\mathscr{Z})$ and the function $\dot{y}^N$ is bounded in $L^2((0,T),\mathscr{Z})$.

PROOF. We first observe that y_n^N and $\dot{y}_n^N$ belongs to $\mathscr{N}$ and that $\mathcal{A}$ and $\mathcal{E}$ are coercive there. Next we test the first equation in (5.4) with $\tilde{z} = \dot{y}_n^N$. We notice that $\langle \mathcal{D}(p_n^N, \dot{y}_n^N), \dot{y}_n^N\rangle \geq 0$ and $\langle \mathcal{B}p_n^N, \dot{y}_n^N\rangle = 0$. Then

$$\frac{1}{\tau_N}\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle + \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \leq \langle l_n^N, \dot{y}_n^N\rangle.$$

Multiplying by $2\tau_N$ we have

$$2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle + 2\tau_N\langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \leq 2\tau_N\langle l_n^N, \dot{y}_n^N\rangle.$$

Now

$$\begin{aligned}2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N\rangle &= \langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle + \langle \mathcal{A}(y_n^N - y_{n-1}^N), (y_n^N - y_{n-1}^N)\rangle \geq \\ &\geq \langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle.\end{aligned}$$

Let $E > 0$ be its coercivity constant of $\mathcal{E}$ in $\mathscr{N}$. Then

$$2\tau_N\langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N\rangle \geq 2\tau_N E\|\dot{y}_n^N\|_{\mathscr{Z}}^2$$

and using Young inequality

$$2\tau_N\langle l_n^N, \dot{y}_n^N\rangle \leq \frac{\tau_N}{E}\|l_n^N\|_{\mathscr{Z}'}^2 + \tau_N E\|\dot{y}_n^N\|_{\mathscr{Z}}^2.$$

Then

$$\langle \mathcal{A}y_n^N, y_n^N\rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N\rangle + \tau_N E\|\dot{y}_n^N\|_{\mathscr{Z}}^2 \leq \frac{\tau_N}{E}\|l_n^N\|_{\mathscr{Z}'}^2. \tag{5.5}$$

Let $A > 0$ the coercivity constant of $\mathcal{A}$ on $\mathscr{N}$. Then adding the inequalities (5.5) we have

$$A\|y_n^N\|_{\mathscr{Z}}^2 + E\sum_{k=1}^n \tau_N\|\dot{y}_k^N\|_{\mathscr{Z}}^2 \leq \langle \mathcal{A}y_n^N, y_n^N\rangle + E\sum_{k=1}^n \tau_N\|\dot{y}_k^N\|_{\mathscr{Z}}^2 \leq \frac{1}{E}\sum_{k=1}^n \tau_N\|l_k^N\|_{\mathscr{Z}'}^2.$$

Using Lemma 5.15 we have that

$$\|y_n^N\|_{\mathscr{Z}} \quad \text{and} \quad \sum_{n=1}^N \tau_N\|\dot{y}_n^N\|_{\mathscr{Z}}^2$$

are bounded uniformly in $N \in \mathbb{N}$ and $n \in \{0, \dots, N\}$. This implies immediately the thesis. □

Using that lemma, up to extracting a subsequence we have that

$$\begin{aligned}\overline{y}^N &\overset{*}{\rightharpoonup} \overline{y} \quad \text{in } L^\infty((0,T),\mathscr{Z})\\ y^N &\overset{*}{\rightharpoonup} y \quad \text{in } L^\infty((0,T),\mathscr{Z})\\ \dot{y}^N &\rightharpoonup z \quad \text{in } L^2((0,T),\mathscr{Z}).\end{aligned}$$

We notice that from $\lambda_n^N = 0$ it follows immediately that $\lambda = 0$.

LEMMA 5.17. We have that $\overline{y} = y \in H^1((0,T),\mathscr{Z})$ and $\dot{y} = z$. Moreover $y(0) = 0$.

PROOF. We prove that $\overline{y}^N - y^N \to 0$ in $L^\infty((0,T),\mathscr{Z})$. Indeed let $t \in [t_{n-1}^N, t_n^N]$. Then

$$\|\overline{y}^N(t_n^N) - y(t_n^N)\|_{\mathscr{Z}} = \|y_n^N - y_n^N - (t_n^N - t)\dot{y}_n^N\|_{\mathscr{Z}} \leq \tau_N\|\dot{y}_n^N\|_{\mathscr{Z}}.$$

It follows that

$$\|\overline{y}^N(t_n^N) - y(t_n^N)\|_{\mathscr{Z}}^2 \leq \tau_N^2\|\dot{y}_n^N\|_{\mathscr{Z}}^2 \leq \tau_N\|\dot{y}^N\|_{L^2((0,T),\mathscr{Z})}^2 \to 0.$$

This establish the equality $\overline{y} = y$. Now let $\phi \in C_c^\infty([0,T))$ and $\zeta \in \mathscr{Z}'$ arbitrary. Then

$$\int_0^T \phi\langle \zeta, \dot{y}^N\rangle dt = -\phi(0)\langle \zeta, y^N(0)\rangle - \int_0^T \dot{\phi}\langle \zeta, y^N\rangle dt.$$

Choosing first $\phi \in C_c^\infty((0,T))$ and exploiting (5.6) we have

$$\int_0^T \phi\langle \zeta, \dot{y}\rangle dt = -\int_0^T \dot{\phi}\langle \zeta, y\rangle dt.$$

This means exactly that $y \in H^1((0,T),\mathscr{X})$ and $\dot{y} = z$. In particular for $\phi \in C_c^\infty([0,T))$ generic we have

$$\int_0^T \phi\langle\zeta,\dot{y}\rangle dt = -\phi(0)\langle\zeta,y(0)\rangle - \int_0^T \dot{\phi}\langle\zeta,y\rangle dt.$$

Then we get also

$$\langle\zeta,y^N(0)\rangle \to \langle\zeta,y(0)\rangle.$$

Since $y^N(0) = 0$ also $y(0) = 0$. □

We now discuss the convergence of p^N.

Lemma 5.18. Up to extracting a further subsequence $u^N \to u$ in $C([0,T],H^1(\Omega)^d)$ and $U^N \to U$ in $C([0,T],L^2(\Omega,\mathrm{DevSym}))$. Moreover $p^N \to p$ in $C([0,T],\mathcal{Q})$ for some p.

Proof. Since $y^N \rightharpoonup y$ in $L^\infty((0,T),\mathscr{X})$ and $\dot{y}^N \rightharpoonup \dot{y}$ in $L^2((0,T),\mathscr{X})$ then also

$$u^N \rightharpoonup u \text{ in } L^\infty(\Omega,H^2) \quad\text{and}\quad \dot{u}^N \rightharpoonup \dot{u} \text{ in } L^2(\Omega,H^2),$$
$$U^N \rightharpoonup U \text{ in } L^\infty(\Omega,H^1) \quad\text{and}\quad \dot{U}^N \rightharpoonup \dot{U} \text{ in } L^2(\Omega,H^1).$$

By Aubin–Lions–Simon lemma (Theorem C.5)

$$u^N \to u \text{ in } C([0,T],H^1) \quad\text{and}\quad U^N \to U \text{ in } C([0,T],L^2).$$

We recall that by (5.3) we have

$$p_n^N = \frac{1}{d}\operatorname{tr}\left[\tau_N\mathfrak{C}\left(E\dot{u}_n^N - \dot{U}_n^N\right)\right] - \frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu_n^N - U_n^N\right)\right].$$

Since $\tau_N \to 0$ and $\dot{y}_n^N$ is bounded in $L^2((0,T),\mathscr{X})$ we have

$$p^N \to p = -\frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] \quad \text{in } L^2((0,T),\mathscr{X}).$$

This concludes the proof. □

Remark 5.19. The fact that $p \in C^0([0,T],\mathcal{Q})$ suggest that there are not bifurcation modes of pressure.

Now we notice that (5.4) implies that for all $\tilde{z} \in L^2((0,T),\mathscr{X})$ we have

$$\int_0^T \langle\mathcal{A}\overline{y}^N,\tilde{z}\rangle dt + \int_0^T \langle\mathcal{E}\dot{y}^N,\tilde{z}\rangle dt + \int_0^T \langle\mathcal{D}(P^N,\dot{y}^N),\tilde{z}\rangle dt + \int_0^T \langle\mathcal{B}p^N,\tilde{z}\rangle dt = \int_0^T \langle l^N,\tilde{z}\rangle dt.$$

By Lemma 5.15 and (5.6) we have

$$\int_0^T \langle l^N,\tilde{z}\rangle dt \to \int_0^T \langle l,\tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{A}\overline{y}^N,\tilde{z}\rangle dt \to \int_0^T \langle\mathcal{A}y,\tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{E}\dot{y}^N,\tilde{z}\rangle dt \to \int_0^T \langle\mathcal{E}\dot{y},\tilde{z}\rangle dt$$
$$\int_0^T \langle\mathcal{B}p^N,\tilde{z}\rangle dt \to \int_0^T \langle\mathcal{B}p,\tilde{z}\rangle dt.$$

Moreover we have

$$\begin{aligned}\int_0^T \|\mathcal{D}(p^N,\dot{y}^N)\|_{\mathscr{X}'}^2 dt &\le \int_0^T\int_\Omega Y(p^N)^2|\partial G(\dot{U}^N)|^2 dxdt \le \\ &\le \int_0^T\int_\Omega (Y(0) + \mathrm{Lip}(Y)|p^N|)^2\|\partial G\|_\infty^2 dxdt \lesssim \\ &\lesssim (1 + \|p^N\|_{L^2((0,T),\mathcal{Q})}^2).\end{aligned}$$

But since p^N converges in $C([0,T],\mathcal{Q})$ then $\mathcal{D}(p^N,\dot{y}^N)$ is bounded in $L^2((0,T),\mathcal{Z}')$. Then up to extracting a further subsequence

$$\mathcal{D}(p^N,\dot{y}^N) \rightharpoonup \delta \text{ in } L^2((0,T),\mathcal{Z}')$$

for some $\delta \in L^2((0,T),\mathcal{Z}')$. Then we have

$$\int_0^T \langle \mathcal{A}y,\tilde{z}\rangle dt + \int_0^T \langle \mathcal{E}\dot{y},\tilde{z}\rangle dt + \int_0^T \langle \delta,\tilde{z}\rangle dt + \int_0^T \langle \mathcal{B}p,\tilde{z}\rangle dt = \int_0^T \langle l,\tilde{z}\rangle dt \tag{5.7}$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. This implies also

$$\langle \mathcal{A}y,\tilde{z}\rangle + \langle \mathcal{E}\dot{y},\tilde{z}\rangle + \langle \delta,\tilde{z}\rangle + \langle \mathcal{B}p,\tilde{z}\rangle = \langle l,\tilde{z}\rangle$$

for almost every $t \in (0,T)$. If we prove that $\delta = \mathcal{D}(p,\dot{y})$ we have that (y,p) is a global in time solution to Problem 5.4 and we have concluded. To this aim let us define $\mathcal{Z}_T = L^2((0,T),\mathcal{Z})$ and $\mathcal{Q}_T = L^2((0,T),\mathcal{Q})$. Let $\mathfrak{J}\colon \mathcal{Q}\times\mathcal{Z} \to \mathbb{R}$ be defined by

$$\mathfrak{J}(p,\dot{y}) = \int_\Omega Y(p)G(\dot{U})dx.$$

Let also $\mathcal{J}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T \to \mathbb{R}$ and $\mathcal{D}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T \to \mathcal{Z}_T'$ defined by

$$\mathfrak{J}_T(p,\dot{y}) = \int_0^T \mathfrak{J}(p,\dot{y})dt \quad \text{and} \quad \langle \mathcal{D}_T(p,\dot{y}),\tilde{z}\rangle = \int_0^T \langle \mathcal{D}(p,\dot{y}),\tilde{z}\rangle dt$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. We notice that for all $y \in \mathcal{Z}_T$ and $p \in \mathcal{Q}_T$ we can identify $\mathcal{D}_T(p,\dot{y}) \in \mathcal{Z}_T'$ with $\mathcal{D}(p,\dot{y}) \in L^2((0,T),\mathcal{Z}')$.

Lemma 5.20. Let $\dot{y} \in L^2((0,T),\mathcal{Z})$, $p \in L^2((0,T),\mathcal{Q})$ and $\delta \in L^2((0,T),\mathcal{Z}')$. Then $\delta = \mathcal{D}(p,\dot{y})$ if and only if

$$\mathfrak{J}_T(p,\tilde{z}) \geq \mathfrak{J}_T(p,\dot{y})dt + \int_0^T \langle \delta,\tilde{z}-\dot{y}\rangle dt \tag{5.8}$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$.

Proof. It is easy to see that $\mathfrak{J}$ is convex in $\dot{y}$ and is Gateaux differentiable in $\dot{y}$ with Gateaux differential

$$\langle D_{\dot{y}}\mathfrak{J}_T(p,\dot{y}),\tilde{z}\rangle = \int_0^T \langle \mathcal{D}(p,\dot{y}),\tilde{z}\rangle dt = \langle \mathcal{D}_T(p,\dot{y}),\tilde{z}\rangle$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. By [13, Proposition 5.3] then $\mathfrak{J}_T$ is sub differentiable and $D_{\dot{y}}^{\text{sub}}\mathfrak{J}_T(p,\dot{y}) = \{D_{\dot{y}}\mathfrak{J}_T(p,\dot{y})\}$. Then $\delta = \mathcal{D}(p,\dot{y})$ if and only if $\delta \in D_{\dot{y}}^{\text{sub}}\mathfrak{J}_T(p,\dot{y})$ but this last inclusion is equivalent to (5.8). □

By that lemma we have

$$\mathfrak{J}_T(p^N,\tilde{z}) \geq \mathfrak{J}_T(p^N,\dot{y}) + \langle \mathcal{D}_T(p^N,\dot{y}^N),\tilde{z}-\dot{y}^N\rangle. \tag{5.9}$$

Now we want to pass to pass to the limit in this inequality. We will need the following lemma.

Lemma 5.21. For each $\dot{y} \in \mathcal{Z}_T$ the functional $\mathfrak{J}_T(\cdot,\dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{Z}_T}$. Moreover for each $p \in \mathcal{Q}_T$ we have that $\mathfrak{J}_T(p,\cdot)$ is convex and continuous. In particular $\mathfrak{J}_T(p,\cdot)$ is weakly lower semicontinuous.

Proof. We have that

$$\begin{aligned}
|\mathfrak{J}_T(p_2,\dot{y}) - \mathfrak{J}_T(p_1,\dot{y})| &\leq \int_0^T \int_\Omega |Y(p_2)-Y(p_1)||G(\dot{U})|dx \leq \\
&\leq \|\partial G\|_\infty \mathrm{Lip}(Y) \int_0^T \int_\Omega |p_2-p_1||\dot{U}|dtdx \lesssim \\
&\lesssim \|p_2-p_1\|_{\mathcal{Q}_T}\|\dot{y}\|_{\mathcal{Z}_T}.
\end{aligned}$$

This proves that for each $\dot{y} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot, \dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{Z}_T}$. The convexity of $\mathcal{J}_T(p, \cdot)$ follows immediately from the convexity of G. Moreover

$$\begin{aligned}|\mathcal{J}_T(p, \dot{y}_2) - \mathcal{J}_T(p, \dot{y}_1)| &\leq \int_0^T \int_\Omega |Y(p)||G(\dot{U}_2) - G(\dot{U}_1)|dx \leq \\ &\leq \|\partial G\|_\infty \int_0^T \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\dot{U}_2 - \dot{U}_1|dtdx \lesssim \\ &\lesssim (1 + \|p\|_{\mathcal{Q}_T})\|\dot{y}_2 - \dot{y}_1\|_{\mathcal{Z}_T}.\end{aligned}$$

This proves the continuity of $\mathcal{J}_T(p, \cdot)$ for each $p \in \mathcal{Q}_T$ fixed. □

Now we have

$$|\mathcal{J}_T(p^N, \tilde{z}) - \mathcal{J}_T(p, \tilde{z})| \leq C\|p^N - p\|_{\mathcal{Q}_T} \to 0$$

for some constant $C > 0$. Moreover

$$\mathcal{J}_T(p^N, \dot{y}^N) = [\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)] + \mathcal{J}_T(p, \dot{y}^N).$$

But since $p^N \top$ in $\mathcal{Q}_T$ and y^N are bounded in $\mathcal{Z}_T$

$$|\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)| \leq C\|y^N\|_{\mathcal{Z}_T}\|p^N - p\|_{\mathcal{Q}_T} \to 0$$

and

$$\liminf_{N\to\infty} \mathcal{J}_T(p, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Then

$$\liminf_{N\to\infty} \mathcal{J}_T(p^N, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Now by construction

$$\begin{aligned}\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{E}\dot{y}^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{D}(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N\rangle dt + \\ + \int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt = \int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt.\end{aligned}$$

But by Lemma 5.15, Lemma 5.18 and (5.6) we have

$$\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt \to \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y}\rangle dt$$

$$\int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt \to \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y}\rangle dt$$

$$\int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt \to \int_0^T \langle l, \tilde{z} - \dot{y}\rangle dt$$

since the convergence of y^N, p^N and l^N is strong. Moreover since

$$\dot{y} \to \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt$$

is continuous and convex in $L^2((0,T), \mathcal{Z})$ it is weakly lower semicontinuous and

$$\liminf_{N\to\infty} \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt \geq \int_0^T \langle \mathcal{E}\dot{y}, \dot{y} - \tilde{z}\rangle dt.$$

This implies that

$$\begin{aligned}&\liminf_{N\to\infty} \langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N\rangle \geq \\ &\geq \int_0^T \langle l, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z} - \dot{y}\rangle dt - \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y}\rangle dt = \\ &= \int_0^T \langle \delta, \tilde{z} - \dot{y}\rangle dt\end{aligned}$$

where the last equality holds for (5.7). Collecting these results when we take the inferior limit for $N \to \infty$ of (5.9) and we get

$$\mathcal{J}_T(p, \tilde{z}) \geq \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt.$$

Since now δ satisfy (5.8) we have by Lemma 5.8 that $\delta = \mathcal{D}(p, \dot{y})$ as desired.

2.5. Example of constitutive relations. In this section we give examples of constitutive relations satisfying the hypotheses of Section 2.1. Let $g\colon [0, \infty) \to \mathbb{R}$ be defined by

$$g(r) = \begin{cases} \frac{r^2}{4c} & \text{if } r \leq c \\ r - \frac{3}{4}c - \frac{c}{2}\log\frac{r}{c} & \text{if } r > c \end{cases}$$

where $c > 0$ so that

$$g'(r) = \begin{cases} \frac{r}{2c} & \text{if } r \leq c \\ 1 - \frac{c}{2r} & \text{if } r > c. \end{cases}$$

We set $G(\dot{U}) = g(|\dot{U}|)$. We let the yield strength be given by

$$Y(p) = \begin{cases} Y_0 & \text{if } p < -p_0 \\ Y_0 + \alpha(p + p_0) & \text{if } p \geq -p_0 \end{cases}$$

where $Y_0 \geq 0$, $p_0 \geq 0$ and $\alpha > 0$. We set $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{D}$, $\mathfrak{E}$ and $\mathfrak{F}$ positive constants and $\mathfrak{C}$ and $\mathfrak{D}$ positive definite isotropic fourth order tensors.

2.6. Further regularity hypotheses. The local existence and uniqueness theorem is obtained under the following additional hypotheses.

(Y') $Y \in C^{1,1}(\mathbb{R})$
(G') $G \in C^{2,1}(\mathrm{DevSym})$

2.7. Local existence and uniqueness and global uniqueness.

Proposition 5.22. The map $\mathcal{F}$ is of class C^1 and the Fréchet derivative is given by

$$\begin{aligned}
\langle D_{\dot{y}}\mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q})\rangle &= \int_\Omega \mathfrak{D}Ev : E\tilde{v}dx + \int_\Omega \mathfrak{E}\,\mathrm{grad}^2 v : \mathrm{grad}^2 \tilde{v}dx + \\
&\quad + \int_\Omega \mathfrak{F}\mathbb{V} \circ \tilde{\mathbb{V}}dx + \int_\Omega Y(p)\partial^2 G(\dot{U})V : \tilde{V}dx \\
\langle D_p\mathcal{F}(\dot{y}, p, y, q)q, (\tilde{z}, \tilde{q})\rangle &= \int_\Omega Y'(p)q\partial G(\dot{U}) : \tilde{V}dx - \int_\Omega q\tilde{\mu}dx \\
\langle D_y\mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q})\rangle &= \int_\Omega \mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \\
&\quad + \int_\Omega \mathfrak{A}\,\mathrm{grad}^2 v \,\vdots\, \mathrm{grad}^2 \tilde{v}dx + \int_\Omega \mathfrak{B}\mathbb{V} \circ \tilde{\mathbb{V}}dx \\
\langle D_t\mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q})\rangle &= -\int_\Omega \dot{f} \cdot \tilde{v}dx - \int_\Omega \dot{h} \cdot \tilde{v}dx.
\end{aligned}$$

Proof. I still need to report in LaTeXthe boring check but it should be true. First we prove the Gateaux differentiability everywhere that ensure also the continuity. Then we prove that the Gateaux differential is continuous. To this aim we use Lemma C.12, the following lemma and the embedding of H^1 in L^6 for $d = 2$ or $d = 3$.

Lemma 5.23. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi\colon \mathbb{R}^m \to \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \to f$ in $L^p(\Omega, \mathbb{R}^m)$ for some $p \in [1, \infty)$. Then $\phi(f_n) \to \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$ for all $q \in [p, \infty)$.

Proof. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \to 0$. Now let $q \in [p, \infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m} |\phi|$ we have

$$\begin{aligned}\|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega\setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leq \\ &\leq \int_{A_n} L|f_n - f|^q dx + 2C|\Omega \setminus A_n|.\end{aligned}$$

But on A_n we have $|f_n - f|^q \leq |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \leq \int_{A_n} L|f_n - f|^p q dx + 2C|\Omega \setminus A_n| \leq L\|f_n - f\|_p^p + 2C|\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \to \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. □

□

We notice that $\mathcal{F}$ does not depend on λ. We call $\xi = (\dot{u}, \dot{U}, \lambda, p)$, $\zeta = (v, V, \mu, q) = (z, q)$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V}, \tilde{\mu}, \tilde{q}) = (\tilde{z}, \tilde{q})$ all belonging to $\mathcal{Z} \times \mathcal{Q}$.

Proposition 5.24. For any $\dot{y} \in \mathscr{Z}$, $p \in \mathscr{Q}$, $y \in \mathscr{Z}$ and $t \in [0, T]$ we have $D_\xi\mathcal{F}(\dot{y}, p, y, t) \in \operatorname{Iso}(\mathscr{Z} \times \mathscr{Q}, \mathscr{Z}' \times \mathscr{Q}')$.

Proof. We define the bilinear form $a\colon \mathscr{Z} \times \mathscr{Z} \to \mathbb{R}$ by

$$\begin{aligned}a(z, \tilde{z}) = \int_\Omega \mathfrak{D}Ev : E\tilde{v}dx - \int_\Omega \mathfrak{C}\frac{\mu}{d}I : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right) dx + \int_\Omega \mathfrak{E}\operatorname{grad}^2 v \,\vdots\, \operatorname{grad}^2 \tilde{v}dx + \\ + \int_\Omega \mathfrak{F}\mathbb{V} \circ \tilde{\mathbb{V}}dx + \int_\Omega Y(p)\partial^2 G(\dot{U})V : \tilde{V}dx.\end{aligned}$$

We define also the bilinear forms $b\colon \mathscr{Q} \times \mathscr{Z} \to \mathbb{R}$ and $c\colon \mathscr{Q} \times \mathscr{Z} \to \mathbb{R}$ by

$$b(p, \tilde{z}) = \int_\Omega (Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu})q dx$$

and

$$c(\tilde{q}, z) = -\int_\Omega \mu\tilde{q}dx = 0.$$

To prove that $D_\xi\mathcal{F}(\dot{y}, p, y, t) \in \operatorname{Iso}(\mathscr{Z} \times \mathscr{Q}, \mathscr{Z}' \times \mathscr{Q}')$ we have to prove that the mixed elliptic problem

$$\begin{aligned}a(z, \tilde{z}) + b(p, \tilde{z}) = \langle l, \tilde{z}\rangle \quad \text{for all } \tilde{z} \in \mathscr{Z} \\ c(\tilde{q}, z) = \langle m, \tilde{q}\rangle \quad \text{for all } \tilde{q} \in \mathscr{Q}\end{aligned}$$

is well posed for all $l \in \mathscr{Z}'$ and $m \in \mathscr{Q}'$. To this aim it is sufficient to check the hypotheses of Theorem C.10. We first check the hypotheses on b. We have

$$\mathscr{N} = \left\{\tilde{z} \in \mathcal{Z} \mid b(p, \tilde{z}) = 0 \text{ for all } p \in \mathscr{Q}\right\} = \left\{\tilde{z} \in \mathcal{Z} \mid \tilde{\mu} = Y'(p)\partial G(\dot{U}) : \tilde{V}\right\}.$$

Notice that $|Y'(p)\partial G(\dot{U}) : \tilde{V}| \leq \operatorname{Lip}(Y)\|\partial G\|_\infty|\tilde{V}| \in L^2$. Now

$$\sup_{q\in\mathscr{Q}\setminus\{0\}} \frac{\int_\Omega (Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu})q dx}{\|q\|_Q} = \|Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu}\|_2.$$

On the other hand since $\tilde{z}_0 = (\tilde{v}_0, \tilde{V}_0, \tilde{\mu}_0) \in \mathscr{N}$ if and only if $\tilde{\mu}_0 = Y'(p)\partial G(\dot{U}) : \tilde{V}_0$ we have

$$\begin{aligned}\|\tilde{z}\|_{\mathscr{Z}/\mathscr{N}} &= \inf_{\tilde{z}_0\in\mathscr{N}} \left[\|\tilde{v} + \tilde{v}_0\|_{1,2} + \|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p)\partial G(\dot{U}) : \tilde{V}_0\|_2\right] = \\ &= \inf_{\tilde{V}_0\in H^1} \left[\|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p)\partial G(\dot{U}) : \tilde{V}_0\|_2\right] \leq \\ &\leq \|\tilde{\mu} - Y'(p)\partial G(\dot{U}) : \tilde{V}\|_2\end{aligned}$$

where the last inequality holds since we can choose $\tilde{V}_0 = -\tilde{V}$. Then we have

$$\sup_{q \in \mathcal{Q}\setminus\{0\}} \frac{\int_\Omega (Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu})q\,dx}{\|q\|_Q} \geq \|\tilde{z}\|_{\mathcal{Z}/\tilde{\mathcal{N}}}.$$

Moreover if $q \in \mathcal{Q}$ is such that

$$b(p, \tilde{z}) = \int_\Omega (Y'(p)\partial G(\dot{U}) \circ \tilde{V} - \tilde{\mu})q\,dx = 0 \quad \text{for all } \tilde{z} \in \mathcal{Z}$$

then choosing $\tilde{V} = 0$ and $\tilde{\mu} \in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on b. The check of the hypotheses on c is similar. In particular we have

$$\mathcal{N} = \left\{ z \in \mathcal{Z} \;\middle|\; c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \mathcal{Q} \right\} = \left\{ z \in \mathcal{Z} \;\middle|\; \mu = 0 \right\}.$$

Now we check the hypotheses on a. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We notice that with that choice of z and $\tilde{z}$ there exist $K > 0$ and $C > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leq K\|z\|_{\mathcal{Z}}$ and

$$a(z, \tilde{z}) \geq \delta\|Ev\|_2^2 + \epsilon\|\mathrm{grad}^2\, v\|_2^2 + \phi\|\mathbb{V}\|_2^2 \geq C\|z\|_{\mathcal{Z}}^2 \geq \frac{C}{K^2}\|\tilde{z}\|_{\mathcal{Z}}^2.$$

This immediately implies that for all $z \in \mathcal{N}$ we have

$$\sup_{\tilde{z} \in \mathcal{Z}} \frac{a(z, \tilde{z})}{\|\tilde{z}\|_{\mathcal{Z}}} \geq \frac{C}{K}\|z\|_{\mathcal{Z}}$$

and if $\tilde{z} \in \tilde{\mathcal{N}}$ satisfy $a(\cdot, \tilde{z}) = 0$ then $\tilde{z} = 0$. Then also the hypotheses on a are verified. This completes the proof. □

By Proposition 5.24 in a neighborhood of every $(\dot{y}_0, p_0, y_0, t_0)$ with $\lambda_0 = 0$ we can explicit $\xi = (\dot{u}, \dot{U}, \lambda, p)$ from the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0.$$

We get

$$(\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \quad \text{and} \quad (\lambda, p) = \mathcal{H}(u, U, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1. Then we can apply Theorem C.16 to get locally in time a unique solution (u, U) of class C^2 to the Cauchy problem

$$\begin{cases} (\dot{u}, U) = \mathscr{G}(u, U, t) \\ u(0) = u_0 \\ U(0) = U_0. \end{cases}$$

Then we set $(\lambda, p) = \mathscr{H}(u, U, t)$ which is of class C^1. It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

Theorem 5.25. Let $(\dot{y}_0, p_0, y_0, t_0) \in \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T]$ with $\lambda_0 = 0$. Then there exists a unique C^1 solution (y, p) to Problem 5.4 defined locally around t_0.

Considering also Theorem 5.5 we obtain global uniqueness.

Corollary 5.26. Assume the hypotheses of Section 2.1 and of Section 2.6. Then there exists a unique $y \in C^1([0, T], \mathcal{Z})$ and $p \in C^1([0, T], \mathcal{Q})$ solution to Problem 5.4.

2.8. Bifurcation analysis.

Remark 5.27. This section is not definitive but contains only the ideas that we have collected in the hope of being able to reach a more complete result regarding the phenomenon of localization of deformation.

Some considerations. Since we have proved in Corollary 5.26 that the solution is unique no bifurcation can occur with nonzero viscosity. As the viscosity approaches to zero we expect that the evolution of the system reveals it extreme sensibility on the problem data when strain localization occur. To explain better the ideas we present the following example that describe what is the behavior that we expect. Consider a perfectly symmetrical specimen under perfectly symmetrical compressive loads. By uniqueness and symmetry in presence of viscosity the specimen will experience a perfectly symmetrical deformation pattern during the entire evolution (a distributed deformation or at most a symmetrical pattern of localized shear bands). However if the loads, the geometry or the material properties of the specimen are not perfectly symmetric then the system experience an unsymmetrical and completely different deformation pattern (for example an unsymmetrical pattern of shear bands). When viscosity decreases, the threshold beyond which perturbations from the condition of perfect symmetry trigger a localization with a significantly altered pattern is reduced. In some sense in the case of null viscosity the system is sensible also to vanishing perturbations. This suggest a lost of uniqueness at a stage of the loading program also in case of perfect symmetry.

We also expect that the bifurcation is actually associated with a jump in the time evolution. The reasons are the following.

(i) When passing to the limit $N \to \infty$ in the Rothe method in Section 2.4 we used the viscosity to get the a priori estimates on $\dot{y}^N$. We where not able to get them without viscosity, using only the $H^1((0,T))$ regularity of the load, as can be done for Von Mises plasticity [22]. This suggest that without viscosity the solution of our model cannot not lies in $H^1((0,T))$ but must lie in $BV((0,T))$.

(ii) Similar models of materials with pressure dependent yield strength exhibit jumps. It is the case of Cam–Clay model studied in [12] or the model studied analytically in [11] coupling the associative Drucker–Prager model with damage.

(iii) We expect that when the viscosity is small the evolution became very fast where the deformation occur. In the limit of null viscosity this became a jump.

Remark 5.28. My desire is to understand better the phenomenon of localization when the viscosity if small but nonzero trying to understand how the sensitivity of the system to imperfection increases with decreasing viscosity. Anyway, if it is useful in some way, I will try to study also the case of zero viscosity. But I would like to avoid the use of the notion of energetic solution. Indeed the choice made by the system at the jump point in the energetic solution approach is determined by a global minimizing approach that seems nonphysical to us. This view is confirmed in [25]. The approach that I could accept is that of balanced viscosity solutions where is the viscosity that drive the evolution at the jumps.

Study on the invertibility of the stiffness operator. In what follows are reported our efforts to understand better the phenomenon of localization. In particular we have confined ourself to the time discrete problem (based on the implicit Euler scheme). We computed the "stiffness operator" (the one that would be assembled in the numerical approximation of the model with Newton method). It seems to became singular during plastic flow in absence of viscosity suggesting that the system reaches indeed a bifurcation point (where the jump occur). This analysis should be intended only as a first trail and not as a final result. We hope that the analysis can be extended also to the the time continuous case.

We consider $\mathfrak{D}_\eta$, $\mathfrak{E}_\eta$ and $\mathfrak{F}_\eta$ as dependent continuously a viscosity parameter $\eta > 0$ and vanishing for $\eta = 0$. We call $\mathcal{F}_\eta$ the corresponding functional. We consider the discrete in time problem (Problem 5.6). We focus on a single time step and to simplify the notation we set $y = y^N_{n-1}$, $t = t^N_{n-1}$, $p = p^N_n$ and $\dot{y} = \tau_N^{-1}(y^N_n - y^N_{n-1})$ and $\tau = \tau_N$. Then $y^N_n = y + \tau\dot{y}$. The incremental problem reduces to find $\dot{y} \in \mathscr{X}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p) = \mathcal{F}_\eta(\dot{y}, p, y + \tau\dot{y}, t + \tau) = 0.$$

We compute the Fréchet differential $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ of $\mathcal{F}^{(y,t)}_{(\tau,\eta)}$ with respect to $(\dot{y}, p)$. It can be written as

$$\langle \mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)(z, q), (\tilde{z}, \tilde{q})\rangle = a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z, \tilde{z}) + b^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q, \tilde{z}) + c^{(\dot{y},p,y,t)}_{(\tau,\eta)}(\tilde{q}, z).$$

We have introduced the bilinear form

$$a^{(\dot y,p,y,t)}_{(\tau,\eta)}(z,\tilde z) = \int_\Omega \mathfrak{D}_\eta Ev : E\tilde v dx + \int_\Omega \tau\mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde v - \tilde V - \frac{\tilde\mu}{d}I\right) dx +$$
$$+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2 v : \mathrm{grad}^2 \tilde v dx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V} \circ \tilde{\mathbb{V}} dx +$$
$$+ \int_\Omega Y(p)\partial^2 G(\dot U) V : \tilde V dx$$
$$b^{(\dot y,p,y,t)}_{(\tau,\eta)}(q,\tilde z) = \int_\Omega q(Y'(p)\partial G(\dot U) : \tilde V - \tilde\mu) dx$$
$$c^{(\dot y,p,y,t)}_{(\tau,\eta)}(q,\tilde z) = -\int_\Omega \tau\mu\tilde q dx.$$

We want to study if $\mathscr{K}^{(y,t)}_{(\tau,\eta)}$ is in $\mathrm{Iso}(\mathscr{Z}\times\mathscr{Q},\mathscr{Z}'\times\mathscr{Q}')$. To this aim we can apply Theorem C.10 to the bilinear forms $a^{(\dot y,p,y,t)}_{(\tau,\eta)}$, $b^{(\dot y,p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot y,p,y,t)}_{(\tau,\eta)}$. Reasoning as in the proof of Proposition 5.24 we have that $b^{(\dot y,p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot y,p,y,t)}_{(\tau,\eta)}$ satisfies the hypotheses. Moreover the sets $\mathscr{N}$ and $\tilde{\mathscr{N}}$ are defined exactly in the same way. We need to check the hypotheses on $a^{(\dot y,p,y,t)}_{(\tau,\eta)}$. We reason again as in the proof of Proposition 5.24. For any $z = (v, V, 0) \in \mathscr{N}$ we consider $\tilde z = (v, V, Y'(p)\partial G(\dot U) : V) \in \tilde{\mathscr{N}}$ and conversely. We have that with that choice of z and $\tilde z$ there exist $K > 0$ such that $\|\tilde z\|_{\mathscr{Z}} \le K\|z\|_{\mathscr{Z}}$. Then

$$a^{(\dot y,p,y,t)}_{(\tau,\eta)}(z,\tilde z) = \int_\Omega \mathfrak{D}_\eta Ev : Ev dx + \int_\Omega \tau\mathfrak{C}\left(Ev - V\right) : \left(Ev - V - \frac{Y'(p)\partial G(\dot U) : V}{d}I\right) dx +$$
$$+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2 v : \mathrm{grad}^2 v dx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V} \circ \mathbb{V} dx +$$
$$+ \int_\Omega Y(p)\partial^2 G(\dot U) V : V dx$$
$$\le \delta_\eta\|Ev\|_2^2 + \tau\gamma\|Ev - V\|_2^2 - 2\tau\kappa\|Ev - V\|_2\|V\|_2 +$$
$$+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot U)\|V\|_2^2.$$

Here

$$\kappa = \frac{|\mathfrak{C}|\mathrm{Lip}(Y)\|\partial G\|_\infty}{2\sqrt{d}} \quad \text{and} \quad \sigma(p,\dot U) = \mathrm{essinf}_\Omega\left[Y(p)\,\mathrm{se}(\partial^2 G(\dot U))\right] \ge 0$$

where se denotes the smallest eigenvalue. Then by Young inequality for all $\theta_1 \in (0, \kappa^{-1}\gamma)$ and $\theta_2 \in (0,\infty)$ we have

$$a^{(\dot y,p,y,t)}_{(\tau,\eta)}(z,\tilde z) \le \delta_\eta\|Ev\|_2^2 + \tau(\gamma - \kappa\theta_1)\|Ev - V\|_2^2 - \tau\frac{\kappa}{\theta_1}\|V\|_2^2 +$$
$$+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot U)\|V\|_2^2.$$
$$\le \left[\delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2)\right]\|Ev\|_2^2 + (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 +$$
$$+ \left\{\phi_\eta + \sigma(p,\dot U) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right]\right\}\|V\|_2^2 +$$
$$+ (\phi_\eta + \tau\beta)l^2\|\mathrm{grad}\, V\|_2^2.$$

The hypotheses on $a^{(\dot y,p,y,t)}_{(\tau,\eta)}$ are satisfied if we can find $\theta = (\theta_1,\theta_2) \in (0,\kappa^{-1}\gamma)\times(0,\infty)$ such that

$$\begin{cases} \delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2) > 0 \\ \phi_\eta + \sigma(p,\dot U) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right] > 0. \end{cases}$$

It $\eta > 0$ then for each θ fixed there exists $\tau > 0$ sufficiently small such that these conditions are satisfied. This connected to the fact that the solution is unique as already established in Section 2.7. When $\eta = 0$ then $\delta_\eta = 0$ and $\phi_\eta = 0$. If $\sigma(p,\dot U) > 0$ then we can choose $\theta_2 < 1$ so that the first inequality is satisfied and for $\tau > 0$ sufficiently small also the second inequality is satisfied. If instead $\sigma(p, \dot{U}) = 0$ then the inequalities does not depend on τ and can be solved only if $\beta\gamma > \kappa^2$. Indeed from the first we have $\theta_2 < 1$ and from the second we get

$$\frac{1}{\theta_2} < 1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1}.$$

Then we can find $\theta_2 \in (0, 1)$ if and only if

$$1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1} > 1 \quad \text{or equivalently} \quad \beta - \frac{\kappa}{\theta_1} > 0.$$

Then we can find $\theta_1 \in (0, 1)$ if and only if $\beta\gamma > \kappa^2$. This means that hardening must be strong enough compared to the sensitivity of the yield strength on pressure (notice that κ is proportional to $\mathrm{Lip}(Y)$).

Remark 5.29. In the case of Von Mises plasticity the stiffness operator is an isomorphism for any positive hardening. This due to the fact that the additional term that comes from the dependence of the yield strength on the plastic pressure is absent.

Intuition about the localization process. We now assume that $\eta = 0$ and that G is a smooth and convex dissipation potential that satisfy $G(\dot{U}) = |\dot{U}| + c$ for all $\dot{U} \in \mathrm{DevSym}$ with $|\dot{U}| > R$ where $c \in \mathbb{R}$ and $R > 0$ are two constants. This gives a plastic force that is independent on the rate and equal to the yield strength when the rate is sufficiently hight. So it is a regularization of the rate independent dissipation potential that preserve the existence of a yield limit. We notice that

$$\partial^2 G(\dot{U}) = \frac{|\dot{U}|^2 I - \dot{U} \otimes \dot{U}}{|\dot{U}|^2} \quad \text{for } |\dot{U}| > R$$

so that $\mathrm{se}(\partial^2 G(\dot{U})) = 0$ there. On the contrary $\mathrm{se}(\partial^2 G(\dot{U}))$ must be very hight for $\dot{U}$ near the origin if G is a very close approximation of the rate independent dissipation potential $|\cdot|$. We now consider a loading process. When the load is still small the plastic force is under the yield limit and $\sigma(p, \dot{U})$ is hight. By the analysis we have just performed this ensure that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible and this suggest that the incremental problem is uniquely solved. Moreover we expect that the deformation rate it is not localized in some region but distributed all over the body. As the load increases $\mathrm{se}(\partial^2 G(\dot{U}))$ decreases (uniformly in all the body). Then $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible only for small time steps τ. Assuming that the hardening is sufficiently small, we can conclude that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ became singular (or is nonsingular only for very small time increments so that the evolution stagnates). This is the signal that we have reached the bifurcation point. Now the system to restore the invertibility of $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ and overcome the singularity point makes $\dot{U}$ small everywhere except that in a small shear band. This cause an elastic unloading elsewhere. Then out of the shear band $\mathrm{se}(\partial^2 G(\dot{U}))$ becomes big again. This makes $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ coercive because out of the shear band the plastic strain is held back by the coercivity of $\partial^2 G(\dot{U})$. In the shear band the plastic strain is held back by the surrounding regions through the plastic strain gradient.

3. Numerical approximation

CHAPTER 6

Large deformation model

This chapter is not updated and incomplete. If I have time I will try to get an existence theorem also for the large strain model but I give the priority to understand better the localization of deformation in the small strain model.

1. Formulation

1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic deformation is a map $\chi\colon \Omega \to \mathbb{R}^d$ satisfying $\det \operatorname{grad} \chi > 0$. We set $F = \operatorname{grad} \chi$, $E = (F^T F - I)/2$ and $M = \operatorname{grad}^2 \chi$. We subdivide $\partial\Omega$ in two disjoint parts, Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $\chi = \overline{\chi}$. The distortion of the microscopic structure is described by the plastic distortion $F_p\colon \Omega \to \mathrm{Lin}^+$. We regard F_p as the elastically relaxed distortion so we define the elastic distortion as $F_e = F F_p^{-1}$. The volumetric distortion of the microscopic structure is described by $J_p = \det F_p$ while the shear distortion by $U_p = J_p^{-1} F_p$. We impose the kinematical constrain $J_p = 1$ and to this aim we introduce the Lagrange multiplier p with the meaning of a plastic pressure. The kinematical constrain can be written also as

$$\int_\Omega (J_p - 1)\tilde{q}\, dx = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

1.2. Statics. We denote with $\tilde{v}$ the virtual rate of u.
Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi\colon \Omega \to \mathbb{R}^d$. We set $v = \dot{\chi}$ and $F = \operatorname{Grad} \chi$ and $L = \dot{F}F^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N. The plastic distortion is a map $F_p\colon \Omega \to \mathrm{Lin}^+$. We set $L_p = \dot{F}_p F_p^{-1}$ and we assume that $L_p = D_p \in \mathrm{Sym}$. We set $U_p = \operatorname{dev} D_p$ and $M_p = \operatorname{Grad} U_p \cdot F_p^{-1}$. The elastic distortion is $F_e = F F_p^{-1}$ and we set $L_e = \dot{F}_e F_e^{-1}$.

Lemma 6.1. It holds $L = L_e + F_e D_p F_e^{-1}$.

Proof. We have $L = \dot{F}F^{-1} = (\dot{F}_e F_p + F_e \dot{F}_p) F_p^{-1} F_e^{-1} = \dot{F}_e F_e^{-1} + F_e \dot{F}_p F_p^{-1} F_e^{-1} = L_e + F_e D_p F_e^{-1}$. □

We add the kinematical constrain $\operatorname{tr} D_p = 0$. To impose this constrain we consider a Lagrange multiplier $p\colon \Omega \to \mathbb{R}$. The constrain can be written in weak form as

$$-\int_\Omega \operatorname{tr} L_p \tilde{q}\, dX = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

1.3. Statics. We choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_\Omega T : \tilde{L}_e\, dX + \int_\Omega \left(T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \vdots \tilde{M}_p \right) dX.$$

Here T is the macrostress, T_p is the microstress which can be assumed deviatoric and K_p is the microhyperstress. We choose an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_\Omega f \cdot \tilde{v}\, dX + \int_{\Gamma_N} h \cdot \tilde{v}\, dA + \int_{\Gamma_D} r \cdot \tilde{v}\, dA.$$

Definition 6.2 (Principle of Virtual Powers). Equilibrium holds when $\tilde{\mathscr{W}} = \tilde{\mathscr{P}}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 6.1 and we write the internal virtual power as

$$\tilde{\mathcal{W}} = \int_\Omega TF^{-T} : \mathrm{Grad}\,\tilde{v}\,dX + \int_\Omega \left(-T_e : \tilde{L}_p + T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \,\vdots\, \tilde{M}_p \right) dX$$

where $T_e = F_e^T T F_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned}\tilde{\mathcal{W}} = &-\int_\Omega \mathrm{Div}(TF^{-T}) : \mathrm{Grad}\,\tilde{v}\,dX + \int_{\partial\Omega} (TF^{-T})N \cdot \tilde{v}\,dA + \\ &+ \int_\Omega \left\{ \left[T_p - \operatorname{dev} T_e - \mathrm{Div}\left(K_p \cdot F_p^{-T} \right) \right] : \tilde{U}_p - \left[\frac{1}{d} \operatorname{tr} T_e + p \right] \operatorname{tr} \tilde{L}_p \right\} dX + \\ &+ \int_{\partial\Omega} \left(K_p \cdot F_p^{-T} N \right) : \tilde{U}_p\,dA.\end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned}&\mathrm{Div}(TF^{-T}) + f = 0 \quad \text{in } \Omega\\ &(TF^{-T})N = r \quad \text{in } \Gamma_D\\ &(TF^{-T})N = h \quad \text{in } \Gamma_N\\ &T_p - \operatorname{dev} T_e - \mathrm{Div}\left(K_p \cdot F_p^{-T} \right) = 0 \quad \text{in } \Omega\\ &\frac{1}{d} \operatorname{tr} T_e + p = 0 \quad \text{in } \Omega\\ &K_p \cdot F_p^{-T} N = 0 \quad \text{in } \partial\Omega.\end{aligned}$$

Collection of useful results

APPENDIX A

Notation

1. Tensors

$\mathrm{Lin} = \mathbb{R}^{d\times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

APPENDIX B

Modeling in continuum mechanics

In this chapter we describe the general procedure to develop a thermodynamically consistent mechanical model. The discussion will be purposely informal since the emphasis of this chapter is on modeling.

1. General procedure to build thermodynamically consistent mechanical models

We subdivide the discussion in three parts corresponding to the treatment of kinematics, statics and constitutive relations.

Kinematics. Consider a general mechanical system. Its kinematical configuration is described by elements $z \in \boldsymbol{Z}$ of a vector space $\boldsymbol{Z}$.

Statics. The statics concerns with the instantaneous equilibrium between the internal and external action insisting on the configuration of the systems. The internal and external actions on the configuration of the system are described as a linear functionals, $\mathcal{W} \in \boldsymbol{Z}^*$ and $\mathcal{P} \in \boldsymbol{Z}^*$ respectively. The equilibrium between internal and external actions is written as $\mathcal{W} = \mathcal{P}$. This can be written as

$$\tilde{\mathcal{W}} \doteq \langle \mathcal{W}, \tilde{w} \rangle = \langle \mathcal{P}, \tilde{w} \rangle \doteq \tilde{\mathcal{P}} \quad \text{for all } \tilde{w} \in \boldsymbol{Z}$$

which is the classical statement of the principle of virtual powers. We refer to the test vector $\tilde{w} \in \boldsymbol{Z}$ as the virtual velocity and to $\tilde{\mathscr{W}}$ and $\tilde{\mathscr{P}}$ as the virtual internal end external power respectively.

Constitutive relations. While the external actions $\mathscr{P}$ are assigned as a function $\mathscr{P} \colon [0, T] \to \boldsymbol{Z}^*$ the internal actions are put in relation with the kinematics of the system through constitutive relations. Since we always neglect inertial effects we assume a constitutive relation giving the internal actions as a set valued map $\mathscr{W} \colon \boldsymbol{Z} \times \boldsymbol{Z} \rightsquigarrow \boldsymbol{Z}$ of z and $\dot{z}$. To obtain constitutive relations compatible with thermodynamics we postulate the existence of a

(i) free energy functional $\Psi \colon \boldsymbol{Z} \to (-\infty, \infty]$ which is differentiable on its domain,
(i) subdifferentiable dissipation potential $\Phi \colon \boldsymbol{Z} \times \boldsymbol{Z} \to [0, \infty]$ such that $\Phi(z, \cdot)$ is convex and $\Phi(z, 0) = 0$,

and we define

$$\mathscr{W}(z, \dot{z}) = \partial \Psi(z) + \partial_{\dot{z}} \Phi(z, \dot{z}).$$

Proposition B.1. Let $z \colon [0, T] \to \boldsymbol{Z}$ be a process satisfying the equilibrium condition

$$\partial \Psi(z(t)) + \partial_{\dot{z}} \Phi(z(t), \dot{z}(t)) \ni \mathcal{P}(t)$$

for every $t \in [0, T]$. Let the dissipation rate be defined as $\mathscr{D} = \langle \mathcal{W}, \dot{z} \rangle - \frac{d}{dt} \Psi(z)$. Then $\mathcal{D} \geq 0$.

Proof. Let $\xi(t) \in \partial_{\dot{z}} \Phi(z(t), \dot{z}(t))$ such that $\partial \Psi(t) + \xi(t) = \mathcal{P}(t)$ for all $t \in [0, T]$. Then

$$\mathcal{D} = \langle \partial \Psi(z), \dot{z} \rangle + \langle \xi, \dot{z} \rangle - \frac{d}{dt} \Psi(z) = \langle \xi, \dot{z} \rangle.$$

But by definition of subdifferential we have $\Phi(z, 0) \geq \Phi(z, \dot{z}) - \langle \xi, \dot{z} \rangle$ so that $\langle \xi, \dot{z} \rangle \geq \Phi(z, \dot{z}) \geq 0$. This completes the thesis. □

Remark B.2. The inequality $\mathscr{D} \geq 0$ is known as the dissipation inequality and has the following meaning: the power $\langle \mathscr{W}, \dot{z} \rangle$ absorbed (or released) by the system is stored reversibly as (or extracted from the) free energy up to a positive irreversible dissipation.

2. Treating constrains with the methods of Lagrange multipliers

Linear constraints can be added in the system by substituting a $\boldsymbol{Z}$ with a subspace. Nonlinear constrains can be handled by choosing appropriately the free energy or the dissipation potential. However sometimes it can be useful to treat the constrain reaction as an additional unknown of the system. This happen especially if it is appropriate to include constitutive relations depending on the reactions.

Kinematics of constrain. Let $\boldsymbol{K}$ be a convex subset of $\boldsymbol{Z}$. We consider the constrain $z \in \boldsymbol{K}$.

Constrain reaction. In addiction to the internal and external actions, when we treat a constrain with the method of Lagrange multipliers we must add also the constrain reaction $\mathcal{R} \in \boldsymbol{Z}^*$. Then the equilibrium condition can be written as

$$\mathcal{W} = \mathcal{P} + \mathcal{R}.$$

We make the assumption that the power expanded by the reaction vanish requiring that $\mathcal{R} \in N_{\boldsymbol{K}}(z)$. The constrain reaction is an additional descriptor of the system together with the kinematical descriptor z.

Constitutive relations. In presence of Lagrange multipliers the constitutive relation for $\mathcal{W}$ can be formulated as

$$\mathcal{W}(z, \dot{z}) = \partial\Psi(z) + \partial_{\dot{z}}\Phi(\mathcal{R}, z, \dot{z}).$$

Here the free energy Ψ is exactly as in the unconstrained case and the dissipation potential Φ is exactly as before but include a dependence on $\mathcal{R}$. The dissipation inequality is still valid in the following version.

Proposition B.3. Let $z\colon [0,T] \to \boldsymbol{Z}$ and $\mathcal{R}\colon [0,T] \to \boldsymbol{Z}^*$ satisfying

$$z(t) \in \boldsymbol{K}, \quad \mathcal{R}(t) \in N_{\boldsymbol{K}}(z(t)) \quad \text{and} \quad \partial\Psi(z(t)) + \partial_{\dot{z}}\Phi(\mathcal{R}(t), z(t), \dot{z}(t)) \ni \mathcal{P}(t) + \mathcal{R}(t)$$

for all $t \in [0,T]$. Let the dissipation rate be defined as $\mathscr{D} = \langle \mathcal{W}, \dot{z} \rangle - \frac{d}{dt}\Psi(z)$. Then $\mathcal{D} \geq 0$.

APPENDIX C

Tools from mathematical analysis

In this chapter we collect some tools of linear and nonlinear analysis used in the main body of this thesis.

1. Some classical results

1.1. A general form of the Gronwall inequality.

Proposition C.1. (Gronwall inequality) Let $z \in L^\infty((0,T))$. Let $C \in \mathbb{R}$ and α, β in $L^1((0,T),[0,\infty))$ such that

$$y(t) \le C + \int_0^t \big(\alpha(s)y(s) + \beta(s)\big)\, ds \quad \text{for a.e. } t \in (0,T).$$

Then

$$z(t) \le e^{\int_0^t \alpha(s)ds}\left(C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} ds\right) \quad \text{for a.e. } t \in (0,T).$$

Proof. Let

$$Z(t) = C + \int_0^t \big(\alpha(s)z(s) + \beta(s)\big)\, ds$$

so that $z(t) \le Z(t)$. Clearly $Z \in \mathrm{AC}([0,T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \le \alpha(t)Z(t) + \beta(t) \quad \text{for a.e. } t \in (0,T).$$

Multiplying by $e^{-\int_0^t \alpha(s)ds}$ one gets

$$\frac{d}{dt}\left(Z(t)e^{-\int_0^t \alpha(s)ds}\right) \le \beta(t)e^{-\int_0^t \alpha(s)ds}$$

and integrating on $(0,t)$

$$Z(t)e^{-\int_0^t \alpha(s)ds} \le C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} ds$$

Then

$$z(t) \le Z(t) \le e^{\int_0^t \alpha(s)ds}\left(C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr}\right) \quad \text{for a.e. } t \in (0,T).$$

□

2. Linear functional analysis

2.1. Other.

Proposition C.2. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz. Let $p \in [1,\infty]$ and let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \to u$ in $L^p(\Omega)$. Moreover if $p > d$ then $u_k \to u$ in $C^{0,\alpha}(\overline{\Omega})$ where $\alpha = 1 - \frac{d}{p}$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u. Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$.

The result when $p > d$ follows from Morrey's theorem and a similar argument. □

2.2. Bochner spaces.

Definition C.3. Let V be a Banach space. We say that $u\colon [0,T] \to V$ is absolutely continuous and we write $u \in AC([0,T],V)$ if there exists $m \in L^1((0,T),(0,\infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r)dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u\colon [0,T] \to V$ and $0 \leq a < b \leq T$ we define the total variation on $[a,b]$

$$\mathrm{Var}(u,[a,b]) = \sup\left\{ \sum_{k=1}^{N} \|u(t_k) - u(t_{k-1})\| \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that u has bounded variation and we write $u \in BV([0,T],V)$ if $\mathrm{Var}(u,[0,T]) < \infty$.

Proposition C.4. Let V be a Banach space. Then $W^{1,1}((0,T),V) \subset AC([0,T],V)$. If V is separable and reflexive the equality holds.

Proof. Let $u \in W^{1,1}((0,T),V)$. Then

$$\|u(t) - u(s)\| = \left\| \int_s^t \dot{u}(r)dr \right\| \leq \int_s^t \|\dot{u}(r)\| dr.$$

This proves that $u \in AC([0,T],V)$ with $m = \|\dot{u}\|$.
Now suppose that V is reflexive. We have that

$$\left\| \frac{u(t+h) - u(t)}{h} \right\| \leq \frac{1}{h} \int_t^{t+h} m(r)dr.$$

By the Lebesgue differentiation theorem the left hand side is bounded as $h \to 0$ for a.e. $t \in (0,T)$. Then for each $h_n \to 0$, up to extracting a subsequence,

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{for some } v \in V.$$

We prove that v is common to each sequence. Since V^* is separable and reflexive then there exists ξ_i dense in V^*. Now for each i the map $t \mapsto \langle \xi_i, u(t)\rangle$ is $AC([0,T])$. Then for a.e. $t \in (0,T)$

$$\lim_{h\to 0} \left\langle \xi_i, \frac{u(t+h) - u(t)}{h} \right\rangle \quad \text{exists for all } i. \tag{C.1}$$

Then if $h_n \to 0$ and $k_n \to 0$ are such that

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{and} \quad \frac{u(t+k_n) - u(t)}{k_n} \rightharpoonup w$$

from (C.1) it follows that $\langle \xi_i, v - w\rangle = 0$ for all i. This implies $v = w$. Up to now we have proved that there exists $v\colon (0,T) \to V$ such that

$$\frac{u(t+h) - u(t)}{h} \rightharpoonup v(t) \quad \text{for a.e. } t \in (0,T).$$

Now one should prove that this convergence is actually strong. This will show automatically that $u \in W^{1,1}((0,T),V)$. □

Theorem C.5 (Aubin–Lions–Simon lemma). Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is compact. Let p and q in $[1,\infty]$. Then the space

$$\{u \in L^p((0,T),V) \mid \dot{u} \in L^q((0,T),E)\}$$

i compactly embedded in

(i) $L^p((0,T),H)$ if $p < \infty$
(ii) $C([0,T],H)$ if $p = \infty$ and $q > 1$.

Proof. See [7, Theorem II.5.16]. □

2.3. Generalized mixed elliptic problems. We start with the following theorem.

THEOREM C.6 (Babuŝka). Let V and Q be reflexive Banach spaces. Let $b\colon Q \times V \to \mathbb{R}$ be bilinear form. Let $Z = \{z \in V \big| b(q,z) = 0 \text{ for all } q \in Q\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q,v) \le M\|q\|_Q\|v\|_V \quad \text{for all } (q,v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q\in Q\setminus\{0\}} \frac{b(q,w)}{\|q\|_Q} \ge \beta\|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q,v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q,w) = \langle f, q\rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \le \frac{1}{\beta}\|f\|_{Q'}.$$

PROOF. *Step 1.* Consider the linear operator $B\colon W \to Q'$ defined by

$$\langle Bw, q\rangle = b(w,q).$$

We notice that

$$|\langle Bw, q\rangle| = \inf_{z\in Z} |b(w+z,q)| \le M\|q\|_Q \inf_{z\in Z}\|w+z\|_V = M\|q\|_Q\|w\|_W$$

so that $B \in \mathscr{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta\|w\|_W \le \sup_{q\in Q\setminus\{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that range(B) is closed. Indeed let $f_n = Bw_n$ be such that $f_n \to f$ in Q. Then f_n is a Cauchy sequence. Now

$$\beta\|w_n - w_m\| \le \sup_{q\in Q\setminus\{0\}} \frac{\langle B(w_n - w_m), q\rangle}{\|q\|_Q} \le \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \to w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in$ range(B).

Step 4. We prove that range(B) $= Q'$. Indeed let $(B^T) = \{q \in Q \big| b(q,v) = 0 \text{ for all } v \in V\} = \{0\}$ by hypotheses. On the other hand $(B^T) = \text{range}(B)^\perp$. Since Q is reflexive range(B) $= (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q'. Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta\|w\|_W \le \sup_{q\in Q\setminus\{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = \sup_{q\in Q\setminus\{0\}} \frac{\langle f, q\rangle}{\|q\|_Q} \le \|f\|_{Q'}.$$

This completes the proof. □

LEMMA C.7. Assume that the hypotheses of Theorem C.6 holds. Then

$$\sup_{w\in W\setminus\{0\}} \frac{b(q,w)}{\|w\|_W} \ge \beta\|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f \in Q' \setminus \{0\}} \frac{\langle f, q \rangle}{\|f\|_{Q'}} = \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leq \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\beta \|w\|_W} \leq \sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\beta \|w\|_W}.$$

This completes the proof. □

Then we get immediately the following proposition.

Corollary C.8. Assume the hypotheses of Theorem C.6. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p, w) = \langle g, w \rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{W'}.$$

Proof. Apply first Lemma C.7 and then Theorem C.6 with the role of W and Q swapped. □

Remark C.9. Assume that $g \in V'$ is such that $\langle g, z \rangle = 0$ for all $z \in Z$. Then we can define g as an element of W'. Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w \rangle = \langle g, v \rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_W = \sup_{w \in W \setminus \{0\}} \frac{\langle g, w \rangle}{\|w\|_W} = \sup_{v \in V \setminus \{0\}} \frac{\langle g, v \rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary C.8 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p, v) = \langle g, v \rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{V'}.$$

The following theorem is the main result of this Section. The result is proved in [28] in the Hilbert spaces setting and we have generalized it for Banach spaces.

Theorem C.10. Let V, $\tilde{V}$, Q and $\tilde{Q}$ be reflexive Banach spaces. Let $b\colon Q \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \left\{ \tilde{z} \in \tilde{V} \middle| b(q, \tilde{z}) = 0 \text{ for all } q \in Q \right\}$ and define $\tilde{W} = \tilde{V}/\tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leq M \|q\|_Q \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geq \beta \|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c\colon \tilde{Q} \times V \to \mathbb{R}$ be a bilinear form. Let $Z = \left\{ z \in V \middle| b(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q} \right\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leq N \|\tilde{q}\|_{\tilde{Q}} \|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q} \in \tilde{Q} \setminus \{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma \|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a\colon V \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \le L\|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V}\setminus\{0\}} \frac{a(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \ge \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$a(u, \tilde{v}) + b(p, \tilde{v}) = \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V}$$
$$c(\tilde{q}, u) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}.$$

Moreover

$$\|u\|_V \le \frac{1}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}$$
$$\|p\|_Q \le \frac{1}{\beta}\left(1 + \frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

Proof. By Theorem C.6 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \le \frac{1}{\gamma}\|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w, \tilde{v}) \le L\|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w, \cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f, \cdot \rangle - a(w, \cdot) \in \tilde{Z}'$. Then by Theorem C.6 the equation

$$a(z, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \le \frac{1}{\alpha}\left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma}\|g\|_{\tilde{Q}'}\right).$$

Now define $u = w + z \in V$ and we notice that

$$a(u, \tilde{z}) = \langle f, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \tag{C.2}$$

and

$$\|u\|_V \le \frac{1}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}.$$

Now we notice by (C.2) we have

$$\langle f, \tilde{z} \rangle - a(u, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) - a(z, \tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark C.9 we have $\langle f, \cdot \rangle - a(u, \cdot) \in \tilde{W}'$ and the equation

$$b(p, \tilde{v}) = \langle f, \tilde{v} \rangle - a(u, \tilde{v}) \quad \text{for all } v \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \le \frac{1}{\beta}\left(\|f\|_{\tilde{V}'} + L\|u\|_V\right) \le \frac{1}{\beta}\left(1 + \frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

This concludes the proof. □

3. Nonlinear functional analysis

3.1. Other.

Lemma C.11. Let X and Y be Banach spaces. Let U an open subset of X and $F\colon U \to Y'$. Let $x \in U$. Assume that there exists $A \in \mathscr{L}(X, Y')$ and for all $v \in X$ there exists $\omega\colon \mathbb{R} \to [0,\infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that
$$\left|\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y\right\rangle\right| \le \omega(\epsilon)\|y\|_Y$$
for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A.

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega\colon \mathbb{R} \to [0,\infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that
$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} = \sup_{y \in Y\setminus\{0\}} \frac{\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y\right\rangle}{\|y\|_Y} \le \omega(\epsilon).$$
Then
$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} \to 0$$
as $\epsilon \to 0$. This is exactly the thesis. □

Lemma C.12. Let X and Y be two Banach spaces. Let A_n and A in $\mathscr{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have
$$|\langle A_n x - Ax, y\rangle| \le \omega_n \|x\|_X \|y\|_Y$$
where $\omega_n \to 0$. Then $A_n \to A$ in $\mathscr{L}(X, Y')$.

Proof. We have
$$\|A_n - A\|_{\mathscr{L}(X,Y')} = \sup_{x \in X\setminus\{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x\in X\setminus\{0\}} \sup_{y \in Y\setminus\{0\}} \frac{\langle A_n x - Ax, y\rangle}{\|x\|_X\|y\|_Y} \le \omega_n \to 0.$$
This concludes the proof. □

Proposition C.13. Let V a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then f is Lipschitz continuous on K.

Proof. By contradiction let u_n and v_n two sequences in K such that
$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \ge n.$$
By compactness of K up to extracting subsequences we can assume that $u_n \to u$ and $v_n \to v$. If $u \ne v$ then
$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \to \frac{|F(u) - F(v)|}{\|u - v\|} < \infty$$
which is a contradiction. If $u = v$ then
$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} = \frac{|\langle DF(u), u_n - v_n\rangle + o(\|u_n - v_n\|)|}{\|u_n - v_n\|} \le \|DF(u)\| + o(1) \to \|DF(u)\| < \infty$$
which is a contradiction. □

3.2. Implicit function theorem.

Theorem C.14. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \mathrm{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying
$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$
Moreover if F is of class C^r with $r \ge 2$ also χ is of class C^r.

Proof. See [29]. □

3.3. Fixed point theorems.

THEOREM C.15 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F\colon C \to C$ continuous and with compact range. Then F admits a fixed point.

PROOF. See [29]. $\square$

3.4. Local existence theorem for Cauchy problems in Banach spaces.

THEOREM C.16. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

PROOF. Since G is of class C^1 there there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V. Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathscr{L}(X,X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L. Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathscr{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathscr{X}} = \sup_{t\in[0,T]} \|x(t)\|_X$. Since $\mathscr{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T\colon B \to C^0([0, T], B)$$

defined by

$$T(x)(t) = x_0 + \int_0^t G(x(s), s)ds.$$

We notice that $\|T(x) - x_0\|_X \leq TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T\colon B \to B$. Moreover for any x_1 and x_2 in X we have

$$\|T(x_2) - T(x_1)\|_{\mathscr{X}} \leq \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leq TL\|x_2 - x_1\|_{\mathscr{X}}.$$

Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2. $\square$

3.5. Browder–Minty theorem. The results of this section are taken from [33] and [34].
Let X be a Banach space and $T\colon X \to X'$ be a nonlinear operator and $f \in X'$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

DEFINITION C.17. Let X be a Banach space and let $T\colon X \to X'$. We say that

(i) T is bounded if maps bounded sets of X into bounded sets of X',
(ii) T is coercive if

$$\frac{\langle T(x), x\rangle}{\|x\|} \to \infty \quad \text{as } \|x\| \to \infty$$

and that
(iii) T is monotone if

$$\langle T(x) - T(y), x - y\rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

LEMMA C.18. Let $T\colon X \to X'$ be continuous and $f \in X'$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x\rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to te equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v\rangle \geq 0.$$

Letting $t \to 0$ we have $\langle T(x) - f, v\rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x\rangle = \langle T(y) - T(x), y - x\rangle \geq 0.$$

Then x is a solution to the variational inequality. □

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition C.19. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has al least one solution.

Proof. Without loss of generality we cn assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x)\cdot x}{|x|} = \frac{T(x)\cdot x}{|x|} - \frac{f\cdot x}{|x|} \geq \frac{T(x)\cdot x}{|x|} - |f|.$$

Now define $G\colon \mathbb{R}^n \to \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x)\cdot x = |x|^2 - T(x)\cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x)\cdot x = T(x)\cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi\colon \mathbb{R}^n \to C$ with $C = \overline{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r\frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H\colon C \to C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \mathring{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x\cdot x = H(x)\cdot x = \frac{r}{|G(x)|}G(x)\cdot x < \frac{r}{|G(x)|}|x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x)\cdot x$ for $|x| \geq r$. This is the a contradiction. □

We are now in position to prove the Browder–Minty theorem.

Theorem C.20. Let X be a reflexive and separable Banach space. Let $T\colon X \to X'$ be bounded, continuous, coercive and monotone. Then for every $f \in X'$ there exists a solution $x \in X$ to the equation $T(x) = f$.

Proof. As anticipated the proof relies on the Galerkin method.
Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X. Let

$$X_n = \operatorname{span}\left\{y_m\right\}_{m=1}^n.$$

We define $f_n \in X_n'$ by $\langle f_n, x\rangle = \langle f, x\rangle$ for all $x \in X_n$. We also define $T_n\colon X_n \to X_n'$ by $\langle T_n(x), y\rangle = \langle T(x), y\rangle$ for all x and y in X_n.
Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X_n'} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x), y\rangle}{\|y\|_{X_n}} \leq \sup_{y\in X\setminus\{0\}} \frac{\langle T(xx), y\rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows form the boundedness of T. Moreover let $x_k \to x$ in X_n. Then

$$\|T_n(x_k) - T_n(x)\|_{X_n'} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x_k) - T_n(x), y\rangle}{\|y\|_{X_n}} \leq \sup_{y\in X\setminus\{0\}} \frac{\langle T(x_k) - T(x), y\rangle}{\|y\|_X} = \\ = \|T(x_k) - T(x)\|_{X'} \to 0$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x\rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x\rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T. By C.19 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.

Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n\rangle}{\|x_n\|_X} \le \frac{\langle T_n(x_n), x_n\rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n\rangle}{\|x_n\|_X} = \frac{\langle f, x_n\rangle}{\|x_n\|_X} \le \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X'. Then up to a subsequence

$$x_n \rightharpoonup x \quad \text{in } X$$
$$T(x_n) \overset{*}{\rightharpoonup} g \quad \text{in } X'$$

for some $x \in X$ and $g \in X'$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m\rangle = \lim_{n\to\infty} \langle T(x_n) - f, y_m\rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n\rangle = \langle T_n(x_n), x_n\rangle = \langle g_n, x_n\rangle = \langle g, x_n\rangle \to \langle g, x\rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \ge \langle T(y) - T(x_n), y - x_n\rangle \to \langle T(y) - g, y - x\rangle = \langle T(y) - f, y - x\rangle.$$

Applying Lemma C.18 we get the thesis. □

Actually to obtain the same existence result the monotonicity assumption can be weakened.

Definition C.21. Let X be a Banach space and let $T\colon X \to X'$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k\to\infty} \langle T(x_k), x_k - x\rangle \le 0$$

then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$.

Theorem C.22. Let X be a separable and reflexive Banach space. Let $T\colon X \to X'$ be continuous, coercive and pseudomonotone. Then for every $f \in X'$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is exactly the same as the one of Theorem C.20 except to Step 5. We know that $x_k \rightharpoonup x$ in X, that $T(x_k) \overset{*}{\rightharpoonup} f$ in X' and $\langle T(x_k), x_k\rangle \to \langle f, x\rangle$. Then in particular

$$\langle T(x_k), x_k - x\rangle = \langle T(x_k), x_k\rangle - \langle T(x_k), x\rangle \to \langle f, x\rangle - \langle f, x\rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$. But

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle = \langle f, x - y\rangle.$$

Then

$$\langle f, x - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. □

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition C.23. Let X be a Banach space. Let $T\colon X \to X'$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with
$$\limsup_{k\to\infty}\langle T(x_k), x_k - x\rangle \le 0.$$
Since T is monotone then
$$\langle T(x_k), x_k - x\rangle \ge \langle T(x), x_k - x\rangle \to 0.$$
This implies also
$$\liminf_{k\to\infty}\langle T(x_k), x_k - x\rangle \ge 0.$$
Then
$$\lim_{k\to\infty}\langle T(x_k), x_k - x\rangle = 0.$$
Now let $y \in X$ and set $y_\epsilon = (1-\epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity
$$0 \le \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon\rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x-y) + x_k - x\rangle.$$
Rearranging the terms we have
$$\epsilon\langle T(x_k), x - y\rangle \ge \langle T(y_\epsilon), x_k - x\rangle - \langle T(x_k), x_k - x\rangle + \epsilon\langle T(y_\epsilon), x - y\rangle.$$
Passing to le limit $k \to \infty$ with $\epsilon > 0$ fixed
$$\epsilon \liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge \epsilon\langle T(y_\epsilon), x - y\rangle.$$
Dividing by ϵ and passing to the limit $\epsilon \to 0^+$ we obtain using the continuity of T that
$$\liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge \langle T(x), x - y\rangle.$$
Eventually
$$\liminf_{k\to\infty}\langle T(x_k), x_k - y\rangle = \lim_{k\to\infty}\langle T(x_k), x_k - x\rangle + \liminf_{k\to\infty}\langle T(x_k), x - y\rangle \ge 0.$$
This proves that T is pseudomonotone. □

The following result is useful.

Proposition C.24. Let X be a Banach space. Let T_1 and T_2 from X into X' be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with
$$\limsup_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - x\rangle \le 0.$$
We claim that
$$\limsup_{k\to\infty}\langle T_i(x_k), x_k - x\rangle \le 0 \tag{C.3}$$
for both $i = 1$ and $i = 2$. By contradiction
$$\limsup_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$
Up to taking a subsequence we can assume that
$$\lim_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$
Then
$$\limsup_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \le -\epsilon < 0. \tag{C.4}$$
Then since T_1 is pseudomonotone
$$\liminf_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \ge 0$$
but this contradicts (C.4). Then (C.3) holds. Since T_i is pseudomonotone then for all $y \in X$ we have
$$\liminf_{k\to\infty}\langle T_i(x_k), x_k - y\rangle \ge \langle T_i(x), x - y\rangle.$$
It follows that
$$\liminf_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - y\rangle \ge \langle T_1(x) + T_2(x), x - y\rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. $\square$

3.6. Lower semicontinuity of integral functionals.

Theorem C.25 (Lower semicontinuity of integral functionals). Let $\Omega \subset \mathbb{R}^d$ open bounded. Let $f\colon \Omega \times \mathbb{R}^n \times \mathbb{R}^m \to (-\infty, \infty]$ such that

(i) $f(\cdot, z, p)$ is measurable for all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$
(ii) $f(x, \cdot, \cdot)$ is continuous for a.e. $x \in \Omega$
(iii) $f(x, z, \cdot)$ is convex for all $x \in \Omega$ and $z \in \mathbb{R}^n$
(iv) there exists $\phi \in L^1(\Omega)$ such that $f(x, z, p) \geq \phi(x)$ for a.e. $x \in \Omega$ and all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$.

Then for all measurable $y\colon \Omega \to \mathbb{R}^n$ and $u\colon \Omega \to \mathbb{R}^m$ is well defined

$$I(y, u) = \int_\Omega f(x, y, u)dx.$$

Moreover if $y_k, y\colon \Omega \to \mathbb{R}^n$ measurable satisfy

$$y_k \to y \quad \text{in measure}$$

and $u_k, u \in L^1(\Omega, \mathbb{R}^m)$ satisfy

$$u_k \to u \quad \text{in } L^1(\Omega, \mathbb{R}^n)$$

then

$$I(y, u) \leq \liminf_{k\to\infty} I(y_k, u_k).$$

4. Convex analysis

Example C.26. Let $f(v) = \frac{1}{2}\|v\|^2$. Then $f^*(\xi) = \frac{1}{2}\|\xi\|^2$. Indeed

$$\langle \xi, v\rangle - \frac{1}{2}\|v\|^2 \leq \frac{2\|\xi\|\|v\| - \|v\|^2}{2} \leq \frac{\|\xi\|^2}{2}.$$

This implies $f^*(\xi) \leq \frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon > 0$ and let $v \in V$ such that $\|v\| = \|\xi\|$ and $\langle \xi, v\rangle \geq \|\xi\|^2 - \epsilon$. Then

$$f^*(\xi) \geq \|\xi\|^2 - \epsilon - \frac{1}{2}\|\xi\|^2 \geq \frac{1}{2}\|\xi\|^2 - \epsilon.$$

By arbitrariness of ϵ we conclude.

Let f and g from V to $(-\infty, \infty]$. Their inf-convolution $f \square g\colon V \to (-\infty, \infty]$ is

$$f \square g(v) = \inf_{w\in V}(f(w) + g(v - w)).$$

Proposition C.27. (inf-convolution duality formula) Let f and g from V to $(-\infty, \infty]$. Then

(i) $(f \square g)^* = f^* + g^*$
(ii) $(f + g)^* \leq f^* \square g^*$
(iii) if f and g are convex and proper and there exists $v \in \operatorname{dom} f \cap \operatorname{dom} g$ and f is continuous at v then $(f + g)^* = f^* \square g^*$.

Proof. *Step 1.* It holds that

$$\begin{aligned}
(f \square g)^* &= \sup_{v\in V}\left(\langle \xi, v\rangle - \inf_{w\in V}\big(f(w) + g(v - w)\big)\right) = \\
&= \sup_{v\in V}\sup_{w\in V}\big(\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)\big) = \\
&= \sup_{w\in V}\sup_{v\in V}\big(\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)\big) = \\
&= \sup_{w\in V}\big(\langle \xi, w\rangle - f(w) + g^*(\xi)\big) = f^*(\xi) + g^*(\xi).
\end{aligned}$$

This establish (i).
Step 2. By the Young–Fenchel inequality

$$f^*(\zeta) + g^*(\xi - \zeta) \geq \langle \xi, v\rangle - f(v) - g(v).$$

Taking the supremum over $v \in V$

$$f^*(\zeta) + g^*(\xi - \zeta) \geq (f + g)^*$$

and taking the infimum over $\zeta \in V$

$$f^* \square g^* \geq (f + g)^*.$$

This proves (ii).

Step 3. Now we prove (iii). If $(f + g)^*(\xi) = \infty$ then (iii) is satisfied at ξ. Then suppose that $(f + g)^*(\xi) \in [-\infty, \infty)$. Since $\operatorname{dom} f \cap \operatorname{dom} g \neq \emptyset$ then actually $(f + g)^*(\xi) = c \in (-\infty, \infty)$. Let

$$A = \{(a, v) \in \mathbb{R} \times V \mid a \leq \langle \xi, v\rangle - g(v) - c\}.$$

Clearly A is nonempty and convex. We show that A and $\operatorname{int}(\operatorname{epi} f)$ are disjoint. Indeed if by contradiction $(a, v) \in A \cap \operatorname{int}(\operatorname{epi} f)$ then

$$f(v) < a \leq \langle \xi, v\rangle - g(v) - c$$

and

$$c < \langle \xi, v\rangle - f(v) - g(v) \leq (f + g)^*(\xi) = c.$$

Then by Hahn–Banach theorem there exists $(\gamma, \zeta) \in \mathbb{R} \times V^*$ such that

$$\gamma a + \langle \zeta, v\rangle \leq \gamma b + \langle \zeta, w\rangle \tag{C.5}$$

for all $(a, v) \in \operatorname{int}(\operatorname{epi} f)$ and $(b, w) \in A$. If we fix $v = w \in \operatorname{dom} f \cap \operatorname{dom} g$, $b \in \mathbb{R}$ sufficiently small such that $(b, v) \in A$ and we let $a \to \infty$, we discover that $\gamma \leq 0$ necessarily. If $\gamma = 0$ then inequality (C.5) and Hahn–Banach theorem imply that $\operatorname{dom} f$ and $\operatorname{dom} g$ are separated by a closed hyperplane. This in turn implies that $\operatorname{int}(\operatorname{dom} f)$ and $\operatorname{dom} g$ are disjoint, but this contradicts the hypotheses so that $\beta < 0$. In this case we set $\theta = |\beta|^{-1} f \zeta$ and from inequality (C.5) we get

$$\begin{aligned}
f^*(\theta) &= \sup \left\{\langle \theta, v\rangle - f(v) \mid v \in V\right\} = \\
&= \sup \left\{\langle \theta, v\rangle - a \mid (a, v) \in \operatorname{epi} f\right\} \leq \\
&\leq \inf \left\{\langle \theta, v\rangle - a \mid (a, v) \in A\right\} = \\
&= c + \inf\{\langle \theta - \xi, v\rangle + g(v) \mid v \in V\} = c - g^*(\xi - \theta).
\end{aligned}$$

Then

$$f^* \square g^*(\xi) \leq f^*(\theta) + g^*(\xi - \theta) \leq c = (f + g)^*(\xi).$$

This concludes the proof. □

Proposition C.28 (properties of positively 1-homogeneous convex functions). Let $f \colon V \to (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0) = 0$ and f is subadditive
(ii) for each $x \in \operatorname{dom} f$ it holds $\partial f(x) = \{\xi \in \partial f(0) \mid f(x) = \langle \xi, x\rangle\}$
(iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

Proof. *Step 1.* Let $v \in \operatorname{dom} f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n \to \infty} f\left(\frac{1}{n} v\right) = \liminf_{n \to \infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2} f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).

Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1} f^*(\xi)$ and this implies $f^*(\xi) = 0$.

Step 3. Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0 \rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^{*}* = \iota_K^* = \sigma_K$. This establishes (iii). □

APPENDIX D

Solution concepts

1. Energetic solutions

1.1. Introduction.

1.2. Assumptions and preliminary results. Assume that

(S1) Y and Z separable and reflexive Banach spaces,

(S2) $\mathscr{Y}$ and $\mathscr{Z}$ weakly closed subsets of Y and Z respectively.

Let $Q = Y \times Z$ and $\mathcal{Q} = \mathscr{Y} \times \mathscr{Z}$. The assumptions on the energy functional are that

(E1) $\mathcal{E}\colon [0,T] \times \mathcal{Q} \to (-\infty, \infty]$,

(E2) for all $t \in [0,T]$ and $E \in \mathbb{R}$ the set $\mathscr{D}_E := \{q \in \mathcal{Q} \mid \mathcal{E}(q,t) \leq E\}$ is bounded and weakly closed

(E3) there exists $\mathscr{D} \subset \mathcal{Q}$ such that $\operatorname{dom} \mathcal{E}(\cdot, t) = \mathscr{D}$ for all $t \in [0,T]$,

(E4) for all $q \in \mathscr{D}$, the function $\mathcal{E}(q, \cdot)$ is $C^1([0,T])$ and
  (i) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q,t)| \leq C_1^E(C_0^E + \mathcal{E}(q,t))$ for all $q \in \mathcal{Q}$ and $t \in [0,T]$,
  (ii) for each $E \in \mathbb{E}$ there exists a modulus of continuity ω^E such that for all $q \in \mathscr{D}_E$ and $t, s \in [0,T]$
$$|\partial_t \mathcal{E}(q,t) - \partial_t \mathcal{E}(q,s)| \leq \omega^E(|t-s|).$$

Notice that (E2) is equivalent to say that $\mathcal{E}$ is weakly lower semicontinuous and has bounded sublevels. Moreover (E4, i) implies

$$\mathcal{E}(q,t_1) + C_0^E \leq e^{C_1^E|t_1 - t_0|}\left(\mathcal{E}(q,t_1) + C_0^E\right) \text{ for all } q \in \mathscr{D} \text{ and } t_0, t_1 \in [0,T].$$

This follows from the following lemma.

Lemma D.1. Let $f \in C^1([0,T])$ and there exists $a \in \mathbb{R}$ and $b > 0$ such that $|\dot{f}(t)| \leq b(f(t) + b)$ for all $t \in [0,T]$. Then for all $t_0, t_1 \in [0,T]$ it holds $f(t_0) + a \leq e^{b|t_1 - t_0|}(f(t_1) + a)$.

Proof. Let $z(t) = f(t) + a$. Then $|\dot{z}| \leq bz$. It follows that $\dot{z} \leq bz$ and

$$z(t) \leq z(t_0) + \int_{t_0}^{t} bz(s)ds \quad \text{for all } 0 \leq t_0 \leq t_1 \leq T.$$

By (C.1) it follows that $z(t_1) \leq e^{b(t_1 - t_0)} z(t_0)$ all for all $0 \leq t_0 \leq t_1 \leq T$. But it holds also $\dot{z} \leq -bz$. If we set $\zeta(t) = z(T-t)$ then $\dot{\zeta} \leq b\zeta$. Then reasoning as before we get $z(t_1) \leq \zeta(T - t_1) \leq e^{b(t_0 - t_1)}\zeta(T - t_0) \leq e^{b(t_0 - t_1)} z(t_0)$ for all $0 \leq t_1 \leq t_0 \leq T$. This concludes the proof. □

The assumptions on the dissipation distance are

(D1) $\mathcal{D}\colon \mathscr{Z} \times \mathscr{Z} \to [0,\infty]$ is a quasidistance, that is $\mathcal{D}(z_0, z_1) = 0$ if and only if $z_0 = z_1$ and $\mathcal{D}(z_0, z_2) \leq \mathcal{D}(z_0, z_1) + \mathcal{D}(z_1, z_2)$ for all $z_1, z_2 \in \mathscr{Z}$,

(D2) $\mathcal{D}$ is weakly lower semicontinuous, meaning that $z_n \rightharpoonup z$ and $\hat{z}_n \rightharpoonup \hat{z}$ implies $\mathcal{D}(z, \hat{z}) \leq \liminf_{n\to\infty} \mathcal{D}(z_n, \hat{z}_n)$.

For a given $q = (y,z) \in \mathcal{Q}$, sometimes we write $\mathcal{D}(q)$ to mean $\mathcal{D}(z)$.

Definition D.2. A sequence $(q_n, t_n) \in \mathcal{Q} \times [0,T]$ is said to be stable if

$$\sup_n \mathcal{E}(q_n, t_n) \leq \infty \quad \text{and} \quad q_n \in \mathcal{S}(t_n)$$

where the stable set is defined by

$$\mathcal{S}(t) = \{q \in \mathcal{Q} \mid \mathcal{E}(q,t) < \infty \text{ and } \mathcal{E}(q,t) \leq \mathcal{E}(\hat{q},t) + \mathcal{D}(q,\hat{q}) \text{ for all } \hat{q} \in \mathcal{Q}\}$$

We add the following assumptions regarding the stable set

(C1) for all stable sequences $(q_n, t_n) \in \mathcal{Q} \times [0,T]$ such that $q_n \rightharpoonup q$ and $t_n \to t$, it holds that $\partial_t \mathcal{E}(q_n, t) \to \partial_t \mathcal{E}(q,t)$,

(C2) the stable sets are closed in the sense that, if $(q_n, t_n) \in \mathcal{Q} \times [0,T]$ is a stable sequence such that $q_n \rightharpoonup q$ and $t_n \to t$, then $q \in \mathcal{S}(t)$.

1.3. Definition of energetic solutions. We define energetic solutions as follows.

Definition D.3. A map $q\colon [0,T] \to \mathcal{Q}$ is called energetic solution if it is stable at every instant, that is

$$q(t) \in \mathcal{S}(t) \text{ for all } t \in [0,T],$$

if $t \mapsto \partial_t \mathcal{E}(q(t),t)$ belongs to $L^1((0,T))$ and the energy identity

$$\mathcal{E}(q(t),t) + \mathrm{Diss}_{\mathcal{D}}(z,[s,t]) = \mathcal{E}(q(0),0) + \int_0^t \partial_t \mathcal{E}(q(s),s)ds$$

holds for all $t \in [0,T]$. Here the dissipation is defined as

$$\mathrm{Diss}_{\mathcal{D}}(z,[s,t]) = \sup\left\{\sum_{k=1}^{N} \mathcal{D}(z(t_{k-1}), z(t_k)) \,\middle|\, s = t_0 < \cdots < t_N = t\right\}$$

for all $0 \leq s < t \leq T$.

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1.4. Existence theorem.

Theorem D.4. (existence of energetic solution) If the hypotheses (S), (E), (D) and (C) holds, then for all $q_0 \in \mathcal{S}(0)$ there exists a bounded energetic solutions q such that $q(0) = q_0$.

1.5. Tools to verify the compatibility conditions.

Proposition D.5. Assume that the following property holds

(PC1) for all stable sequence (q_k, t_k) such that $(q_k, t_k) \rightharpoonup (q,t)$ and for all $\hat{q} \in \mathcal{Q}$, there exist a sequence $\hat{q}_k$ such that

$$\limsup_{k\to\infty} \left(\mathcal{E}(\hat{q}_k, t_k) + \mathcal{D}(q_k, \hat{q}_k)\right) \leq \mathcal{E}(\hat{q},t) + \mathcal{D}(q,\hat{q}).$$

Then for all stable sequence (q_k, t_k) such that $(q_k, t_k) \rightharpoonup (q,t)$ it holds that $\lim_{k\to\infty} \mathcal{E}(q_k, t_k) = \mathcal{E}(q,t)$. Moreover (C2) holds.

Assume that in addiction to (PC1) also the following property holds

(PC2) for all $\epsilon > 0$ and $E \in \mathbb{R}$, there exists $\delta > 0$ such that for all $t, s \in [0,T]$ and $q \in \mathcal{Q}$ the implication

$$\mathcal{E}(q,t) \leq E \text{ and } |t-s| < \delta \text{ implies } |\partial_t \mathcal{E}(q,t) - \partial_t \mathcal{E}(q,s)| < \epsilon$$

holds.

Then (C1) holds.

2. Balanced-viscosity solutions

2.1. Introduction. Let

V a separable and reflexive Banach space.

Consider an evolution $u\colon [0,T] \to V$ and the doubly nonlinear equation for the evolution

$$\partial_u \mathcal{E}(u(t),t) + \partial\Phi(\dot u(t)) \ni 0 \quad \text{for } t \in (0,T).$$

Here $\mathcal{E}$ is the energy and Φ is the dissipation potential. In application usually $\mathcal{E}(u,t) = \Psi(u) - \langle \mathcal{F}(t), u\rangle$ where Ψ is the free energy and $\mathcal{F}$ is the external action. For a given $\eta > 0$ consider the time-rescaled evolution $u\colon (0,\eta^{-1}T) \to V$ and the time-rescaled equation

$$\partial_u \mathcal{E}(u(t),\eta t) + \partial\Phi(\dot u(t)) \ni 0 \quad \text{for } t \in (0,\eta^{-1}T). \tag{D.1}$$

For $\eta \to 0$ this describe the situation in which the external actions are mush slower with respect to the characteristic relaxation time of the system. Equation (D.1) can be written as

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot u_\eta(t)) \ni 0 \quad \text{for } t \in (0,T) \tag{D.2}$$

where $u_\eta(t) = u(\eta^{-1}t)$ and $\Phi_\eta(v) = \eta^{-1}\Phi(\eta v)$.

2.2. Assumptions. The energy $\mathcal{E}$ is supposed to satisfy the following assumptions.

$\mathcal{E} \in C^1(V \times [0,T])$

for all $E > 0$ the sublevel $D_E = \{u \in V \mid \sup_{t\in[0,T]} \mathcal{E}(u,t) \le D_E\}$ is weakly compact

($\mathcal{E}$.3) there exists $C > 0$ such that $|\partial_t \mathcal{E}(u,t)| \le C(1 + \mathcal{E}(u,t))$ for all $u \in V$ and $t \in [0,T]$

The dissipation potential is defined by

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$ its rescaled version is

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

Here $\mathcal{R}$ is a rate-independent dissipation potential and $\mathcal{V}$ is a rate-dependent dissipation potential that models viscosity. It is assumed that $\mathcal{R}$ and $\mathcal{V}$ satisfy the following assumptions.

(Φ.1) $\mathcal{R}$ and $\mathcal{V}$ from V into $[0,\infty)$ are continuous, convex and strictly positive in $V \setminus \{0\}$

(Φ.2) $\mathcal{R}$ is positively 1-homogeneous

$\mathcal{V}(v) = \phi(\|v\|)$ with $\phi \in C^1([0,\infty))$ convex such that $\phi(0) = 0$, $\phi'(0) = 0$, $\phi(r) > 0$ for $r > 0$ and $\lim_{r\to\infty} \phi'(r) = \infty$

Notice that from Proposition C.28 it holds $\mathcal{R}(0) = 0$. Moreover notice that ϕ can be extended to an even $C^1(\mathbb{R})$ function.

2.3. Preliminary results regarding the dissipation potential. Let $K = \partial\mathcal{R}(0)$. By Proposition C.28 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$. Notice that there exists a constant $M > 0$ such that $\mathcal{R}(v) \le M\|v\|$ for all $v \in V$. Indeed by contradiction assume that there exist $v_n \in V \setminus \{0\}$ such that $\mathcal{R}(v_n) \ge n\|v_n\|$. Since $\mathcal{R}$ is positively homogeneous the inequality implies $\mathcal{R}(n^{-1}v_n/\|v_n\|) \ge 1$ so $\liminf_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) \ge 1$. But $n^{-1}v_n/\|v_n\| \to 0$ so by continuity of $\mathcal{R}$ it follows $\lim_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) = 0$. This is the desired contradiction.

Since Φ_1 is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta\to 0^+} \Phi_\eta(v) = \inf_{\eta>0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

It simple to check that $\Phi_\eta^*(\xi) = \eta^{-1}\Phi^*(\xi)$. It is possible to obtain a more explicit representation formula for Φ^*. By the inf-convolution duality formula

$$\Phi_\eta^*(\xi) = \left(\mathcal{R} + \mathcal{V}_\eta\right)^*(\xi) = \inf_{\zeta\in V}\left(\mathcal{R}^*(\zeta) + \mathcal{V}_\eta^*(\xi-\zeta)\right) = \eta^{-1}\inf_{\zeta\in K}\mathcal{V}^*(\xi-\zeta) = \eta^{-1}\min_{\zeta\in K}\mathcal{V}^*(\xi-\zeta). \tag{D.3}$$

Last equality holds true because, since K is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

Lemma D.6. The function ϕ^* satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.

Proof. Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. □

Using that lemma and (D.3)

$$\Phi^*_\eta(\xi) = \eta^{-1} \min_{\zeta\in K} \phi^*(\|\xi - \zeta\|). \tag{D.4}$$

2.4. Absolutely continuous and BV evolutions. Let $\mathcal{R}\colon V \to [0,\infty)$ satisfying (Φ.1) and (Φ.2). We say that a map $u\colon [0,T] \to V$ is $\mathrm{AC}_{\mathcal{R}}([0,T],V)$ if there exists $m \in L^1((0,T),[0,\infty))$ such that

$$\mathcal{R}(u(t_1) - u(t_0)) \le \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \le t_0 < t_1 \le T. \tag{D.5}$$

Lemma D.7. Let $u \in \mathrm{AC}_{\mathcal{R}}([0,T],V)$. Then

$$\mathcal{R}[\dot u](t) \doteq \lim_{h\to 0} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right)$$

exists for a.e. $t \in (0,T)$ and $\mathcal{R}[\dot u] \in L^1((0,T))$. Moreover for all m as in (D.5)

$$\mathcal{R}[\dot u](t) \le m(t) \quad \text{for a.e. } t \in (0,T).$$

Proof. For any $s \in (0,T)$ and $t \in (0,T)$ let $\delta_s(t) = \mathcal{R}(u(t) - u(s))$. Then

$$\delta_s(t_1) - \delta_s(t_0) \le \mathcal{R}(u(t_1) - u(t_0)) \le \int_{t_0}^{t_1} m(s)ds \quad \text{for all } 0 \le t_0 < t_1 \le T.$$

The second inequality holds since $\mathcal{R}$ is sub additive by Proposition C.28 so

$$\delta_s(t_1) = \mathcal{R}(u(t_1) - u(s)) \le \mathcal{R}(u(t_1) - u(t_0)) + \mathcal{R}(u(t_0) - u(s)) = \mathcal{R}(u(t_1) - u(t_0)) + \delta_s(t_0).$$

Then the map $t \mapsto \delta_s(t) - \int_0^t m(r)dr = \beta_s(t)$ is non increasing on $(0,T)$ and in particular it is differentiable a.e. on $(0,T)$. This in turns implies that δ_s is differentiable a.e. on $(0,T)$ and

$$\dot\delta_s(t) \le \delta(t) \doteq \liminf_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) \quad \text{for a.e. } t \in (0,T).$$

Note that

$$0 \le \delta(t) \le \liminf_{h\to 0^+} \frac{1}{h}\int_t^{t+h} m(s)ds = m(t) \quad \text{for a.e. } t \in (0,T)$$

and this implies $\delta \in L^1((0,T))$. Since $\delta_s(t) = \beta_s(t) + \int_0^t m(r)dr$ and β_s is non increasing then the singular part of the distributional derivative of δ_s is non positive and

$$\mathcal{R}(u(t) - u(s)) = \delta_s(t) \le \int_s^t \dot\delta_s(r)dr \le \int_s^t \delta(r)dr. \tag{D.6}$$

Let

$$\gamma(t) \doteq \limsup_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right).$$

By (D.6) then $\delta(t) \le \gamma(t) \le \delta(t)$. This proves that

$$\delta^+(t) \doteq \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \delta(t) \quad \text{exists for a.e. } t \in (0,T).$$

Applying the previous argument to $t \mapsto u(T-t)$ and $v \mapsto \mathcal{R}(-v)$ in place of u and $\mathcal{R}$ respectively, follows that

$$\delta^-(t) = \lim_{h\to 0^-} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t)-u(t-h)}{h}\right) \quad \text{exists for a.e. } t \in (0,T)$$

and satisfy the conditions

$$\mathcal{R}(u(t)-u(s)) \leq \int_s^t \delta^-(r)dr.$$

and $\delta^-(t) \leq m(t)$ for a.e. $t \in (0,T)$. This means that δ^+ and δ^- coincides both with the minimal $m \in L^1((0,T),(0,\infty))$ for which (D.5) holds. □

Given a map $u\colon [0,T] \to V$ we define the total variation on $[a,b] \subset [0,T]$ as

$$\mathrm{Var}_{\mathcal{R}}(u,[a,b]) = \sup\left\{ \sum_{k=1}^N \mathcal{R}(u(t_k)) - \mathcal{R}(u(t_{k-1})) \,\middle|\, a = t_0 < \cdots < t_N = b \right\}.$$

We say that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$ if $\mathrm{Var}_{\mathcal{R}}(u,[0,T]) < \infty$.

Lemma D.8. It holds that $\mathrm{AC}([0,T],V) \subset \mathrm{AC}_{\mathcal{R}}([0,T],V)$ and $\mathrm{BV}([0,T],V) \subset \mathrm{BV}_{\mathcal{R}}([0,T],V)$.

Proof. Recall that there exists $M > 0$ such that $\mathcal{R}(v) \leq M\|v\|$. Let $u \in AC([0,T],V)$. Then

$$\mathcal{R}(u(t_1)-u(t_0)) \leq M\|u(t_1)-u(t_0)\| \leq \int_{t_0}^{t_1} Mm(s)ds.$$

This shows that $u \in \mathrm{AC}_{\mathcal{R}}([0,T],V)$. Next let $u \in \mathrm{BV}([0,T],V)$. Then

$$\mathrm{Var}_{\mathcal{R}}(u,[0,T]) \leq M\mathrm{Var}(u,[0,T]).$$

This implies that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$. □

2.5. A chain rule formula for the energy functional.

Proposition D.9. Let $u \in AC([0,T],V)$. Then $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$ and

$$\frac{d}{dt}\mathcal{E}(u(t),t) = \langle \partial_u \mathcal{E}(u(t),t), \dot{u}(t)\rangle + \partial_t \mathcal{E}(u(t),t) \quad \text{for a.e. } t \in (0,T). \tag{D.7}$$

Proof. *Step 1.* Since u is continuous then $K = u([0,T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in K. Then up to a subsequence $t_n \to t$ and by continuity $u_n = u(t_n) \to u(t)$.
Step 2. Since $\mathcal{E} \in C^1(V \times [0,T])$ then by Proposition C.13 it is Lipschitz in $K \times [0,T]$. Let $L > 0$ be the Lipschitz constant.
Step 3. Let $0 \leq s < t \leq T$ and $m \in L^1((0,T),(0,\infty))$ as in Definition C.3. Then

$$|\mathcal{E}(u(t),t) - \mathcal{E}(u(s),s)| \leq L\left(\|u(t)-u(s)\| + |t-s|\right) \leq L\int_s^t (m(r)+r)\,dr.$$

This proves that $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$.
Step 4. By Proposition C.4 it follows that u is a.e. differentiable in $(0,T)$. This implies formula (D.7). □

2.6. Existence results in presence of positive viscosity.

Proposition D.10. Let $u_0 \in V$. Then there exists $u_\eta \in \mathrm{AC}([0,T],V)$ such that $u_\eta(0) = u_0$ and

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for a.e. } t \in (0,T).$$

Moreover the identity

$$\mathcal{E}(u_\eta(t),t) + \int_s^t \Big(\Phi_\eta(\dot{u}_\eta(r)) + \Phi_\eta^*(-\partial_u\mathcal{E}(u_\eta(r),r)\Big)\,dr = \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(r),r)dr \tag{D.8}$$

holds for every $0 \leq s < t \leq T$.

Proof. I still have to report the proof of existence. We prove the identity (D.8). By the properties of subdifferentials

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\langle \partial_u\mathcal{E}(u_\eta(t),t), \dot{u}_\eta(t)\rangle \quad \text{for a.e. } t \in (0,T).$$

By (D.7) we have

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\frac{d}{dt}\mathcal{E}(u_\eta(t),t) + \partial_t\mathcal{E}(u_\eta(t),t) \quad \text{for a.e. } t \in (0,T).$$

Integrating this identity in (s,t) we get the thesis. □

2.7. Contact dissipation potential. The dissipation contact potential $\mathfrak{p}\colon V \times V^* \to [0,\infty)$ is

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\Phi_\eta(v) + \Phi^*_\eta(\xi)\right) = \inf_{\eta>0} \eta^{-1}\left(\Phi(\eta v) + \Phi^*(\xi)\right).$$

By (D.4)

$$\begin{aligned}\mathfrak{p}(v,\xi) &= \inf_{\eta>0}\left(\mathcal{R}(v) + \eta^{-1}\phi(\eta\|v\|) + \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|)\right) = \\ &= \mathcal{R}(v) + \inf_{\eta>0}\left[\eta^{-1}\min_{\zeta\in K}\left(\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)\right)\right] = \\ &= \mathcal{R}(v) + \min_{\zeta\in K}\inf_{\eta>0}\frac{\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)}{\eta}\end{aligned}$$

where last equality holds because the minimum point does not depends on η. But for all r and γ in $\mathbb{R}$

$$\inf_{\eta>0}\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} \geq \gamma r$$

and the identity holds for η such that $\gamma = \phi'(\eta r)$. Then

$$\mathfrak{p}(v,\xi) = \mathcal{R}(v) + \|v\|\min_{\zeta\in K}\|\xi-\zeta\|. \tag{D.9}$$

Let $\mathfrak{f}\colon V \times V \times [0,T] \to [0,\infty)$ defined by

$$\mathfrak{f}(u,v,t) = \mathfrak{p}(v, -\partial_u\mathcal{E}(u,t)) = \mathcal{R}(v) + \mathfrak{e}(u,t)\|v\|$$

where $\mathfrak{e}\colon V \times [0,T] \to [0,\infty)$ is defined by

$$\mathfrak{e}(u,t) = \min_{\zeta\in K}\|\partial_u\mathcal{E}(u,t) + \zeta\|.$$

Notice that $\mathfrak{e}(u,t) = 0$ if and only if $-\partial_u\mathcal{E}(u,t) \in K$.

2.8. Parametrized balanced-viscosity solutions. Let $u_\eta \in \mathrm{AC}([0,T],V)$ be the solution to the doubly nonlinear equation

$$\partial_u\mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot u_\eta(t)) \ni 0 \quad \text{for } t\in(0,T)$$

The identity (D.8) implies

$$\mathcal{E}(u_\eta(t),t) + \int_s^t \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dr \leq \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(t),t)dr \tag{D.10}$$

for all $0 \leq s < t \leq T$.

Lemma D.11. There exists a constant $C>0$ such that

$$\int_0^T \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dt \leq C \quad \text{for all } \eta > 0.$$

Proof. Since $\mathfrak{f}\geq 0$ by ($\mathcal{E}$.3) then inequality (D.10) implies

$$\mathcal{E}(u_\eta(t),t) \leq \mathcal{E}(u_0,0) + \int_0^t \partial_t\mathcal{E}(u(s),s)ds \leq \mathcal{E}(u_0,0) + C\int_0^t (1+\mathcal{E}(u_\eta(s),s))ds.$$

By ($\mathcal{E}$.3) and Gronwall inequality $-1 \leq \mathcal{E}(u_\eta(t),t) \leq C_1$ for all $t\in[0,T]$ and $\eta>0$. Then (D.10) implies

$$\int_0^T \mathfrak{f}(u_\eta(t),\dot u_\eta(t),t)dt \leq 1 + \mathcal{E}(u_0,0) + CT(1+C_1).$$

This concludes the proof. □

Let $\boldsymbol{s}_\eta \colon [0,T] \to [0,S_\eta]$ defined by

$$\boldsymbol{s}_\eta(t) = t + \int_0^t \mathfrak{f}(u_\eta(s), \dot{u}_\eta(s), s)ds \quad \text{and} \quad S_\eta = \boldsymbol{s}_\eta(T). \tag{D.11}$$

Let also

$$\boldsymbol{t}_\eta = \boldsymbol{s}_\eta^{-1} \colon [0,S_\eta] \to [0,T] \quad \text{and} \quad \boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta) \colon [0,S_\eta] \to V.$$

To write the time-rescaled energy identity we introduce the map $\mathfrak{F}_\eta \colon V\times[0,T]\times V\times(0,\infty) \to [0,\infty)$ defined by

$$\mathfrak{F}_\eta(u,t,v,a) = a\Phi_\eta(v/a) + \Phi_\eta^*(-\partial_u\mathcal{E}(u,t)) - a\partial_t\mathcal{E}(u,t) = \mathcal{R}(v) + \mathfrak{E}_\eta(u,t,v,a) - a\partial_t\mathcal{E}(u,t),$$

where $\mathfrak{E}_\eta \colon V\times[0,T]\times V\times(0,\infty) \to [0,\infty)$ is given by

$$\mathfrak{E}_\eta(u,t,v,a) = a\mathcal{V}_\eta(v/a) + a\Phi_\eta^*(-\partial_u\mathcal{E}(u,t)) = a\mathcal{V}_\eta(v/a) + \frac{a}{\eta}\min_{\zeta\in V^*}\phi^*(\|\partial_u\mathcal{E}(u,t)+\zeta\|) =$$

$$= a\mathcal{V}_\eta(v/a) + \frac{a}{\eta}\phi^*\left(\min_{\zeta\in V^*}\|\partial_u\mathcal{E}(u,t)+\zeta\|\right) = a\mathcal{V}_\eta(v/a) + \frac{a}{\eta}\phi^*(\mathfrak{e}(u,t)).$$

Then (D.8) becomes

$$\mathcal{E}(\boldsymbol{u}_\eta(t),\boldsymbol{t}_\eta(t)) + \int_s^t \mathfrak{F}_\eta(\boldsymbol{u}_\eta(r),\boldsymbol{t}_\eta(r),\dot{\boldsymbol{u}}_\eta(r),\dot{\boldsymbol{t}}_\eta(r))dr = \mathcal{E}(\boldsymbol{u}_\eta(s),\boldsymbol{t}_\eta(s)) \quad \text{for all } 0\le s<t\le S_\eta.$$

Identity (D.11) implies

$$1 = \dot{\boldsymbol{t}}_\eta(s) + \mathfrak{f}(\boldsymbol{u}_\eta(s),\dot{\boldsymbol{u}}_\eta(s),\boldsymbol{t}_\eta(s)) \quad \text{for a.e. } s\in(0,S_\eta).$$

Now let $\mathfrak{F}\colon V\times[0,T]\times V\times[0,\infty) \to [0,\infty]$ defined by

$$\mathfrak{F}(u,t,v,a) = \mathcal{R}(v) + \mathfrak{E}(u,t,v,a) - a\partial_t\mathcal{E}(u,t)$$

where $\mathfrak{E}\colon V\times[0,T]\times V\times[0,\infty) \to [0,\infty]$

$$\mathfrak{E}(u,t,v,a) = \begin{cases} \iota_K(-\partial_u\mathcal{E}(u,t)) & \text{if } a>0 \\ \mathfrak{e}(u,t)\|v\| & \text{if } a=0. \end{cases}$$

Definition D.12. We say that a pair $(\boldsymbol{u},\boldsymbol{t})\colon [0,S]\to V\times[0,T]$ is an admissible parametrized curve if

(i) $\boldsymbol{t}$ is non decreasing and absolutely continuous, $\boldsymbol{u}\in \mathrm{AC}_{\mathcal{R}}([0,S],D_E)$ for some $E>0$
(ii) $\boldsymbol{u}$ locally absolutely continuous (with respect to $\|\cdot\|$) in the open set

$$G = \{s\in[0,S] \mid \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))>0\} = \{s\in[0,S] \mid -\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))\notin K\}$$

(iii) the estimate

$$\int_0^S \mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s)ds = \int_0^S \mathcal{R}[\dot{\boldsymbol{u}}](s)ds + \int_G \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|ds < \infty$$

holds.

For every admissible parametrized curve and for a.e. $s\in[0,S]$ is well defined

$$\mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](s) = \begin{cases} \mathcal{R}[\dot{\boldsymbol{u}}](s) + \iota_K(-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))) - \dot{\boldsymbol{t}}(s)\partial_t\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(s)) & \text{if } \dot{\boldsymbol{t}}(s)>0 \\ \mathcal{R}[\dot{\boldsymbol{u}}](s) + \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| & \text{if } \dot{\boldsymbol{t}}(s)=0. \end{cases}$$

where is implicit that if $s\notin G$ then $\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|=0$.

We denote with $\mathscr{A}([0,S],V\times[0,T])$ the set of all admissible parametrized curves and we say that $(\boldsymbol{u},\boldsymbol{t})$ is

(i) non degenerate if $\dot{\boldsymbol{t}}(s)+\mathcal{R}[\dot{\boldsymbol{u}}](s)>0$ for a.e. $s\in[0,S]$
(ii) surjective if $\boldsymbol{t}(0)=0$ and $\boldsymbol{t}(S)=T$
(iii) $\boldsymbol{m}$-normalized with $\boldsymbol{m}\in L^\infty((0,S),(0,\infty))$ if $(\boldsymbol{u},\boldsymbol{t})$ fulfills

$$\dot{\boldsymbol{t}}(s) + \mathcal{R}[\dot{\boldsymbol{u}}](s) + \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| = \boldsymbol{m}(s) \quad \text{for a.e. } s\in(0,S)$$

Definition D.13. A parametrized balanced-viscosity solution is a non degenerate and surjective curve $(\boldsymbol{u}, \boldsymbol{t}) \in \mathscr{A}([0,S], V \times [0,T])$ satisfying

$$\mathcal{E}(\boldsymbol{u}(t), \boldsymbol{t}(t)) + \int_s^t \mathfrak{F}[\boldsymbol{u}, \boldsymbol{t}, \dot{\boldsymbol{u}}, \dot{\boldsymbol{t}}](r)dr = \mathcal{E}(\boldsymbol{u}(s), \boldsymbol{t}(s)) \quad \text{for all } 0 \le s < t \le S.$$

Lemma D.14. Let $\boldsymbol{t}_n \in \mathrm{AC}([0,S],[0,T])$ non decreasing and surjective. Let $E > 0$ and $\boldsymbol{u}_n \in \mathrm{AC}([0,S], D_E)$. Assume that there exists $L > 0$ such that

$$\dot{\boldsymbol{t}}_n(s) + \mathfrak{f}(\boldsymbol{u}_n(s), \dot{\boldsymbol{u}}_n(s), \boldsymbol{t}_n(s)) \le L \quad \text{for a.e. } s \in (0,S) \tag{D.12}$$

Then up to a extracting a subsequence, $(\boldsymbol{u}_n, \boldsymbol{t}_n)$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u}, \boldsymbol{t}) \in \mathscr{A}([0,S], D_E \times [0,T])$. Moreover

$$\dot{\boldsymbol{t}}(s) + \mathfrak{f}[\boldsymbol{u}, \dot{\boldsymbol{u}}, \boldsymbol{t}](s) \le L \quad \text{for a.e. } s \in (0,S)$$

and the following lower semicontinuity properties holds

$$\liminf_{n\to\infty} \int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))dr \ge \int_s^t \mathcal{R}[\dot{\boldsymbol{u}}](r)dr$$

$$\liminf_{n\to\infty} \int_s^t \mathfrak{e}(\boldsymbol{u}_n(r), \boldsymbol{t}_n(r))\|\dot{\boldsymbol{u}}_n(r)\|dr \ge \int_s^t \mathfrak{e}(\boldsymbol{u}(r), \boldsymbol{t}(r))\|\dot{\boldsymbol{u}}(r)\|dr$$

$$\liminf_{n\to\infty} \int_s^t \mathfrak{E}_{\eta_n}(\boldsymbol{u}_n(r), \boldsymbol{t}_n(r), \dot{\boldsymbol{u}}_n(r), \dot{\boldsymbol{t}}_n(r))dr \ge \int_s^t \mathfrak{E}(\boldsymbol{u}(r), \boldsymbol{t}(r), \dot{\boldsymbol{u}}(r), \dot{\boldsymbol{t}}(r))dr$$

for every $0 \le s < t \le S$ and $\eta_n \in (0,T)$ with $\eta_n \to 0$.

Proof. *Step 1.* From the fact that $\boldsymbol{t}_n(0) = 0$ and that $\dot{\boldsymbol{t}}_n$ are uniformly bounded (by (D.12)), it follows that $\boldsymbol{t}_n$ are uniformly Lipschitz continuous. Then by Ascoli–Arzelà up to extracting a subsequence, $\boldsymbol{t}_n$ converges uniformly to some $\boldsymbol{t} \in C^{0,1}([0,T])$. VERIFICARE CHE $\boldsymbol{t}$ SODDISFA LE PROPRIETÀ RICHIESTE.

Step 2. Since $\mathcal{R}$ is continuous and D_E is weakly compact we can prove that for all $\delta > 0$ there exists $M_\delta > 0$ such that

$$|v - w| \le \delta + M_\delta \mathcal{R}(v - w).$$

Indeed by contradiction assume that there is some $\delta > 0$ such that for all n there exist v_n and w_n with

$$|v_n - w_n| \ge \delta + n\mathcal{R}(v - w).$$

Since D_E is weakly compact, we can assume that $v_n \rightharpoonup v$ and $w_n \rightharpoonup w$. Up to extracting a subsequence we can also assume that $v_n \to v$ and $w_n \to w$ in H. QUI VA SISTEMATA LA SITUAZIONE. IL MOTIVO È CHE SERVIREBBE CONVERGENZA FORTE NELLA (SEMI)NORMA IN CUI $\mathcal{R}$ È CONTINUA. Then

$$|v - w| \ge \delta + \limsup_{n\to\infty} n\mathcal{R}(v_n - w_n).$$

This is a contradiction because necessarily $v \ne w$ so that $\mathcal{R}(v - w) > 0$ and in this case

$$\limsup_{n\to\infty} n\mathcal{R}(v_n - w_n) = \infty.$$

Let $\omega\colon [0,\infty) \to [0,\infty)$ defined by $\omega(r) = \inf_{\delta>0}(\delta + M_\delta r)$. Notice that ω is non decreasing and $\lim_{r\to 0}\omega(r) = 0$.

Step 3. By Jensen inequality and assumption (D.12)

$$\mathcal{R}(\boldsymbol{u}_n(t) - \boldsymbol{u}_n(s)) = |t - s|\mathcal{R}\left(\frac{1}{|t - r|}\int_s^t \dot{\boldsymbol{u}}_n(r)dr\right) \le \int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))drl \le L|t - s|.$$

Then

$$|\boldsymbol{u}_n(t) - \boldsymbol{u}_n(s)| \le \omega(\mathcal{R}(\boldsymbol{u}_n(t) - \boldsymbol{u}_n(s))) \le \omega(L|t - s|).$$

This implies that $\boldsymbol{u}_n$ are equicontinuous with respect to $|\cdot|$. Then up to extracting a subsequence

$$\lim_{n\to\infty} \sup_{s\in[0,S]} |\boldsymbol{u}(s) - \boldsymbol{u}_n(s)| = 0$$

for some $\boldsymbol{u} \in \mathrm{AC}_{|\cdot|}([0,T], V)$. It follows thatB □

Theorem D.15 (existence of parametrized solutions). Let $u_\eta \in \mathrm{AC}([0,T],V)$ be solutions to the doubly nonlinear equation (D.2) with fixed initial condition $u_0 \in V$. Consider non decreasing and surjective absolutely continuous time rescalings $\boldsymbol{t}_\eta \colon [0,S] \to [0,T]$ and let $\boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta)$. Suppose that as $\eta \to 0$

$$\text{(D.13)} \qquad \boldsymbol{m}_\eta = \dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta) \overset{*}{\rightharpoonup} \boldsymbol{m} \quad \text{in } L^\infty((0,S))$$

for some $\boldsymbol{m} \in L^\infty((0,S),(0,\infty))$. Then there exists a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_\eta, \boldsymbol{t}_\eta)$ converges uniformly on $[0,S]$ to a parametrized balanced viscosity solution $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V \times [0,T])$. Moreover $(\boldsymbol{u},\boldsymbol{t})$ is $\boldsymbol{m}$-normalized.

Proof. Arguing as in the proof of Lemma D.11 one finds that there exists $E > 0$ such that $\boldsymbol{u}_\eta(s) \in D_E$ for all $\eta > 0$ and $s \in [0,S]$. Since (D.13) holds, there exists $L > 0$ such that $\|\dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta)\|_\infty \le L$. Then the hypotheses of Lemma D.14 are satisfied and we can find a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_{\eta_n}, \boldsymbol{t}_{\eta_n})$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V \times [0,T])$. Since $\mathcal{E} \in C^1(V \times [0,T])$ then

$$\text{(D.14)} \qquad \mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \to \mathcal{E}(\boldsymbol{u}(s), \boldsymbol{t}(s)) \quad \text{for all } s \in [0,S].$$

Now, since $\dot{\boldsymbol{t}}_\eta$ are bounded in $L^\infty((0,S))$, up to extracting a subsequence it holds that $\dot{\boldsymbol{t}}_{\eta_n} \overset{*}{\rightharpoonup} \dot{\boldsymbol{t}}$ in $L^\infty((0,T))$. In particular $\dot{\boldsymbol{t}}_{\eta_n} \rightharpoonup \dot{\boldsymbol{t}}$ in $L^1((0,T))$. Then by Lemma D.16

$$\text{(D.15)} \qquad \liminf_{n\to\infty} \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r))dr \ge \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}(r), \boldsymbol{t}(r))dr.$$

Now taking the inferior limit as $n \to \infty$ of the energy identity

$$\mathcal{E}(\boldsymbol{u}_{\eta_n}(t), \boldsymbol{t}_{\eta_n}(t)) + \int_s^t \mathfrak{F}_{\eta_n}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r), \dot{\boldsymbol{u}}_{\eta_n}(r), \dot{\boldsymbol{t}}_{\eta_n}(r))dr \le \mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \quad \text{for all } 0 \le s < t \le T$$

taking into account (D.14), (D.15) and the lower semicontinuity estimates of Lemma D.14

$$\mathcal{E}(\boldsymbol{u}(t), \boldsymbol{t}(t)) + \int_s^t \mathfrak{F}[\boldsymbol{u}, \boldsymbol{t}, \dot{\boldsymbol{u}}, \dot{\boldsymbol{t}}](r)dr \le \mathcal{E}(\boldsymbol{u}(s), \boldsymbol{t}(s)) + \int_s^t \partial_t \mathcal{E}(\boldsymbol{u}(r), \boldsymbol{t}(r))dr$$

for all $0 \le s < t \le T$. ADESSO MANCA DA OTTENERE L'ALTRA STIMA (E VOLENDO MOSTRARE CHE SODDISFA LA GIUSTA NORMALIZZAZIONE). □

Lemma D.16. Let $f_n, f, m_n, m \colon (0,T) \to [0,\infty]$ be measurable functions that satisfy

$$\liminf_{n\to\infty} f_n(s) \ge f(s) \quad \text{for a.e. } s \in (0,T)$$
$$m_n \rightharpoonup m \quad \text{in } L^1((0,T)).$$

Then

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \ge \int_0^T f(s)m(s)ds.$$

Proof. Let $g_k(s) = \inf_{n\ge k} \min\{f_n(s), n\}$. Notice that $g_k \in L^\infty((0,T))$. Fix k and notice that for all $n \ge k$

$$\int_0^T f_n(s)m_n(s)ds \ge \int_0^T g_k(s)m_n(s)ds.$$

Taking the inferior limit as $n \to \infty$

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \ge \int_0^T g_k(s)m(s)ds.$$

Now g_k is an increasing sequence and

$$\lim_{k\to\infty} g_k(s) = \lim_{k\to\infty} \inf_{n\ge k} \min\{f_n(s), n\} = \liminf_{n\to\infty} f_n(s) \ge f(s) \quad \text{for a.e. } s \in (0,S).$$

The passing to the limit $k \to \infty$ and using monotone convergence the proof is concluded. □

2.9. An example. Let $V = L^2(\Omega)$ with $\Omega \subset \mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v| dx \quad \text{and} \quad \mathcal{V}(v) = \frac{1}{2}\int_\Omega |v|^2 dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v) - \mathcal{R}(w)| \leq \int_\Omega ||v| - |w||\, dx \leq \int_\Omega |v - w| dx \leq C\|v - w\|_2.$$

It holds that K is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi \in V^* = L^2(\Omega)$ satisfies $\|\xi\|_\infty \leq 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v \in V} \int_\Omega (\xi v - |v|)\, dx \leq \sup_{v \in V} \int_\Omega (\xi v - |v|)\, dx \leq \sup_{v \in V} \int_\Omega (|\xi| - 1)|v| dx \leq 0$$

and $\mathcal{R}^*(\xi) \geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon > 0$ and $w \in L^1(\Omega)$ such that $\langle \xi, w\rangle \geq 1 + \epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta > 0$ there exist $v \in L^2(\Omega)$ such that $\|w - v\| \leq \delta$. Then for all $\lambda > 0$

$$\mathcal{R}^*(\xi) \geq \langle \xi, \lambda v\rangle - \|\lambda v\|_1 \geq \lambda\left(\langle \xi, w\rangle - \|w\|_1 - \delta(1 + \|\xi\|)\right) \geq \lambda\left(\epsilon - \delta(1 + \|\xi\|)\right).$$

Choosing δ such that $\epsilon - \delta(1 + \|\xi\|) > 0$ and letting $\lambda \to \infty$ we get $\mathcal{R}(\xi) = \infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (D.3)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta} \min_{\zeta \in K} \|\xi - \zeta\|_2^2 = \frac{1}{2\eta}\int_\Omega \min_{|\zeta| \leq 1} |\xi - \zeta|^2 dx.$$

By (D.9) it follows

$$\mathfrak{p}(v, \xi) = \|v\|_1 + \|v\|_2 \min_{\zeta \in K} \|\xi - \zeta\|_2.$$

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